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جامعة الإمارات العربية المتحدة
United Arab Emirates University

United Arab Emirates University

College of Engineering

Department of Civil and Environmental Engineering

**DURABILITY OF GLASS FIBER-REINFORCED POLYMER BARS
IN SEAWATER-CONTAMINATED CONCRETE**

Abdelrahman A. E. Alsallamin

This thesis is submitted in partial fulfilment of the requirements for the degree of
Master of Science in Civil Engineering

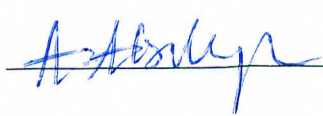
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May 2017

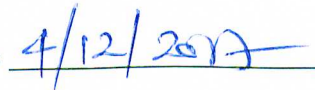
Declaration of Original Work

I, Abdelrahman A. E. Alsallamin, the undersigned, a graduate student at the United Arab Emirates University (UAEU), and the author of this thesis entitled “*Durability of Glass Fiber-Reinforced Polymer Bars in Seawater-Contaminated Concrete*”, hereby, solemnly declare that this thesis is my own original research work that has been done and prepared by me under the supervision of Dr. Tamer El Maaddawy, in the College of Engineering at UAEU. This work has not previously been presented or published, or formed the basis for the award of any academic degree, diploma or a similar title at this or any other university. Any materials borrowed from other sources (whether published or unpublished) and relied upon or included in my thesis have been properly cited and acknowledged in accordance with appropriate academic conventions. I further declare that there is no potential conflict of interest with respect to the research, data collection, authorship, presentation and/or publication of this thesis.

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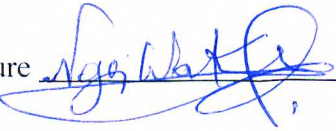
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Abstract

This research aims to investigate the durability performance and microstructure characteristics of two different types of glass fiber-reinforced polymer (GFRP) bars in severe environment. GFRP bars encased in seawater-contaminated concrete were immersed in tap water for 5, 10, and 15 months at temperatures of 20, 40, and 60°C. Half of the specimens were conditioned under a sustained load of 25% of their ultimate strength whereas the other half was conditioned without load. Following conditioning, the GFRP bars were retrieved then tested to failure under uniaxial tension. Microstructure analysis was performed by employing differential scanning calorimetry (DSC), Fourier transform infrared (FTIR) spectroscopy, scanning electron microscopy (SEM), and matrix digestion using nitric acid.

Type I GFRP bars, with the lower moisture uptake, exhibited insignificant strength reductions in the range of 2 to 15% when conditioned without load. Their Type II counterparts exhibited higher moisture uptake, higher hydroxyl ions, lower matrix retention, and thus, substantial strength reductions in the range of 19 to 50% were recorded. The extent of degradation was more sensitive to the conditioning temperature rather than conditioning duration. A decrease in the glass transition temperature (T_g) of both types of GFRP bars was recorded, indicating matrix plasticization. Results of SEM highlighted matrix disintegration and fiber debonding after conditioning.

Specimens conditioned under a sustained load exhibited higher moisture absorption than that of their counterparts conditioned without load. None of the loaded specimens conditioned at 20°C were creep-ruptured during conditioning. The presence of the sustained load during conditioning at 20°C for 15 months reduced the tensile strength retention by approximately 14 and 5% for Type I and Type II GFRP bars, respectively. In contrast, many bars were creep-ruptured and significant reductions in the tensile strength retention were recorded due to the presence of the sustained load during conditioning at the higher temperatures of 40 and 60°C.

The accelerated aging test data along with the Arrhenius concept were employed to develop a durability design model that can predict the tensile strength retention of both types of GFRP bars in moist seawater-contaminated concrete.

Keywords: Accelerated aging, GFRP, Concrete, Durability, Microstructure.

Title and Abstract (in Arabic)

قدرة تحمل قضبان البوليمر المقوى بالألياف الزجاجية مع الزمن في الخرسانة المختلطة بمياه البحر

الملخص

يهدف هذا البحث إلى التحقيق في أداء المتانة والخصائص المجهريّة لنوعين مختلفين من قضبان البوليمر المقوى بالألياف الزجاجية (GFRP) في بيئة قاسية. تم غمر قضبان البوليمر المقوى بالألياف الزجاجية المغطاة في الخرسانة المختلطة بمياه البحر في مياه الصنبور لمدة ٥ و ١٠ و ١٥ شهراً عند درجات حرارة ٢٠ و ٤٠ و ٦٠ درجة مئوية. نصف العينات كانت تحت تحميل مستمر أثناء المعالجة بما يقدر ب ٢٥٪ من قوتها القصوى في حين أن النصف الآخر كان دون تحميل. بعدما تمت المعالجة الأولية، تم استخراج قضبان البوليمر المقوى بالألياف الزجاجية من الخرسانة ثم اختبرت بالشد أحادي المحور وصولاً إلى انهيار العينة. تم إجراء تحاليل مجهرية باستخدام تفضلية المسح الكالوري (DSC) والتحويل الطيفي لفورية الأشعة فوق الحمراء (FTIR) ومسح المجهر الإلكتروني (SEM) وتآكل النسيج باستخدام حمض النتريك. النوع الأول من القضبان أظهر انخفاض في امتصاص الرطوبة، وكذلك انخفاض ضئيل في قوة التحمل تتراوح بين ٢ و ١٥٪ في حالة عدم التعرض للتحميل أثناء المعالجة. أما النوع الثاني من القضبان أظهرت ارتفاع في امتصاص الرطوبة ومستويات عالية من أيونات الهيدروكسيل (OH) وانخفاض في تآكل النسيج مما أدى إلى انخفاضات كبيرة في قوة التحمل تتراوح بين ١٩ و ٥٠٪. هذا التدهور في قوة التحمل كان أكثر حساسية لدرجة الحرارة بدلاً من المدة الزمنية. تم ملاحظة انخفاض في درجة الحرارة الانتقالية للزجاج (T_g) لكلا النوعين من القضبان مما يدل على لدونة النسيج. أظهرت نتائج المسح المجهر الإلكتروني تفكك في الأنسجة والألياف الزجاجية. العينات المعرضة للتحميل أثناء المعالجة أظهرت قابلية أعلى لامتصاص الرطوبة. لم تتعرض العينات المحملة في درجة حرارة ٢٠ درجة مئوية لتمزق زحف طويل الأمد. أدى وجود الحمل المستمر أثناء المعالجة عند ٢٠ درجة مئوية لمدة ١٥ شهراً إلى تقليل نسبة الاحتفاظ بقدرة تحمل الشد بحوالي ١٤ و ٥٪ بالنسبة إلى قضبان النوع الأول والنوع الثاني على التوالي. على النقيض من ذلك، تم تسجيل العديد من حالات تمزق زحف طويل الأمد وتم ملاحظة انخفاض كبير في قدرة تحمل الشد بسبب وجود التحميل المستمر أثناء المعالجة في درجات حرارة ٤٠ و ٦٠ درجة مئوية. تم استخدام البيانات جنباً إلى جنب مع مفهوم أرينيوس لتطوير نموذج تصميم المتانة الذي من خلاله يمكن التنبؤ بقدرة تحمل الشد لكلا النوعين من القضبان في الخرسانة المختلطة بمياه البحر الرطبة.

مفاهيم البحث الرئيسية: اختبارات المتانة المتسارعة، قضبان البوليمر المقوى بالألياف الزجاجية، خرسانة، متانة، مجهرية

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Dedication

To my beloved parents and family

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List of Abbreviations and Symbols

f_{20} :	Stress at 20% of the Tensile Strength
f_{50} :	Stress at 50% of the Tensile Strength
ϵ_{20} :	Strain at 20% of the Tensile Strength Tested
ϵ_{50} :	Strain at 50% of the Tensile Strength
a and b:	Regression Constants
A:	Cross Sectional of Concrete Cylinder in cm^2
a:	The Exposed Area of the Specimen, in mm^2
ALK.S.:	Alkaline Solution
AS:	Acid Solution
ASTM:	American Society for Testing and Materials
BFRE:	Basalt Fiber Reinforced Polymer
c.c:	Collected Volume of Water in (m^3)
D:	Diameter of Concrete Cylinder in (mm)
d:	The Density of the Water in g/mm^3
DIW:	Deionized Water
DSC:	Differential Scanning Calorimetry
E_a :	Activation Energy
E_f :	Tensile Modulus Of Elasticity,
f_c' :	Compressive Strength
FRP:	Fiber Reinforced Polymer
FTIR:	Fourier Transform Infrared Spectroscopy
f_u :	Tensile Strength
GFRP:	Glass Fiber Reinforced Polymer
GPa:	Giga Pascal

h:	Thickness of Specimen (Mm)
HP-SWSSC:	High Performance Sweater and Sea Sand Concrete
I:	Absorption
I _o :	Current (Amperes) Immediately after Voltage is Applied
I _t :	Current (Amperes) at T Min after Voltage is Applied
K:	Degradation Rate (1/Time)
KBr:	Potassium Bromide
MPa:	Mega Pascal
N/A:	Not Applicable
N-SWSSC:	Normal Sweater and Sea Sand Concrete
P:	Applied Load in (N)
pH:	Potential of Hydrogen
R:	Universal Gas Constant
RCPT:	Rapid Chloride Penetration Test
SEM:	Scanning Electron Microscope
SL:	Sustained Load
SS:	Salt Solution
SW:	Seawater
T:	Temperature in Kelvin
t:	Exposure Time in Days or Months
Temp.:	Temperature
Tg:	The Glass Transition Temperature
TSF:	Time Shift Factor
TSR:	Tensile Strength Retention
TW:	Tap Water

UL:	Unloaded
UPV:	Ultra-Pulse Velocity
V_f :	Volume of Fiber
Y:	Tensile Strength Retention (%)
Z:	Impedance Measured by The Device (Ω)
ρ :	Resistivity ($\Omega\cdot\text{cm}$)
τ :	Fitted Parameter

Chapter 1: Introduction

1.1 Problem Statement

Many of the existing infrastructures in Gulf region are located in coastal cities of the Arabian Gulf. The severe environment of these cities accelerates corrosion of the steel reinforcement in concrete. Corrosion damage and associated cracking result in severe safety hazards and large financial losses. The use of non-corrosive materials as reinforcement in concrete structures would prolong their service life and reduce the maintenance cost. Glass fiber-reinforced polymer (GFRP) bars have a great potential to replace the traditional steel reinforcement and eliminate corrosion problems. Nevertheless, the durability performance of GFRP bars in concrete subjected to seawater splash at elevated temperatures is questionable. There is a need to develop a realistic durability design model that can predict the tensile strength retention of GFRP bars in such a harsh environment. The durability performance and microstructural characteristics of GFRP bars in moist seawater-contaminated concrete should be rigorously assessed before GFRP can be routinely used as reinforcement in concrete structures exposed to severe environment.

1.2 Goals and Objectives

The literature survey (described in chapter 2) showed wide and significant variations in the tensile strength reduction of GFRP bars caused by environmental exposure. Numerous studies were conducted on durability of GFRP bars in simulated concrete pore-solutions but few tests were performed on GFRP bars within the actual concrete environment. The long-term performance of GFRP bars in seawater-

contaminated concrete needs to be investigated in order to facilitate the development of a realistic durability design model of GFRP reinforcing bars in regions of severe environment. The main goal of this research work is to examine the durability of GFRP bars in moist seawater-contaminated concrete. The specific objectives are to:

- Examine the durability performance and microstructure characteristics of two different types of commercially-produced GFRP bars conditioned in moist seawater-contaminated concrete.
- Investigate the effect of varying the conditioning time, temperature and presence of a sustained load on the extent of degradation of the conditioned GFRP bars.
- Develop realistic durability design models that can predict the tensile strength retention of both types of GFRP in moist seawater-contaminated concrete.

1.3 Methodology and Approach

A comprehensive literature review has been conducted to summarize the available experimental studies of GFRP bars in concrete. The geometrical and mechanical properties of three replicate unconditioned GFRP bar samples have been evaluated. These properties include cross-sectional properties, tensile strength, modulus and rupture strain. The cross-sectional and tensile properties are evaluated using the ACI 440.3R-12 [1].

The GFRP bars in seawater-contaminated concrete have been subjected to accelerated ageing. The following parameters are adopted in the study:

- Temperature: 20, 40, and 60°C
- Time of exposure: 5, 10, and 15 months
- Surrounding media: moist seawater-contaminated concrete

- Loading condition during accelerate aging: unloaded versus **sustained load of 25% of tensile strength**
- Type of GFRP bars: two types of commercially-produced GFRP bars.

Following conditioning, the GFRP bars have been tested to failure under axial tension to determine the retention in tensile strength and modulus. The microstructure characteristics of GFRP bars have been evaluated by conducting Scanning Electron Microscope (SEM), Differential Scanning Calorimetry (DSC), and Fourier Transform Infrared Spectroscopy (FTIR), and matrix digestion using nitric acid. The moisture absorption of GFRP bars were obtained as well.

The Arrhenius concept, described in chapter 5, has been employed along with the laboratory data of the accelerated aging tests to develop a durability design model that can predict the long-term performance of GFRP bars in moist seawater-contaminated concrete.

1.4 Study Contribution

The corrosion of steel in concrete structures needs costly repair and maintenance, in Canada the cost of repairs of multistory parking garages is estimated to be around 6 billion CDN\$ dollar, while in United States it ranges between 50 and 100 billion US\$ [2,3]. Because of their high strength to weight ratio, light weight, and high corrosion resistance, GFRP bars can be considered as an ideal solution to eliminate corrosion problems in concrete [4,5,6]. GFRP bars are, however, vulnerable to degradation in tensile properties when subjected to alkaline or acidic solution, moisture/water, and elevated temperatures [7,8,9].

Although GFRP bars have a great potential to replace conventional steel and overcome corrosion problems in offshore structures and bridge decks in coastal cities

or seaports, evaluation of their durability performance and microstructural characteristics when conditioned in moist seawater-contaminated concrete has received little attention. This research aims at filling this gap and providing insight into the durability performance of two different types of commercially-produced GFRP bars in severe environment. Three conditioning tanks with built-in heaters and thermostat were fabricated for conditioning of GFRP bars in unloaded condition. Tensile strength retentions of conditioned GFRP bars were measured to evaluate their durability performance. The tensile strength results are supplemented by rigorous microstructural analysis. The interaction between the void content, moisture uptake, matrix retention, fiber-matrix debonding, tensile strength retention, and the increase in hydroxyl ions caused by hydrolysis is elucidated.

A total of 12 sustained loading steel frames have been designed and fabricated at UAEU. Each frame can apply a constant sustained load to three replicate specimens at a time during conditioning. GFRP bars conditioned under a sustained load were monitored and any creep-ruptured bars were recorded. The effect of conditioning under a sustained load on the rate of degradation in mechanical properties of GFRP was investigated. New durability design models that can predict the tensile strength retention of the two types of GFRP bars in moist seawater-concrete were developed.

1.5 Organization of the Report

This thesis is divided into six chapters as follows:

Chapter 1: A brief introduction is given about the problem statement, followed by the research objectives, significance and organization of the thesis.

Chapter 2: A detailed literature review on various topics on the use of GFRP in engineering structures, durability of GFRP, and factors affecting the durability of GFRP is provided.

Chapter 3: In this chapter, details of the experimental work, sample preparation, test set-up and instrumentation are explained.

Chapter 4: Results of all tests are presented and discussed in this chapter. The effects of test variables on the tensile strength retention of both GFRP bar types are presented and discussed.

Chapter 5: The accelerated aging test results along with the Arrhenius concept were employed to develop a realistic durability design model of both types of GFRP bars conditioned in moist seawater-contaminated concrete.

Chapter 6: Main conclusions of the work along with recommendations for future research in the area of durability of GRP are presented.

Chapter 2: Literature Review

2.1 Introduction

This chapter summarizes findings of available research work published in the literature on durability of GFRP reinforcing bars. Factors affecting the durability of GFRP have been identified. The effects of test variables on durability performance of GFRP are compiled and discussed in this chapter.

2.2 Background

Literature review is done on some researches related to my point of study which is durability of glass fiber-reinforced polymer bars in seawater-contaminated concrete; some researches focused on studying the effect of one or two parameters but none of them studied the effect of all parameters in one study. For each paper, I included the title of the article, author, type or diameter of FRP, total number of samples used in the study, properties of FRP such as volume of fiber, modules of elasticity..., level of sustained load, conditioning regime, time of exposure, number of replicate samples, and microstructures characteristics. Table 2.1 summarizes available previous studies of durability of GFRP bars. The main parameters affecting the durability of GFRP bars were discussed.

Table 2.1: Summary of previous studies

Reference		Wang et. al [10]	Benmokrane et. al [11]	Fang et. al [12]	Gang et. al [13]
No. of Specimens		45 BFRP, 39 GFRP	N/A	N/A	155
Properties of FRP	Type	BFRP/GFRP	GFRP	GFRP	BFRP
	Diameter (mm)	6	12	N/A	6
	Modulus of Elasticity (GPa)	BFRP (93.1-110) GFRP (76)	4, 3.6, and 3	N/A	46
	Ultimate Tensile Strength (MPa)	BFRP (3800-4840) GFRP (2200)	65,82, and 90	N/A	1398
	V _f	BFRP (65) GFRP (63)	79, 84, and 79	N/A	N/A
Level of Sustained Load		N/A	N/A	N/A	10 to 60%
Elevated temperature (°C)		32,40,48, and 55	60	Room temperature	25, 40, and 55
Surrounding Media		N-SWSSCHP-SWSSC	Concrete	Water, Sweater	DIW, ALK.S, SS, AS
Time of Exposure (Days)		21, 42, and 63	42, 125, and 210	30, 60, 90, 120, 150, and 180	21, 42, and 63
No. of Replicate Samples		N/A	N/A	N/A	5
Microstructure Analysis	Moisture uptake	No	Yes	Yes	No
	SEM	Yes	Yes	Yes	Yes
	Matrix digestion analysis	No	No	No	No
	FTIR	No	Yes	Yes	No
	DSC	No	Yes	Yes	No
Durability Model		Yes	No	No	No

Table 2.1: Summary of previous studies (Cont.)

Reference		Davalos et. al [14]	Sen et. al [15]	Nkurunziza et. al [16]
No. of Specimens		N/A	36	20
Properties of FRP	Type	GFRP type I & II	CFRP	GFRP
	Diameter (mm)	9.3	8	9.5
	Modulus of Elasticity (GPa)	Type I, 46 Type II, 49	44.4	40
	Ultimate Tensile Strength (MPa)	856 841	821	658
	V _f	70	N/A	75
Level of Sustained Load		0, 2000–2600 $\mu\epsilon$	0, 10, 25	25% and 29% - 38%
Elevated temperature (°C)		20, 40, 50, and 60	N/A	25
Surrounding Media		Water for 7 days	N/A	DIW (pH = 7.0) Alk. (pH = 12.8)
Time of Exposure (Days)		30, 90, 150, 210, and 270	30, 90, 180, 270	417 (10,000 hr.)
No. of Replicate Samples		N/A	9	5
Microstructure Analysis	Moisture uptake	No	No	No
	SEM	Yes	Yes	No
	Matrix digestion analysis	No	No	No
	FTIR	No	No	No
	DSC	No	No	No
Durability Model		Yes	No	No

Table 2.1: Summary of previous studies (Cont.)

Reference		Chen et. al [17]	Robert et. al [18]	Al-Salloum et. al [19]	Serbescu et. al [20]
No. of Specimens		N/A	65	150	132
Properties of FRP	Type	GFRP type I & II	GFRP	GFRP	BFRP Type I and II
	Diameter (mm)	9.53	12.7	12	6, 10
	Modulus of Elasticity (GPa)	N/A	46.3	60.4	N/A
	Ultimate Tensile Strength (MPa)	925 771	768	1478	N/A
	V _f	70	77.9	N/A	75
Level of Sustained Load		No Load	No Load	No Load	No Load
Elevated temperature (°C)		20, 40, and 60	23, 40, 50	25, 50, hot humid, and dry humid	20, 40, 60
Surrounding Media		pH = 13.6, 12.7	Tap Water	TW, SW, Alk., Gulf Area, Riyadh Area	Water, Pore Sol., Alk.
Time of Exposure (Days)		60, 90, 120, 240	60, 120, 180, 240	180, 360, 540	5, 8, 42, 210
No. of Replicate Samples		N/A	5	5	5
Microstructure Analysis	Moisture uptake	No	No	No	No
	SEM	No	Yes	Yes	No
	Matrix digestion analysis	No	No	No	No
	FTIR	No	Yes	No	No
	DSC	No	Yes	No	No
Durability Model		Yes	Yes	No	Yes

Table 2.1: Summary of previous studies (Cont.)

Reference		Debaiky et. al [21]	Robert et. al [22]	Robert and Benmokrane [23]	Li et. al [24]
No. of Specimens		N/A	78	N/A	N/A
Properties of FRP	Type	GFRP	GFRP	GFRP	BFRP
	Diameter (mm)	9.5, 12.7, 16	9.3	19	7
	Modulus of Elasticity (GPa)	40, 42, 42	38.5	47.6	50.3
	Ultimate Tensile Strength (MPa)	658, 639, 580	608	728	899
	V _f	N/A	74.5	65.4	72
Level of Sustained Load		19-29	No Load	No Load	No Load
Elevated temperature (°C)		20, 42-73	23, 40, 60, 80	23, 40, 50	20, 40, 60, 80
Surrounding Media		DI. Water, Alkaline	Distilled Water	Water	Water, Alkaline
Time of Exposure (Days)		30, 120, 240, 420	40, 100, 120	60, 120, 180	N/a
No. of Replicate Samples		5	6	N/A	N/A
Microstructure Analysis	Moisture uptake	No	Yes	No	Yes
	SEM	Yes	Yes	Yes	Yes
	Matrix digestion analysis	No	No	No	No
	FTIR	Yes	No	Yes	No
	DSC	Yes	No	Yes	No
Durability Model		No	No	No	No

2.3 Durability Factors

2.3.1 Effect of Varying Temperature

Wang et al. [10] indicated that the degradations were much accelerated at higher temperatures, after 63 days of exposure to moist normal seawater sea-sand concrete (N-SWSSC) at 32, 40, 48 and 55°C, the tensile strength retentions of basalt fiber-reinforced polymer (BFRP) bars were 92.7%, 81.7%, 59.1% and 26.0%, respectively. After the 63 days of exposure to moist high performance seawater sea-sand concrete (HP-SWSSC) at 32, 40 and 55°C, the tensile strength retentions of BFRP were 97.9%, 90.2% and 77.4%, respectively. After 63 days of exposure to moist N-SWSSC solution at 32, 40 and 55°C, the tensile strength retentions of GFRP were 87.4%, 90.8% and 80.1%, respectively. When the surrounding environment was moist HP-SWSSC, the tensile strength retentions of GFRP were 97.9%, 94.1% and 89.6%, respectively.

Gang et al. [13] indicated that the varying temperature was playing a significant rule in GFRP durability. The rate of degradation was faster at the high temperatures. After 21 days of conditioning, the tensile strength retention was 99.1% at 25°C then dropped to 82.0% at 55°C.

Davalos et al. [14] indicated that the elevated temperature accelerated the degradation in tensile strength. The retention in tensile strength for loaded GFRP bars in concrete beams at 20°C for 30 days was 97% while it was about 87% at temperature of 60°C. The effect of temperature was very clear in loaded GFRP bars in concrete beams at 20°C and 60°C for 210 days where the retentions in the tensile strength were about 80% and 49%, respectively. The presence of sustained load during conditioning slightly reduced the tensile strength retention by approximately 3% when the conditioning temperature was 20°C. In contrast, the presence of sustained load during

conditioning did not reduce the tensile strength retention when the conditioning temperature was 40°C or 60°C.

Chen et al. [17] reported that high temperature had an effect on the durability of the GFRP and it was used to accelerate the degradation of GFRP. Tensile strength retention was about 82% when the GFRP was subjected to a temperature of 20°C for 60 days and it dropped to 52% when it was subjected to 60°C under the same condition and same time of exposure.

Robert et al. [18] indicated that the variation in tensile strength was minor when the temperature increased from 40 to 50°C after 240 days of conditioning in tap water. The tensile strength reduction was in the range of 10% to 16%. Tensile strength reductions of 16, 10, and 9% were recorded after 8 months of conditioning in tap water at 50, 40, and 23°C, respectively. Increasing the temperature increased the water's rate of diffusion and accelerated chemicals reactions causing degradation. The absorption of water can lead to a degradation at the fiber/matrix interface, leading to a loss in the ultimate tensile strength.

Al-Salloum et al. [19] reported that increasing the temperature to 50°C resulted in a faster degradation in the bars leading to a decrease in the tensile strength. Moisture and temperature were the main parameters affecting the durability of composite materials. It was noticed that the moisture absorbed by the composites combined with the temperature of exposure induced stresses in the material which consequently damaged the fiber and matrix and their interface and decreased the strength of GFRP material with time. The tensile strength retention was about 94.7% when the GFRP bars were conditioned in tap water at 25°C for 6 months. When the temperature increased to 50°C, the tensile strength retention dropped to 80.3%.

Serbescu et al. [20] reported that when the temperature increased to 60°C, significant drop in strength was observed, tensile strength retention was about 94.5% when the GFRP was subjected to tap water at 20°C for 1000 h, while it was only 39% at 60°C under the same conditioning.

Debaiky et al. [21] reported a maximum of 11% reduction in the tensile strength (compared to the guaranteed strength) of GFRP bars after exposure to an alkaline solution at 60°C and under a sustained load.

Robert et al. [22] reported that the effect of temperature was the most affecting factor as compared to other factors, such as time or sustained load. The retention of the flexure strength at a temperature of 23°C was 97.5%, while at 80°C it was 80.7%.

Robert and Benmokrane [23] indicated that after 240 days of water immersion of preloaded GFRP bars embedded in mortar, the tensile strength retentions were 95, 92, and 90% at 23, 40, and 50°C, respectively.

Li et al. [24] reported that the water uptakes of the composite bars were 0.1, 0.26, 0.32 and 0.56% after 6 months of immersion in distilled water at 20, 40, 60 and 80°C, respectively. When immersed in an alkaline solution at the same conditioning temperatures, the maximum water uptakes were 0.14, 0.20, 0.47 and 0.63%, respectively.

2.3.2 Effect of Surrounding Media

Surrounding media or environment is very important key parameter that could affect the durability of the FRP. Surrounding media can be water, air, alkaline solution, acid solution, concrete environment, or salt solution, even the concentration of the solution may affect the durability.

Wang et al. [10] reported that normal seawater sea-sand concrete (N-SWSSC) and high-performance seawater sea-sand concrete (HP-SWSSC) environments caused damages to BFRP and GFRP bars. Data suggested that the N-SWSSC environment caused more damage to BFRP and GFRP bars than the HP-SWSSC environment. This was attributed to the greater alkali-ion content of the N-SWSSC than that of HP-SWSSC.

Fang et al. [12] concluded that the value of the T_g of unaged specimens was 78.5°C, whereas the values of specimens immersed in water and seawater for 6 months were 76.2 and 76.5, respectively. The decrease in T_g was due to the effect of water plasticization. Immersion in water and seawater significantly affected the mechanical properties of GFRP. The tensile strength was 382 MPa for specimens immersed in water, while it was 390 MPa for specimens immersed in seawater for 6 months.

Gang et al. [13] indicated that changing the surrounding media (deionized water, salt, and acid solution) affected tensile strength retention of GFRP bars. The effect was very clear when the GFRP was immersed in an alkaline solution (2g Ca(OH)_2 , 0.9g NaOH, and 4.2g KOH in 1L of DW) for 42 days where the strength retention was about 88.9%. When GFRP bars were immersed in an acid solution (1.58 g concentrated sulfuric acid with a mass fraction of 98.3% in 1L of DW – pH = 1.5), the strength retention was 93.1. The deionized water and salt solution media (24.53g NaCl, 5.02g MgCl_2 , 4.09g Na_2SO_4 and 1.16g CaCl_2 in 1 L of DW) had same effect on the durability of GFRP where a tensile strength retention of 94.4% was recorded.

Davalos et al. [14] reported that saturated concrete environment (natural alkaline exposure) was more aggressive to GFRP than conditioning in open air. GFRP bars embedded in saturated concrete exhibited a tensile strength retention of 87.0% after

150 days of exposure at 20°C. The strength retention was 99.0% in open air. This occurred because water was acting like a soluble for the alkaline concrete.

Sen et al. [15] indicated that the reduction in strength of GFRP was due to diffusion of an alkali solution through the vinyl ester resin that was used in the pultrusion process. All failures occurred within the part of the specimen constantly exposed to an alkaline solution, indicating that alkali attack was the main cause of degradation.

Nkurunziza et al. [16] concluded that alkaline solution tended to have more harmful effects on the bars than de-ionized water at higher stress levels because the level of stress in the bars controls the formation of micro cracks in the resin matrix. The residual tensile strength of the bars after extended exposure to de-ionized water was almost unchanged. GFRP specimens were subjected to two different environments (deionized water and alkaline solution) for 10,000h and with two levels of sustained load (25% and 38% of ultimate strength). Results showed that the alkaline solution affected the tensile strength at all levels of sustained loads. The tensile strength retention of GFRP bar specimens subjected to alkaline solution under a sustained load level of 25% was 84.4% while it was 92.7% when the surrounding environment was deionized water.

Chen et al. [17] reported that GFRP bars were susceptible to attack by water and acidic and alkaline solutions. The most severe degradation was observed in alkaline solutions. The main attack mechanisms include etching, leaching, and embrittlement, the matrices of GFRP bars were intended to protect the fibers from harmful agents, but hydrolysis, plasticization, and swelling due to alkaline solution may lead to degradation of the matrix itself. The tensile strength retention decreased as the pH level of the surrounding media increased.

Robert et al. [18] indicated that the durability of GFRP in tap water was less affected than those exposed to simulated concrete pore solution. The losses in the tensile strength of the GFRP bars aged in an alkaline solution were higher than those aged in moist concrete.

Al-Salloum et al. [19] reported that regardless of the type or period of exposure, all tested GFRP bars had the same mode of failure and had almost linear stress – strain relationships up to failure. A total of 8 conditioning environments were used in this study; TW, SW, DW, and ALK which refer to tap water, seawater, deionized water, and alkaline solution, respectively. R and 50 refer to room temperature and 50°C, respectively. RF and JF refer to Reyadh and Jubail field, respectively. The maximum loss in the tensile strength of the tested GFRP bars was observed in the bars exposed to TW50 and ALK50 environments where the average loss was about 24.48% and 24.05%, respectively, of the initial strength after 18 months of exposure. For the specimens in the TWR, laboratory environments, and the two field environments; RF and JF, almost no reduction in the tensile strength was recorded after 18 months of exposure. The tensile strength retention was in the range of 94.8% to 99.8%. Some specimens were subjected to dry/wet cycle with presence of seawater at elevated temperature of 50 °C and the retention was about 90.3%. The SEM results showed that the matrix around the glass fibers in both ALK50 and TW50 specimens were significantly deteriorated. However, there was almost no deterioration in the glass fibers. This explains the significant losses recorded in both tensile strength and fracture strain and minor losses recorded in the tensile modulus.

Serbescu et al. [20] reported that immersion of composite samples in water (pH 7) at room temperature did not have any significant effect on their mechanical properties. When the pH value increased to that of concrete; the bars lost slightly more strength.

High alkalinity solution was more aggressive than the concrete environment though it resembled the plastic concrete conditions and this value of pH was expected to cause deterioration and promote embrittlement. The effect of alkaline solution was severe; and the highest strength retention was 55% for BFRP bars conditioned at 60°C and pH13 for 1000 hr. The strength retention was 92.5% when the surrounding media had a pH value of 7.

Debaiky et al. [21] indicated that the attack of OH ions led to loss of structural integrity of the glass fiber in alkaline environment. The tensile strength retention of GFRP bars immersed in water for two months at an average temperature of 72°C was 96%. After the same time of exposure but in alkaline solution with pH value of 12.7 at an average temperature of 64°C, the tensile strength retention was 88%. The effect of alkaline environment was very clear despite the varying temperature.

Li et al. [24] reported that a remarkable hydrolysis of resin was found and resulted in the exposure of basalt fibers when BFRP bars were conditioned at 60 °C and immersed in an alkaline solution. For the water immersion environment, the resin on the rebar surface did not show such degradation.

2.3.3 Effect of Sustained Load

Gang et al. [13] reported that the presence of a **sustained load during conditioning affected** the rate of degradation in the tensile strength of FRP bars. Some specimens were creep-ruptured under the sustained load during conditioning. The presence of sustained load level of 20% of ultimate strength during conditioning had minor and negligible effects on the degradation rate. When the stress level reached 40% of ultimate strength the degradation rate was accelerated. Creep rupture took place when the sustained stress level increased to 60% of ultimate strength during conditioning.

The tensile strength retention of the specimens that were not loaded during conditioning was 90.3% after 21 days of exposure in alkaline solution at 40°C. When the sustained load level was 60% of ultimate strength, the tensile strength retention dropped to 85.1%.

Davalos et al. [14] indicated that the reduction in the tensile strength for non-loaded GFRP bars embedded in concrete beams then immersed in curing tanks for 150 days was about 10%. Similar GFRP bars conditioned in loaded concrete beams exhibited a tensile strength reduction of about 20%.

Sen et al. [15] reported that the level of sustained load during conditioning affected the durability of GFRP bars. Three levels of sustained loads were used (0%, 10%, and 15% of tensile strength). Reduction of strength with time exposure of one month for GFRP bars was about 50% (unloaded), 60% (loaded 10%), and 100% (loaded 15%). Some GFRP bar specimens were creep-ruptured when loaded by 15% of tensile strength at ages of 1, 3, and 9 months.

Nkurunziza et al. [16] reported that the presence of a sustained load level of 25% or 38% of tensile strength during the accelerated aging test duration (10,000 hr) had no effect on the residual modulus of elasticity of the tested GFRP bars.

Robert and Benmokrane [23] used four different tensile stress levels of 20, 40, 60, and 80% of the theoretical ultimate tensile strength (854 MPa) to initiate cracks and micro cracks in polymer and glass fibers. High stress level (more than 60% of the ultimate tensile strength) led to fiber cracking, resulting in an increase in moisture uptake at saturation.

2.3.4 Effect of Time of Exposure

Wang et al. [10] reported that the tensile strengths of BFRP and GFRP bars decreased with an increase in the exposure period at all temperatures. The tensile strength retentions were 94.2, 88.7, and 81.7% for BFRP bars exposed to N-SWSSC environment at 55°C for 21, 42, and 63 days, respectively, while they were 95.3, 93, and 90.2% for BFRP bars exposed to HP-SWSSC environment at 55°C for 21, 42, and 63 days, respectively. The tensile strength retentions were 91.5, 90, and 81% for GFRP bars exposed to N-SWSSC at 55°C for 21, 42, and 63 days, respectively, while they were 93, 92, and 91% for BFRP bars exposed to HP-SWSSC at 55°C for 21, 42, and 63 days, respectively.

Benmokrane et al. [11] indicated that both the polyester and epoxy GFRP bars had similar flexural strength reductions after 5000 h of immersion (25% and 23%, respectively), while the vinyl-ester GFRP bars returned a lower reduction of 17%. These observations demonstrated that the bond between the GFRP fibers and polyester resin before and after conditioning was lower than that between the glass fibers and the vinyl-ester or epoxy resin. The flexural strength of the polyester GFRP bars was significantly affected by the accelerated aging (25% reduction after 5000 h).

Fang et al. [12] reported that the moisture absorption increased initially then decreased with immersion time because of the hydrolysis reaction. After 6 months of aging time, the T_g of the specimens after immersion in water and seawater decreased by 2.9% and 2.5%, respectively.

Gang et al. [13] reported that time of exposure had an effect no matter what the surrounding environment or the sustained load level was. Increasing the exposure time decreased the tensile strength retention of conditioned FRP bars. When FRP bars were

was exposed to an alkaline solution at a temperature of 40°C for 21 days, the retention in tensile strength was 90.3% while it was 85% after 63 days of conditioning.

Davalos et al. [14] indicated that time of exposure affected the tensile strength retention. The retention in tensile strength for GFRP bars loaded in concrete beams at 20 °C for 30 days was 98%, while at the same temperature for 270 days of exposure time it was 82%.

Sen et al. [15] indicated that increasing the time of exposure from one to three months increased the strength reduction from 50% to 63%. Further increase in the conditioning time up to a total time of 9 months did not result in a further increase in the tensile strength.

Chen et al. [17] reported that the tensile strength of GFRP bars decreased with an increase in exposure time at all temperatures. The tensile strength retentions of GFRP1 ($f_u = 925$ MPa) at a temperature of 20°C for ages of 60, 90, 120, 240 days were 81, 63, 57, and 43% respectively. The tensile strength retentions of GFRP2 ($f_u = 771$ MPa) at a temperature of 20°C for ages of 60, 70, 90, 120 days were 98, 96, 95, and 90%, respectively.

Robert et al. [18] reported an increase in the reduction in the ultimate tensile strength with an increase in the immersion duration. The results showed that the longer the time of immersion, the larger the loss of resistance. When FRP samples were exposed to a temperature of 50°C, the tensile strength retention dropped from 97% at age of 60 days to 83% at age of 240 days.

Debaiky et al. [21] reported that the tensile strength retention of 16 mm diameter GFRP bars subjected to tap water reduced from 96% to 90% when the exposure time increased from one month to two months.

Robert et al. [22] reported tensile strength retentions of 97.5, 94.6, and 93.0% at 40, 100, and 120 days of conditioning, respectively at a temperature of 23°C. At a temperature of 40°C, the strength retentions were 95.2, 92.1, and 90.1% at ages of 40, 100, and 120 days, respectively. At a temperature of 60°C, the strength retention were 88.5, 85.6, and 82.0% at ages of 40, 100, and 120 days, respectively.

Robert and Benmokrane [23] concluded that the time of exposure affected the durability of GFRP but the effect was not as high as the effect of the conditioning temperature. The tensile strength retention at an age of 60 days and a temperature of 23°C was 99%, while it was 94.8% at an age of 240 days under the same conditions.

Chapter 3: Experimental Program

3.1 Introduction

This chapter presents details of the experimental program adopted in this study. GFRP bars encased in seawater-contaminated concrete were placed in conditioning tanks without sustained load. Another group of concrete-encased GFRP specimens were conditioned under a sustained load. Properties of surrounding concrete such as, pH, compressive strength, tensile strength, UPV, concrete resistivity, RCPT, and water permeability were measured and reported. Microstructural characteristics of GFRP bars were evaluated using scanning electron microscopy (SEM), matrix digestion using nitric acid, differential scanning calorimetry (DSC), and fourier transform infrared (FTIR) spectroscopy. Following conditioning, GFRP bars were retrieved then tested to failure under uniaxial tension. The degradation in tensile properties due to conditioning was investigated.

3.2 Test Program

The test matrix is given in Table 3.1. Two groups of GFRP bars were tested in this study; GFRP group Type I and Type II. Three replicate GFRP specimens from each bar type were tested without conditioning to act as a benchmark and these are considered control specimens. Fifty-four concrete encased GFRP specimens were subjected to accelerated aging without load whereas Fifty-four specimens were conditioned under a sustained load of 25% of the initial tensile strength of the GFRP bars. All conditioned specimens were surrounded by moist seawater-contaminated concrete during the accelerated aging. The test variables were the time of conditioning:

5, 10, and 15 months and the temperature of the surrounding water: 20, 40 and 60°C.

Three replicate samples were used for each testing condition.

Table 3.1: Test matrix

GFRP bar type	Surrounding media	Loading state during conditioning	Temperature [°C]	Time of exposure [month]	No. of replicate samples
Type I	Control				3
	Moist seawater-contaminated concrete	No load	20	5	3
				10	3
				15	3
			40	5	3
				10	3
				15	3
			60	5	3
				10	3
				15	3
		Sustained load	20	5	3
				10	3
				15	3
			40	5	3
				10	3
				15	3
			60	5	3
				10	3
				15	3
Type II	Control				3
	Moist seawater-contaminated concrete	No load	20	5	3
				10	3
				15	3
			40	5	3
				10	3
				15	3
			60	5	3
				10	3
				15	3
		Sustained load	20	5	3
				10	3
				15	3
			40	5	3
				10	3
				15	3
			60	5	3
				10	3
				15	3

Samples, taken from unloaded GFRP specimens, were prepared to evaluate their microstructure as shown in Table 3.2. Some loaded GFRP specimens were ruptured during conditioning under a sustained load. Samples were taken from these ruptured specimens for subsequent microstructural evaluation as shown in Table 3.3. Samples of moisture absorption, SEM, and matrix digestion using nitric acid were prepared as solid particles, while, samples of DSC and FTIR were likely powder as shown in Figure 3.2.

Table 3.2: Test matrix of microstructure for unloaded samples

GFRP type	Temperature [°C]	Time of exposure [month]	Moisture uptake	SEM	Matrix digestion analysis	FTIR	DSC
Type I	Control	0	✓	✓	✓	✓	✓
	20	5	✓	N/A	✓	✓	✓
		10	✓	N/A	✓	✓	✓
		15	✓	✓	✓	✓	✓
	40	5	✓	✓	✓	✓	✓
		10	✓	✓	✓	✓	✓
		15	✓	✓	✓	✓	✓
	60	5	✓	N/A	✓	✓	✓
		10	✓	N/A	✓	✓	✓
		15	✓	✓	✓	✓	✓
Type II	Control	0	✓	✓	✓	✓	✓
	20	5	✓	N/A	✓	✓	✓
		10	✓	N/A	✓	✓	✓
		15	✓	✓	✓	✓	✓
	40	5	✓	✓	✓	✓	✓
		10	✓	✓	✓	✓	✓
		15	✓	✓	✓	✓	✓
	60	5	✓	N/A	✓	✓	✓
		10	✓	N/A	✓	✓	✓
		15	✓	✓	✓	✓	✓

Table 3.3: Test matrix of microstructure for loaded samples

GFRP type	Temperature [°C]	Time of exposure [month]	Moisture uptake	SEM	Matrix digestion analysis	FTIR	DSC
Type I	Control	0	✓	✓	✓	✓	✓
	40	3.8	✓	✓	✓	✓	✓
	60	2.5	✓	✓	✓	✓	✓
	60	6.7	✓	✓	✓	✓	✓
	60	7.9	✓	✓	✓	✓	✓
	60	9.6	✓	✓	✓	✓	✓
Type II	Control	0	✓	✓	✓	✓	✓
	40	8.6	✓	✓	✓	✓	✓
	40	13.7	✓	✓	✓	✓	✓
	60	4.9	✓	✓	✓	✓	✓
	60	8.6	✓	✓	✓	✓	✓
	60	12.3	✓	✓	✓	✓	✓



Figure 3.1: GFRP test samples (solid and powder)

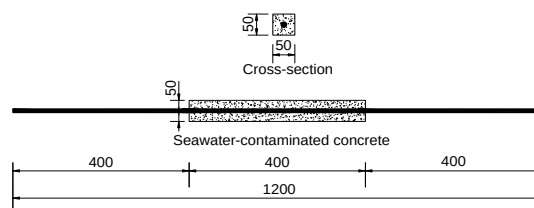
3.3 GFRP bars

Two types of GFRP bars made of high strength continuous glass fibers impregnated in epoxy resin were utilized in this study. The two GFRP bar types had ribs on the surface. Type I GFRP bars had inner and outer diameters of 7.2 and 8 mm, respectively, whereas those of Type II were 8 and 9 mm. The average cross-sectional area of each type was determined according to the test method specified by the ACI 440.3R-12 [1]. Type I GFRP had an average cross-sectional area of 45 mm² whereas

Type II had an average area of 57 mm^2 . The void contents of Type I and II GFRP bars were determined as 0.1% and 0.23%, respectively, as per ASTM D3171 and D2734-16 [25, 26]. The mass fraction of glass fibers, determined by matrix digestion using nitric acid, in accordance with ASTM D3171 [25], was 78.3% for Type I and 75.5% for Type II. It is customary that the fiber content calculated as per ASTM D3171 [25] includes fibers and fillers. Thermogravimetry analysis (TGA) was also performed following ASTM E1868-10 to verify fiber contents determined by matrix digestion using nitric acid. TGA resulted in respective fiber contents of 79% and 75% by mass [27]. The difference in fiber content determined by both methods is in the range of 0.5% to 0.7%, which falls within the limits given in Section 14 of the ASTM E1868. The as-received tensile strength and modulus, determined according to the test method specified by the ACI 440.3R-12 [1], were $816 \pm 15 \text{ MPa}$ and $53 \pm 3 \text{ GPa}$ for Type I and $1321 \pm 25 \text{ MPa}$ and $53 \pm 2 \text{ GPa}$ for Type II, respectively.

3.4 Fabrication and Test Specimens

The GFRP test specimens were 1200 mm long with the middle third surrounded by seawater-contaminated concrete, with a cross-section of $50 \times 50 \text{ mm}$, to represent concrete subjected to seawater splash in field condition. All GFRP samples were marked first (400 mm from one side) and then inserted in the wooden form to be casted as shown in Figure 3.2 and 3.3.



(a) Schematic



(b) Before concrete casting

Figure 3.2: Test Specimen (a) Schematic; (b) Before concrete casting

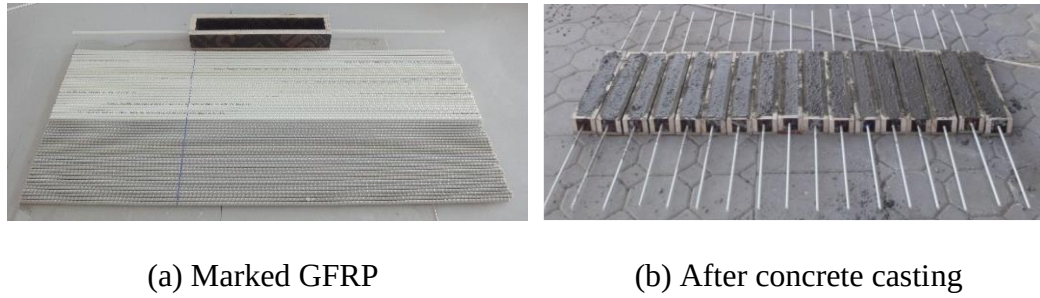
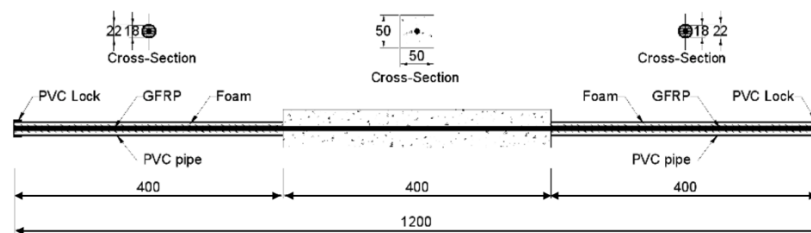


Figure 3.3: GFRP after casting (a) Marked GFRP; (b) After concrete casting

3.4.1 Unloaded Specimens

Prior to conditioning, polyvinyl chloride (PVC) pipes were installed around the exposed parts of the GFRP bars as shown in Figure 3.4, the inside space was filled with foam then blocked/sealed carefully at each end to protect these regions during conditioning to prevent water from coming in. GFRP-reinforced concrete specimens were immersed in temperature-controlled water tanks in unloaded condition (Figure 3.5). The tanks were custom-designed to maintain constant elevated temperature environment (three elevated temperatures of 20, 40, and 60°C with three accelerated aging of 5, 10, and 15 months). Middle third of samples was surrounded by seawater contaminated concrete. The concrete was provided around the test region to resemble actual field conditions. Following conditioning, the GFRP bars were retrieved carefully from the concrete. The PVC pipes were removed and end grips were installed.



(a) Schematic



(b) After PVC was installed

Figure 3.4: Polyvinyl chloride (PVC) installation (a) Schematic; (b) After PVC was installed

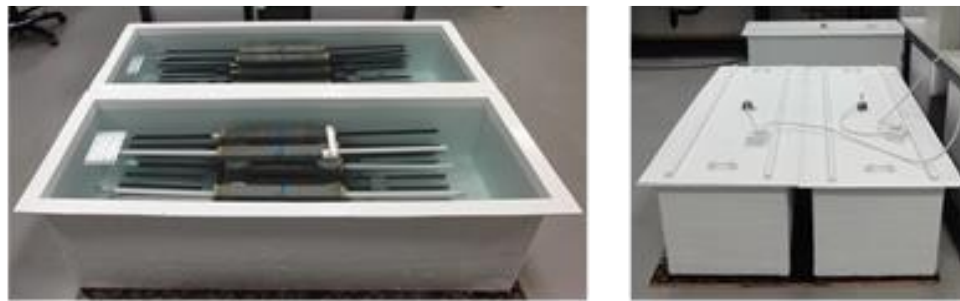


Figure 3.5: Specimens under accelerated aging

3.4.2 Loaded Specimens

Half of specimens were subjected to accelerated aging under a sustained load of 25% of the initial tensile strength of the GFRP bars. These specimens are shown in Figure 3.6. End grips were installed. The end grips consisted of a steel pipe, 450 mm long, with inner and outer diameters of 25 and 34 mm, respectively. Two hooks were attached to both ends to be subjected to tension frames as shown in Figure 3.7; these hooks were removed away after the conditioning was done. Steel loading frames were designed and fabricated at the UAEU for this purpose (Figure 3.8). A total of twelve loading frame were fabricated; four frames were located inside the lab to maintain the room temperature of 20°C and the other 8 frames were located outside and connected to two heaters to maintain the temperature of 40, and 60°C as shown in Figure 3.9.

Each loading frame applied a constant sustained load to three replicate GFRP samples while being exposed to accelerated aging of 5, 10, and 15 months. The surrounding tanks were custom-designed and each tank had three replicate samples. The tank was sealed from all sides to prevent water leakage as shown in Figure 3.10.

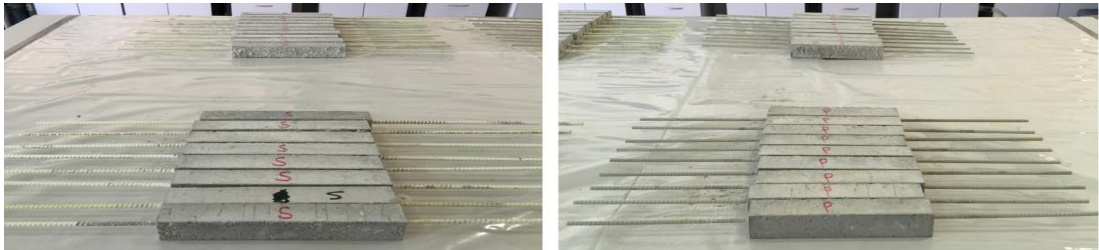
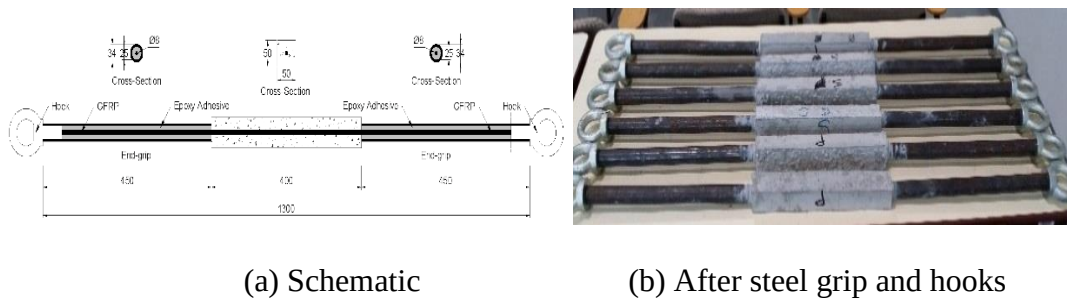


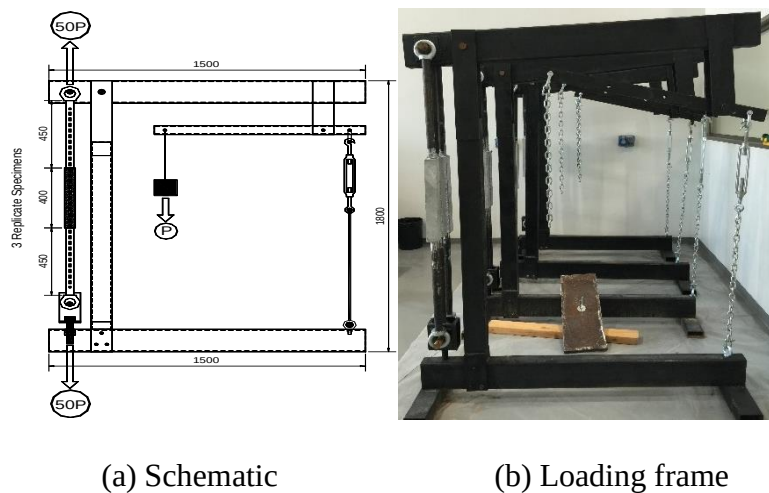
Figure 3.6: Loaded specimens before installation of steel grip and hooks



(a) Schematic

(b) After steel grip and hooks

Figure 3.7: End grips with hooks installation (a) Schematic; (b) After steel grips and hooks



(a) Schematic

(b) Loading frame

Figure 3.8: Sustained loading system (a) Schematic; (b) Loading frame

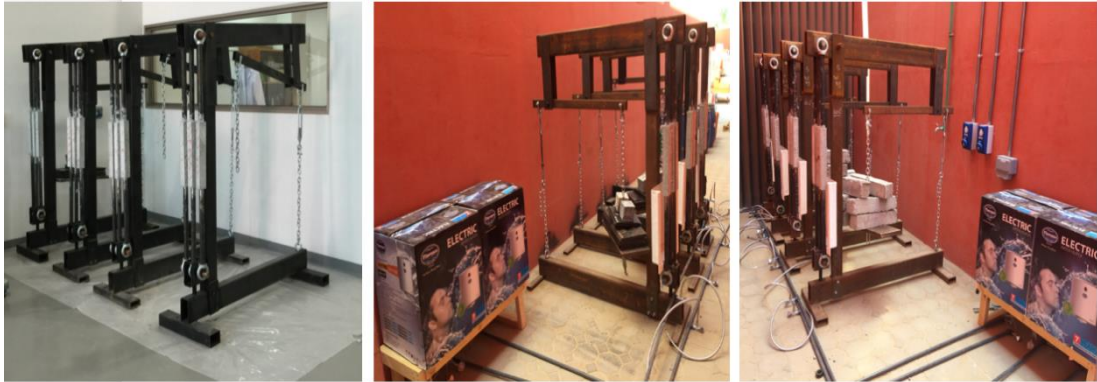


Figure 3.9: Sustained loading frames (20°C; 40°C; and 60°C)

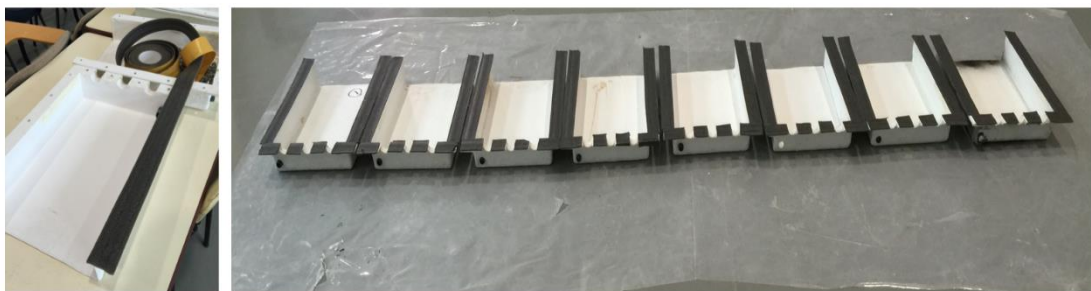
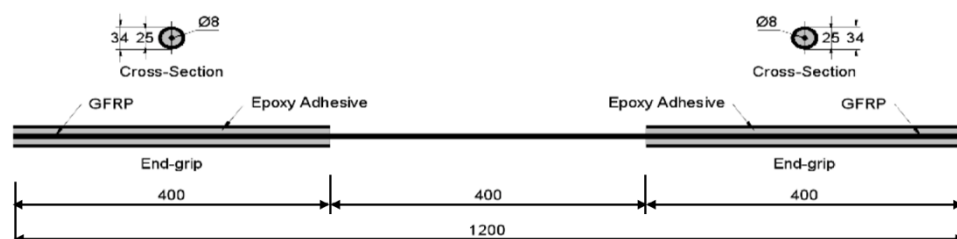


Figure 3.10: Sealed curing tanks used in sustained loading frames

3.4.3 End Grips

The end grips consisted of a steel pipe, 400 mm long, with inner and outer diameters of 25 and 34 mm, respectively as shown in Figure 3.5. The steel pipe was cleaned carefully and randomly threaded from inside to make the inner surface rough. The pipe was then installed at each end of the specimen then filled with epoxy to maintain adequate bond between the GFRP bar and the inner surface of the pipe.



(a) Schematic



(b) After steel grip was installed

Figure 3.11: End grips installation for tensile testing (a) Schematic; (b) After steel grip was installed

Epoxy resin commercially known as Sikadur 30 LP®, which consisted of two components, mixed with the ratio of 3:1 by weight was used to bond the GFRP bars to the threaded steel pipes. The mixed matrix was injected into the steel pipes using a cartridge gun. Figure 3.12 shows materials and tools used for preparation and application of the epoxy adhesive used. Pipes used in the end grips of the unloaded specimens were roughened from inside to insure the full bonding between epoxy and steel pipes, while these for loaded specimens were roughened from inside and threaded from one side to attach the steel hook before epoxy adhesive was installed as shown in Figure 3.13. Two plastic rings were used on both sides of the specimen to make sure that the GFRP bar will be centered and has no contact with the steel grip from inside as shown in Figure 3.14.



(a) Sikadur LP®



(b) Mixer used



(c) Sika cartridge gun

Figure 3.12: Materials and tools used for epoxy application (a) Siakdur LP®; (b) Mixer used; (c) Sika cartridge gun



Figure 3.13: Roughened and Threaded steel grip



Figure 3.14: GFRP specimens with plastic rings

3.5 Properties of Surrounding Concrete

The mix proportions of the concrete used in the present study are given in Table 3.4. The cement was ordinary Type I Portland cement. The coarse aggregate was natural crushed stone with a nominal size of 10 mm. Two types of fine aggregates, crushed natural stone and dune sand, mixed in 1:1 by mass, were used. The mixing water was seawater obtained from the Arabian Gulf.

Table 3.4: Concrete mix proportions for one cubic meter

Cement [kg]	Fine aggregate [kg]	Coarse aggregate [kg]	Seawater [kg]	w/c
400	580	1160	200	0.5

A concrete mix was designed and casted in Concrete Laboratory of UAE University, concrete mix was used to simulate the alkaline environment around the GFRP bars. Seawater is water that has a very high percentage of dissolved salts comparing with tap water; seawater in the world's oceans has a salinity of about 3.5% (one liter of seawater has around 35 grams of dissolved salts). Seawater is considered as an aqueous solution containing a variety of dissolved solids and gases. Seawater was used in concrete mix design to represent the coastline of the gulf region, sample was brought from Dubai and chemical analysis was conducted. Chemical analysis was conducted in College of Science Laboratories – United Arab Emirates University, the most important elements in seawater are Chloride and Sodium, and we can notice that they appear in seawater in large scale (23700 ppm and 13700 ppm respectively) see Table 3.5.

Table 3.5: Chemical Analysis of Seawater

Elements	Parts per million
Cl – Chloride (ASTM D512 – 12)	23700
Na – Sodium (ASTM D3561 – 11)	13700
Mg – Magnesium (ASTM D511 – 14)	1670
SO ₄ – Sulfate (ASTM D516 – 11)	475
Ca – Calcium (ASTM D511 – 14)	437
K – Potassium (ASTM D3561 – 11)	429

Quality of concrete was tested for specimens in the conditioning tanks and the specimens which subjected to loading frame. The pH, compressive strength, tensile strength, UPV, concrete resistivity, sorbitivity, RCPT, and water permeability were conducted.

3.5.1 pH Value

pH stands for the power of hydrogen; it is measurement of hydrogen ions concentration and the total pH scale ranges from 1 to 14, pH value of 7 is considered as neutral and pH value less than 7 is considered as acid while pH value greater than 7 is considered as basic or alkaline. The surrounding media or environment will affect durability of GFRP.

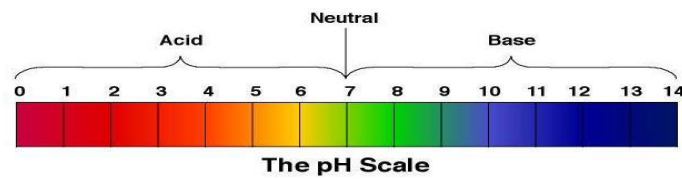


Figure 3.15: pH Scale

The concrete sample was crushed first to form a powder and then mixed with distilled water (pH7), the pH meter was used to determine the exact pH value as shown in Figure 3.16. The pH value of the concrete was on average of 12.4 indicating that the surrounding environment around the GFRP bars is alkaline according to ASTM E70-07 [28].

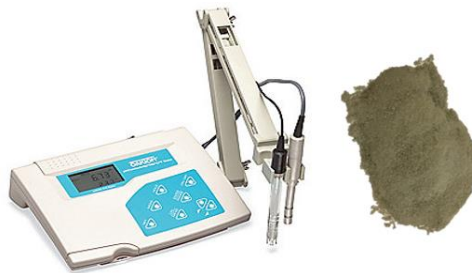


Figure 3.16: pH Meter and concrete powder

3.5.2 Compressive Strength

Compressive strength is the capacity of material or structure to withstand loads. It can be measured by plotting applied force against deformation. All Concrete cylinders were smoothed and two steel caps were used at top and bottom of the cylinders to

insure distributed load over the cross-sectional area of concrete cylinders. Compressive strength of concrete was conducted in Civil & Environmental Engineering Laboratory – United Arab Emirates University. An automated Autocon 2000 compressive strength machine with capacity of 2000 kN was used (Figure 3.17) according to ASTM C39/C39M-17A [29]. The average compressive strength of provided concrete was 43 MPa. Compressive strength of concrete was calculated according to Equation 3.1.

$$f'_c = \frac{P}{A} \quad (3.1)$$

Where:

P : Applied load in “N”

A : Cross sectional area of concrete cylinder



Figure 3.17: Automated Machine (2000 kN)

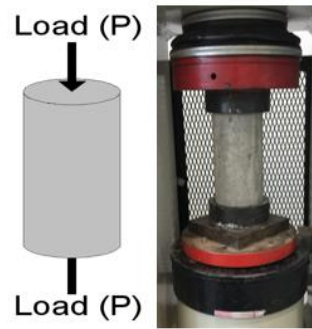


Figure 3.18: Compression of concrete cylinder

3.5.3 Splitting Test

Concrete is very vulnerable to tensile cracking because of different kind of effects and applying loading itself and therefore tensile strength is very important, tensile strength is very low compared to compressive strength. Tensile strength of concrete was conducted in Civil & Environmental Engineering Laboratory – UAE University according to ASTM C496/C496M-11 [30]. Load was applied to both sides and tensile strength was calculated according to Equation 3.2. The tensile strength was on average 3.15 MPa

$$f_t = \frac{2P}{\pi DL} \quad (3.2)$$

Where:

P : Applied load in “N”

D : Diameter of concrete cylinder in “mm”

L : Length of concrete cylinder in “mm”

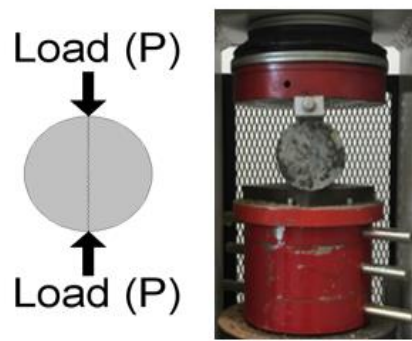


Figure 3.19: Splitting test of concrete cylinder

3.5.4 Ultrasonic Pulse Velocity Test (UPV)

UPV is one of the test methods for evaluation of structural integrity. It tells us the condition of the concrete structure if there are voids, cracks, or honeycombs. UPV device consists of connection cable, velocity meter, lubricant gel, transmitter, and receiver, see Figure 3.21 according to ASTM C597-16 [31]. UPV measures the time needed to transit an ultrasonic pulse through the concrete between sender (transmitter) and receiver. The reading of UPV depends on the quality of the concrete (in terms of uniformity, density, or homogeneity)



Figure 3.20: UPV instrumentation and testing

The transmitter and the receiver were plated by conductive gel and then attached to the concrete cylinders following the direct method. The UPV value was calculated according to Equation 3.3.

$$Velocity (m/s) = \frac{\text{Distance between transmitter and receiver (m)}}{UPV \text{ reading } (\mu s)} \quad (3.3)$$

The UPV value was on average 5.5 km/s. According to Table 3.6, an ultimate conclusion can be stated about an excellent condition of the concrete quality for the tested cylinders [32].

Table 3.6: Concrete Quality According to UPV value [32]

Pulse Velocity (km/s)	Concrete Quality (Grading)
Above 4.5	Excellent
3.5 to 4.5	Good
3.0 to 3.5	Medium
Below 3.0	Doubtful

3.5.5 Bulk Concrete Resistivity ($k\Omega.cm$)

In concrete materials, the electrical resistivity is correlated well with important durability parameters such as permeability, diffusivity and in general the micro-structure characteristics of concrete. It is a nondestructive device used for measuring the electrical resistivity of concrete specimens. The beauty of this test is that measurement can easily be made on the same concrete samples that are currently used for the compressive strength testing of concrete. Giatec RCON2™ device was used to perform this test; it is fast, accurate, and flexible.

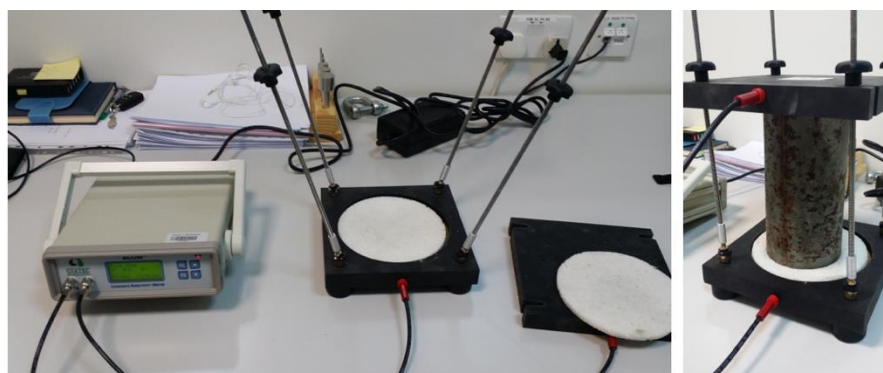


Figure 3.21: Concrete cylinder attached to conductivity holder

It applies a small alternating current at intended frequencies and measures the voltage between the two ends of the concrete specimen. Specimen was placed between conductivity holder with sponge coated with gel at top and bottom, it was connected to data logger and computer to record the impedance according to ASTM C1760-12 [33]. The impedance can be calculated from measured voltage and applied current values then concrete resistivity was calculated according to Equation 3.4.

$$\rho = \frac{A}{L} * Z \quad (3.4)$$

Where:

ρ : Resistivity “ $\Omega \cdot \text{cm}$ ”

A : Cross-sectional area of the specimen “ cm^2 ”

L : Length of the specimen “ cm ”

Z : Impedance measured by the device “ Ω ”

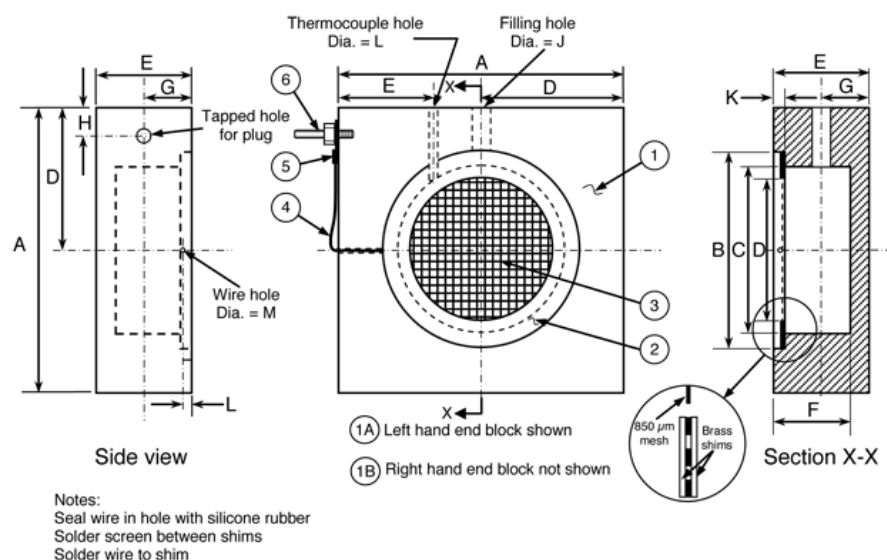
Bulk resistivity was on average 8 $\text{k}\Omega \cdot \text{cm}$. Table 3.7 states the correlation between the bulk electrical resistivity and durability performance of concrete [34], bulk resistivity results showed moderate chloride penetration that increases the severity and probability of corrosion for this concrete mix if conditions of the corrosion existed (water, oxygen ... etc).

Table 3.7: Correlation between bulk resistivity & Chloride Penetration [34]

Chloride Penetration	28 - days Bulk Resistivity
High	< 5
Moderate	5 -10
Low	10 - 20
Very Low	20 - 200
Negligible	> 200

3.5.6 Rapid Chloride Penetration Test (RCPT)

According to ASTM C1202-12, water saturated 50 mm thick; 100 mm diameter concrete specimens subjected to a 60 v applied DC voltage for 6 hours using the apparatus and the cell arrangement as shown in Figure 3.23 [35].



Dimensions, mm

A	B	C	D	E	F	G	H	J	K	L	M
150	105	89	75	50	41	25	15	10	6	5	2.5

Parts List

Item No.	Quantity	Description	Specification
1A 1B	1 1	Cell block end	Poly (methyl methacrylate)
2	4	Brass shim	0.5 mm thick
3	2	Brass screen	850 μ m (No. 20) mesh
4	2	Solid copper wire	2 mm (14 gauge) nylon cladding
5	2	Ring terminal	For 2 mm (14 gauge) wire
6	2	Banana plug	6.4 mm with threaded stud

Figure 3.22: RCPT cell Arrangement [35]

The specimens were fit in the chamber with the required brass as well as rubber oaring. The record time was set as 15 minutes and the log time as 6 hours and the current of 60V is passed continuously. The readings of corresponding cells were recorded at every record time with its initial readings.

The concrete samples (50mm thick & 100mm diameter) were coated with epoxy coat as shown in Figure 3.24 to make sure that the penetration will occur through the

surface only. Sample were placed in the special ring and sample was checked that it was fitted inside the rubber case. Two sides were filled with the chemical solutions (Red side was filled with NaOH 0.3% - 12g/l & the Black side was filled with NaCl 3% - 30g/l). Cells were connected to the data logger (using wires) and the data logger started record the reading (Current & Colombes) every 15 minutes (Figure 3.25).



Figure 3.23: Epoxy coating of RCPT disks



Figure 3.24: Rapid Chloride Penetration Test

The value of the passed charge was on average 1427 coulombs indicating that the chloride ion penetrability was low [35].

Table 3.8: Chloride ion penetrability [35]

Charged Passed (Coulombs)	Chloride Ion Penetrability
> 4000	High
2000 – 4000	Moderate
1000 – 2000	Low
100 – 1000	Very Low
< 100	Negligible

3.5.7 Concrete Permeability

A fully automated apparatus as shown in Figure 3.26 is designed to carry out water permeability tests on concrete specimen max. 160 mm diameter with maximum height of 160 mm. Varnish the side areas of the sample using epoxy resin, so that the surfaces are waterproof and water cannot pass through. The catalyst must be added to the resin in a 40 % percentage, Figure 3.27.

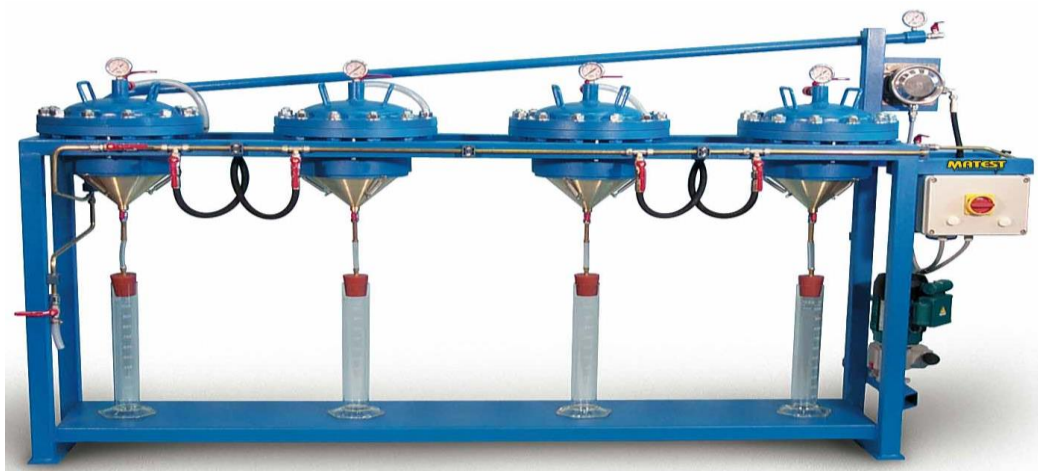


Figure 3.25: Water Permeability Machine



Figure 3.26: Coated Samples

After coating was done; the cover of the special housing cell was opened and sample was placed inside the special housing by pushing it towards the bottom until it comes in touch with the flange. The sealing rubber was closed (tight the small bolts inside the cell to make sure that the water will come only through the surfaces, not the

sides), the cell cover was placed on top and closed as shown in Figure 3.28. The pressure regulator is calibrated at the factory to keep the cells' pressure at 30 Bar. The pressure can be adjusted, the pressure used in this test the pressure was 20 Bar. The pump was started using main switch and the water was collected through the sample and the time was recorded using stopwatch. The coefficient of permeability was calculated according to Equation 3.5.

$$K = \frac{c.c * h}{A * t * P} \quad (3.5)$$

Where:

c.c: Collected volume of water in (m³)

h: Thickness of specimen (mm)

A: surface area of the specimen (cm²)

t: time to permeate (s)

P: hydrostatic pressure in cm of water column

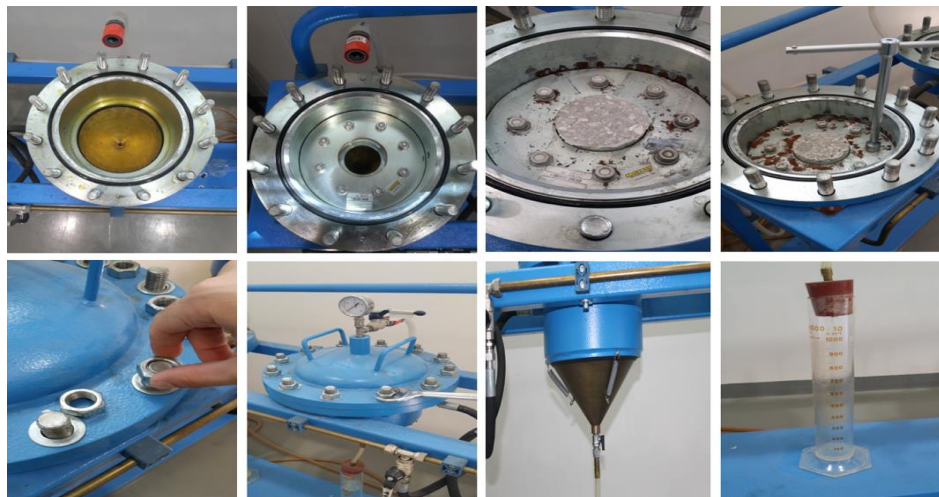


Figure 3.27: Test procedure of permeability test of concrete

The average permeability coefficient (*k*) of concrete was on average 1×10^{-7} m/s.

3.5.8 Moisture Absorption

The moisture uptake due to conditioning was assessed in accordance with ASTM D570-98E1 [36]. Naturally, the change in mass before and after immersion represents the water absorbed during conditioning. However, the GFRP reinforcing bars may have experienced mass dissolution due to degradation mechanisms including hydrolysis reaction. In order to account for this mass loss, conditioned specimens were dried in an oven at 100°C for 24 hours. The recorded oven-dried mass was compared to the initial mass of GFRP samples. Equation 3.6 was used to determine the corrected moisture uptake by mass.

$$\text{Moisture uptake}(\%, \text{by mass}) = 100X \frac{\text{Conditioned mass} - \text{Oven dried mass}}{\text{initial mass}} \quad (3.6)$$

3.5.9 Scanning Electron Microscope (SEM)

Microstructure and morphological changes of control and conditioned GFRP samples were examined using a JEOL-JSM 6390A scanning electron microscope (SEM) as shown in Figure 3.29. Specimens were cut, polished, and coated with a thin gold layer to ensure conductivity during testing (Figure 3.30). The obtained micrographs were used to identify degradation in fibers, polymer, and fiber-matrix interface.



Figure 3.28: JEOL-JSM 6390A (SEM)

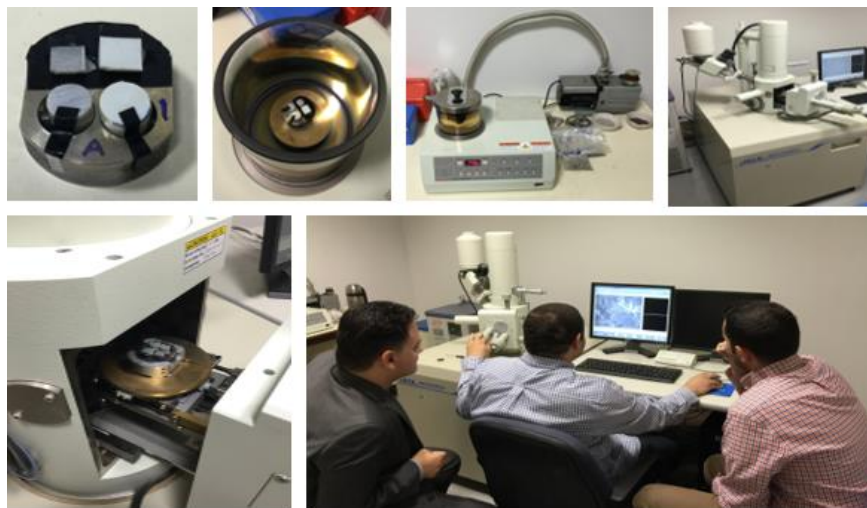


Figure 3.29: Gold coating and sample testing procedure

3.5.10 Matrix Digestion using Nitric Acid

To evaluate the fiber and matrix contents of conditioned GFRP reinforcing bars, matrix digestion using nitric acid was utilized, according to ASTM D3171 [25]. While the chemical reaction of GFRP bars with nitric acid results in digestion and dissolution of the epoxy resin, the fibers remain unaffected. Powder samples of 0.5–1.5 g was collected and mixed with 50 mL of 70% aqueous nitric acid. The mixture was then placed in a controlled temperature bath for 6 hours at 80°C. The contents were

filtered and washed with distilled water prior to drying in an oven for 1 hour at 100°C.

Equation 3.7 expresses the matrix content in terms of mass measured.

$$\text{Matrix Content (\%, mass)} = 100X \frac{\text{Initial mass} - \text{Final mass}}{\text{initial mass}} \quad (3.7)$$

3.5.11 Differential Scanning Calorimetry (DSC)

Powdered GFRP samples were examined by differential scanning calorimetry (DSC) using a DSC Q2000 calorimeter equipped with a refrigerated cooling system as shown in Figure 3.31. Specimens, on the range of 10 mg, were sealed in aluminum pans and prepared for analysis. DSC curves were obtained by heating the samples from 20°C to 225°C at a heating rate of 5°C/min. The glass transition temperature (T_g) was determined according to ASTM E1356-08 standard [37]. Two scans were performed for each specimen. The first scan was used to compare the T_g of conditioned samples to the unconditioned control. A decrease in T_g was indicative of a plasticizing effect. The second scan was useful to identify the mechanism of degradation. If the T_g of conditioned sample, after the second scan, was in the same range as that of the control, it indicated a reversible plasticizing effect due to the moisture absorption. However, a lower T_g compared to the control signified an irreversible chemical degradation.



Figure 3.30: DSC Q2000 calorimeter

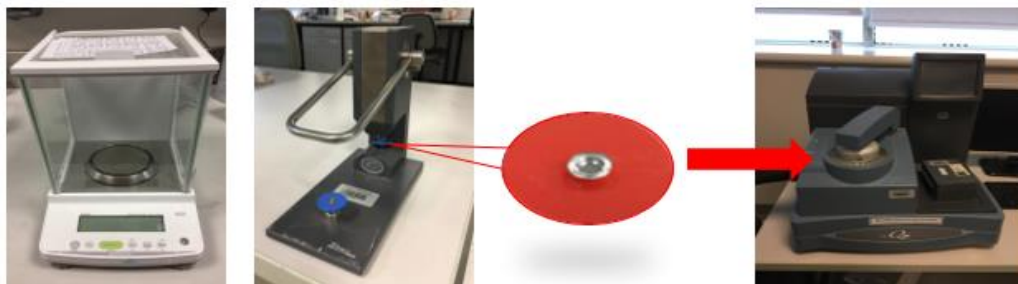


Figure 3.31: GFRP powder preparation and testing

3.5.12 Fourier Transform Infrared Spectrometer (FTIR)

The mixed powder was analyzed from 400 to 4000 cm^{-1} at a resolution of 1 cm^{-1} using a Varian 3100 FT-IR spectrometer as shown in Figure 3.32. Fourier transform infrared (FTIR) spectroscopy was performed on GFRP samples to measure the degradation due to hydrolysis reaction. Powdered specimens were interground with potassium bromide (KBr) at a powder sample: KBr = 1:4, by weight as shown in Figure 3.34.



Figure 3.32: Varian 3100 FT-IR spectrometer



Figure 3.33: Procedure of FTIR test of powder samples

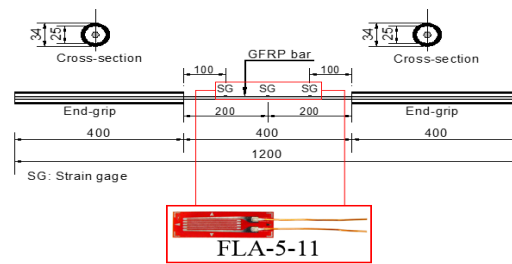
3.5.13 Tensile Test Set-Up and Instrumentations

The test region of the GFRP bar, middle 400 mm, was instrumented with three strain gages as shown in Figure 3.35 to monitor the strains during tensile testing. One strain gage was installed at the midpoint of the test region. The other strain gages were installed at a distance 100 mm away from each end of the test region. The specimens were tested to failure under uniaxial tension in displacement control at a rate of 1.5 mm/min. A data acquisition system was used to capture the load and strain readings. A test in progress is shown in Figure 3.36.

The tensile strength of GFRP bars, f_u , was calculated by dividing the maximum load by the average cross-sectional area of the GFRP bar. The tensile modulus of elasticity, E_f , was calculated from the stress-strain response according to Equation 3.8 as per the ACI 440.3R-12 [1].

$$E_f = \frac{f_{50} - f_{20}}{\epsilon_{50} - \epsilon_{20}} \quad (3.8)$$

Where f_{50} is the stress at 50% of the tensile strength, f_{20} is the stress at 20% of the tensile strength, ϵ_{50} is the strain at 50% of the tensile strength, and ϵ_{20} is the strain at 20% of the tensile strength.



(a) Schematic



(b) Photo

Figure 3.34: Strain gages configurations

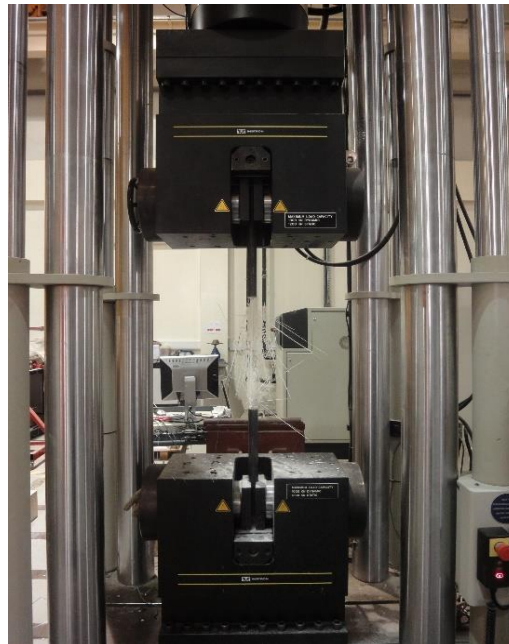


Figure 3.35: A test in progress

Chapter 4: Results and Discussions

4.1 Introduction

Test results of GFRP specimens are presented in this chapter. Crack width of the concrete surrounding the loaded specimens are reported. Failure mode of tested specimens is described. Results of moisture uptake, tensile strength retention, modulus of elasticity, SEM, matrix digestion, FTIR, DSC of the tested specimens are presented and discussed.

4.2 Crack Width

Concrete surrounding loaded GFRP bars were cracked after loading and a crackscope instrument was used to measure the crack width (Figure 4.1). Figure 4.2 shows the crack pattern. Four transverse cracks developed at a spacing in the range of 70 to 100 mm. Two crack patterns were observed. In one pattern, the transverse cracks went through the entire width of the cross section. The cracks in the second pattern stopped at the midpoint of section. The crack width was measured in “mm” for all loaded specimens prior to conditioning and after conditioning.

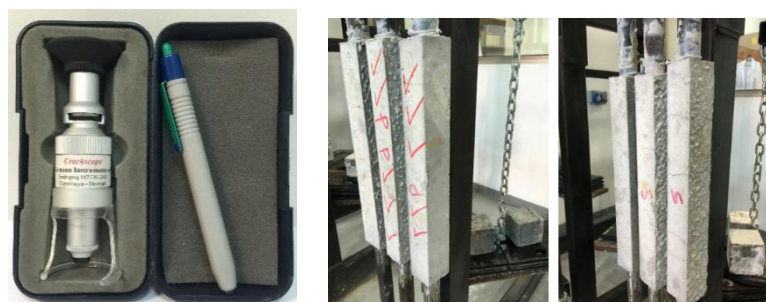


Figure 4.1: Crackscope and Cracked sections of middle third of concrete

The crack widths are listed in Table 4.1. The three replicate specimens of Type I GFRP conditioned at 60°C were ruptured suddenly during conditioning under sustained load, and hence, the crack width after conditioning was not measured. The

author did not measure the crack width after conditioning for the specimens conditioned for 5 months at 20°C. The crack width measured in the concrete surrounding the loaded specimens was on average 0.32 mm prior to conditioning and 0.52 mm after 15 months.

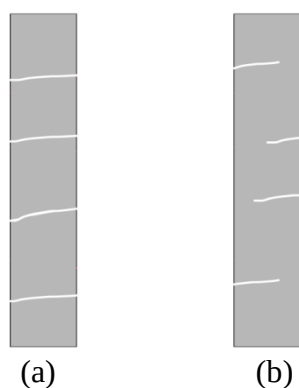


Figure 4.2: Crack pattern (a) pattern I; (b) pattern II

Table 4.1: Crack width for concrete specimen prior and after conditioning

GFRP Type	Temperature (°C)	Exposure Time (months)	Crack Width (mm)					
			Prior to Conditioning			After Conditioning		
			Min.	Max.	Avg.	Min.	Max.	Avg.
Type I	20	5	0.15	0.50	0.27	-	-	-
		10	0.10	0.50	0.27	0.20	0.60	0.40
		15	0.15	0.45	0.27	0.20	0.50	0.35
	40	5	0.20	0.40	0.27	0.20	0.40	0.29
		10	0.15	0.40	0.26	0.30	0.50	0.36
		15	0.15	0.50	0.31	0.30	0.70	0.44
	60	5	0.10	0.50	0.31	0.40	0.70	0.53
		10	0.20	0.50	0.34	-	-	-
		15	0.20	0.50	0.29	0.40	0.60	0.50
Type II	20	5	0.20	0.50	0.35	-	-	-
		10	0.10	0.40	0.23	0.30	0.60	0.49
		15	0.10	0.60	0.36	0.40	0.70	0.51
	40	5	0.20	0.50	0.29	0.20	0.80	0.44
		10	0.10	0.50	0.25	0.20	0.60	0.39
		15	0.20	0.60	0.36	0.50	0.90	0.63
	60	5	0.30	0.70	0.51	0.50	0.80	0.63
		10	0.10	0.40	0.26	0.20	0.60	0.43
		15	0.40	0.80	0.51	0.60	0.90	0.70

4.3 Failure Mode

All specimens failed by violent rupture of fibers accompanied by debonding at the fiber-matrix interface (Figure 4.3). None of the tested GFRP bars exhibited premature failure at the end grips. Photos of unconditioned and conditioned GFRP specimens after tensile testing are shown in Figure 4.4(a) and (b), respectively.

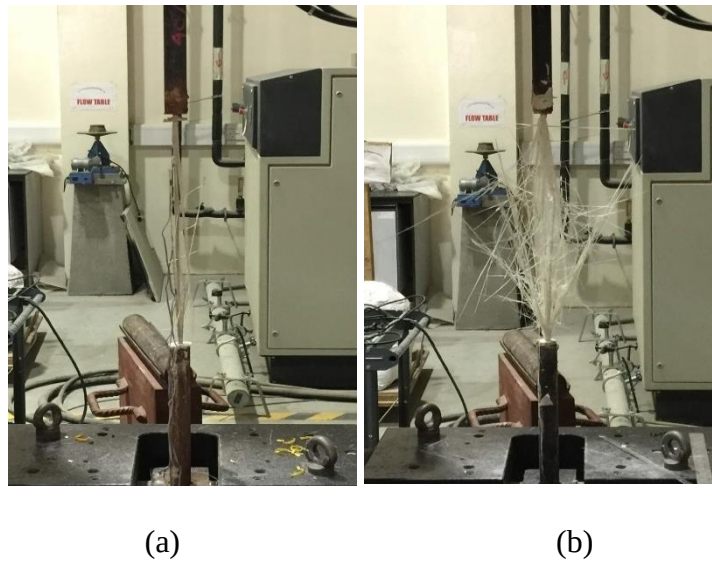


Figure 4.3: Failure mode of tested GFRP bars (a) Type I; (b) Type II



Figure 4.4: Photos of tested GFRP specimens: (a) unconditioned samples, (b) conditioned samples.

4.4 Result and Discussion

4.4.1 Unloaded Specimens

4.4.1.1 Moisture Uptake

Generally, the moisture could penetrate into a composite material by the flow of water into microgaps between polymer chains, capillary transport into microcracks at the fiber-matrix interface, and through matrix microcracks that formed during the compounding process [38]. For both types of GFRP bars, conditioning at 20°C for 5 months did not result in any moisture uptake. GFRP bars conditioned at 20°C exhibited, however, some moisture uptake at 10 and 15 months of exposure. GFRP bars conditioned at the higher temperatures of 40 and 60°C exhibited a moisture uptake from the beginning of the accelerated aging test. They had higher moisture uptake than that of their counterparts conditioned at 20°C at all times of exposure.

From Figure 4.5, it can be seen that increasing the conditioning temperature resulted in more water absorption due to a higher diffusion rate and formation of microcracks. The moisture uptake increased rapidly within the first 10 months of exposure, particularly for the specimens conditioned at the higher temperatures of 40 and 60°C. Further increase in the time of conditioning resulted in no or insignificant additional increase in the moisture uptake. This suggested that moisture absorption had possibly reached an equilibrium state after 10 months of conditioning.

For any conditioning regime, Type II GFRP bars experienced higher moisture uptake than that exhibited by their Type I counterparts. Type II GFRP bars had higher void content than that of Type I, which resulted in higher moisture absorption during conditioning. The increased moisture uptake exhibited by Type II GFRP bars facilitated progression of the hydrolysis reaction (see FTIR results). Hydrolysis causes matrix softening and impairs the bond at the fiber-matrix interface.

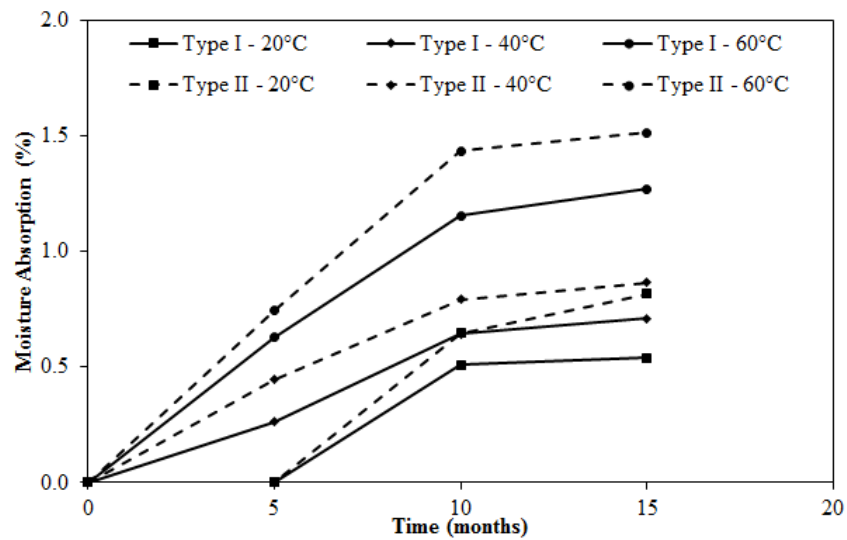


Figure 4.5: Moisture uptake of GFRP specimens conditioned without load

4.4.1.2 Tensile Strength Retention

The GFRP bars of both types exhibited an almost linear stress-strain response up to failure. The tensile strengths of conditioned samples are compared to those of the control unconditioned counterparts in Figure 4.6. The specimen designation shown in this figure consists of four characters. The first two characters refer to the conditioning duration (5M, 10M, and 15M) whereas the last two characters refer to the conditioning temperature (20C, 40C, and 60C).

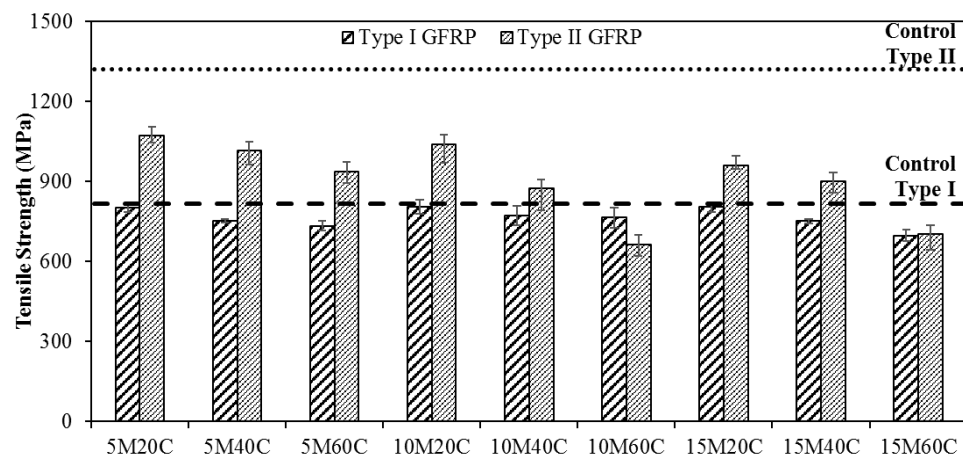


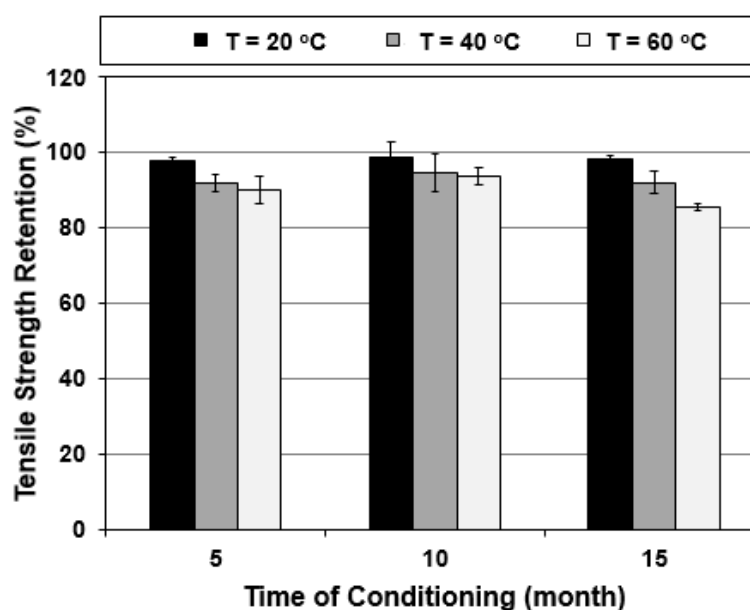
Figure 4.6: Tensile properties of unloaded specimens

Tensile strength retention of Type I GFRP bars are shown in Figure 4.7(a). Type I GFRP exhibited insignificant tensile strength reductions of 2, 8, and 10% after 5 months of conditioning at 20, 40, and 60°C, respectively. Subsequently, the tensile strength retention of Type I bars remained almost unaltered till the end of the accelerated aging test, except for the specimens conditioned at 60°C for 15 months. These specimens experienced a tensile strength reduction of 15%. The tensile strength retention of Type I GFRP bars tended to decrease with an increase in the temperature.

Tensile strength retention of Type II GFRP bars are shown in Figure 4.7(b). Type II GFRP bars experienced inferior durability performance in moist seawater-contaminated concrete than that exhibited by Type I. At 5 months of accelerated aging, Type II bars experienced tensile strength reductions of 19, 23, and 29% at 20, 40, and 60°C, respectively. The tensile strength retention decreased as the time of conditioning increased from 5 to 10 months at all temperatures. At 10 months of conditioning, Type II bars experienced tensile strength reductions of 21, 34, and 50% at 20, 40, and 60°C, respectively. The reduction in the tensile strength was more pronounced at the higher temperatures. This demonstrates that hydrolysis is accentuated at high temperature, which would result in the disintegration of chemical bonds at the fiber-matrix interface and separation between the fiber and matrix. Increasing the conditioning time to 15 months further decreased the tensile strength retention of Type II GFRP bars conditioned at 20°C where a strength reduction of 27% was recorded. The tensile strength retentions of Type II bars subjected to 15 months of conditioning at 40 and 60°C were almost equal to those of their counterparts subjected to 10 months of conditioning. It seemed that the chemical bonds at the fiber-matrix interface was severely weakened after 10 months of conditioning at high temperatures by the

hydrolysis reaction to the extent that an additional time of conditioning did not further reduce the strength retention.

Although Type II GFRP bars showed higher tensile strength prior to conditioning than that of Type I, the durability performance of the former was inferior compared with that of the latter. The tensile strength reduction could be due to matrix softening and/or fiber-matrix interfacial debonding, both caused by hydrolysis. Type II GFRP bars had higher void content and moisture uptake than those of Type I. The increased moisture absorption exhibited by Type II GFRP bars facilitated progression of the hydrolysis reaction (see FTIR results) and impaired the bond at the fiber-matrix interface (see SEM results). This could explain why Type II GFRP bars exhibited inferior durability performance than that of Type I. It should be also noted that the difference in the diameter of both GFRP bar types used in the present study is approximately 10%. This minor difference in bar diameter is too small to have an effect on the rate of degradation.



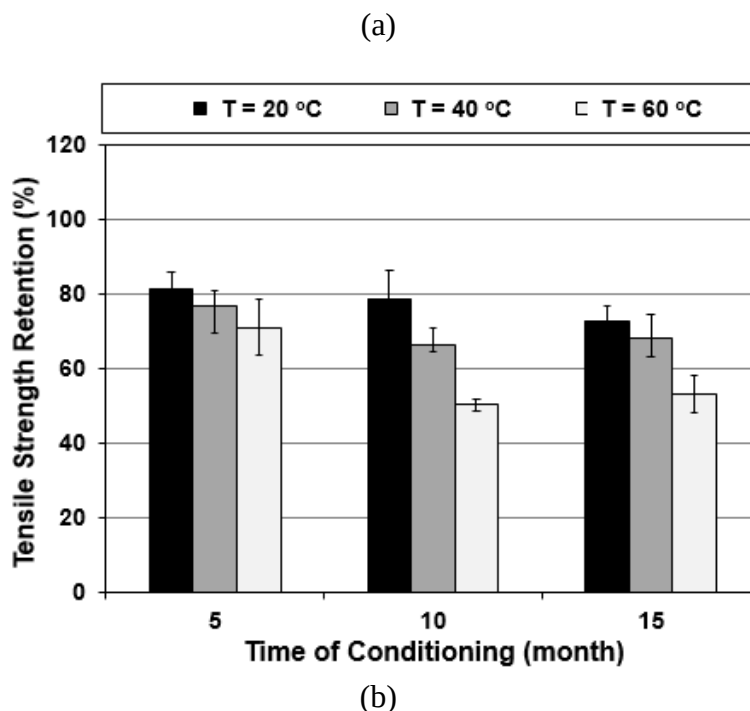


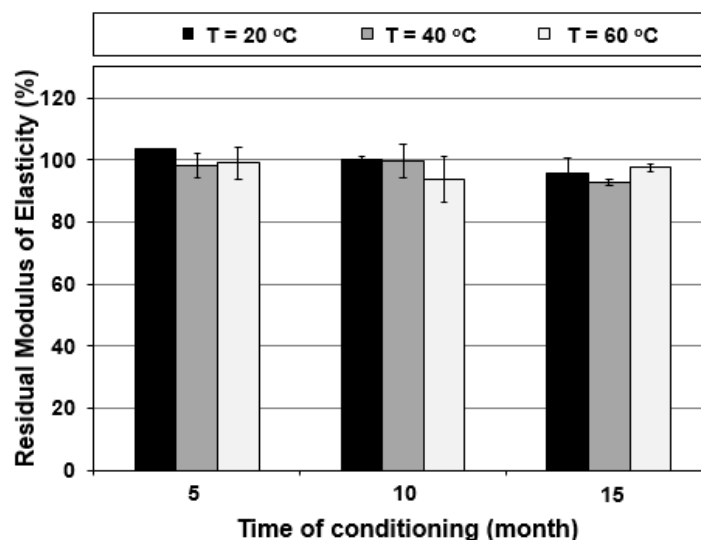
Figure 4.7 : Tensile strength retention of unloaded GFRP bars: (a) Type I, (b) Type II

4.4.1.3 Modulus of Elasticity

The effect of accelerated ageing on the modulus of elasticity of both types of GFRP bars is shown in Figure 4.8. The GFRP bar Type I exhibited insignificant increase in the tensile modulus after 5 months of exposure at 20°C whereas the modulus remained unchanged at the higher temperatures of 40 and 60°C. At 10 months of conditioning, the tensile modulus of GFRP bar Type I was unchanged at 20 and 40°C but slightly decreased by 6% at the higher temperature of 60°C. At 15 months of conditioning, the tensile modulus of GFRP bar Type I was almost the same at 20 and 60°C, but slightly decreased by 7% at 40 °C. The ingress of seawater into the GFRP bars causes swelling and plasticization/softening of the matrix. Swelling of the matrix increases the tensile modulus by improving the mechanical adhesion between the fiber and matrix. Plasticization/softening of the matrix reduces the tensile modulus. The initial slight

increase in the tensile modulus at 20°C can be ascribed to the swelling effect whereas the subsequent reduction in the modulus can be attributed to the plasticization effect.

Accelerated aging of GFRP bar Type II increased the tensile modulus of elasticity despite the reduction in the tensile strength. Increasing the temperature and time of exposure further improved the tensile modulus of GFRP bar Type II. The increase in tensile modulus of GFRP bar Type II could be due to reaction of water with the matrix, which would break down the molecular weight of the matrix, thus making it stiffer [37]. Another possible reason is that for some GFRP composites subjected to prolonged times of conditioning in seawater at elevated temperatures, the water could reduce the mobility of polymer chains, and hence the matrix becomes stiff and cannot absorb energy [39]. Although this phenomenon would render the physical interaction and mechanical adhesion ineffective at the fiber-matrix interface, thus reducing the tensile strength, the matrix becomes stiffer and the tensile modulus of the composite increases. This could explain why the tensile modulus of GFRP bar Type II increased after conditioning despite the reduction in the tensile strength. At 15 months of conditioning, there was a slight decrease in tensile modulus of 6, 4, and 1% at 20, 40, and 60°C respectively.



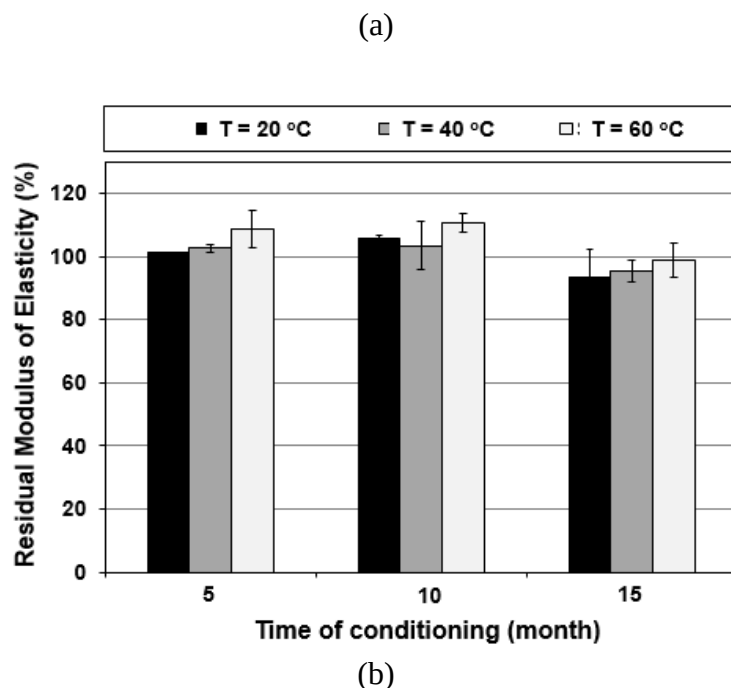


Figure 4.8: Residual modulus of elasticity of unloaded specimens (a) GFRP Type I, (b) GFRP Type II

4.4.1.4 SEM Analysis

SEM tests were performed to examine the effect of conditioning temperature on the morphology of GFRP bars conditioned in water for 15 months. Figure 4.9 and 4.10 present respective longitudinal and cross-sectional micrographs of unconditioned Type I GFRP bar and those conditioned at 20°C, 40°C, and 60°C. Figure 4.9(a) highlights the matrix adhesion to the fibers with good bonding at fiber-matrix interface in the unconditioned control sample. After 15 months of conditioning at 20°C, separation at the interface is detected as shown in Figure 4.9(b). The associated gap width could reach 2 μm . This degradation could result from crack propagation from the matrix to the interface as opposed to directly into the fiber [39]. Specimens conditioned at 40°C are depicted in Figure 4.9(c). A wider gap is visible at the interface; it is on the order of 5 μm . Further, the effect of increasing the temperature from 40°C to 60°C is studied in Figure 4.9(d). Although the process does not widen

the crack, it produces a relatively smoother fiber surface. It is an indication of matrix deterioration due to hydrolysis which accelerates in a hot water environment. Such damage to the matrix leads to the non-uniform distribution of load among fibers [40], resulting in lower tensile strength as presented in Figure 4.6. Cross-sectional micrographs of Figure 4.10 show that while fibers were passive to conditioning, the surrounding matrix suffered of degradation. Samples exposed to higher temperatures of 40°C and 60°C displayed signs of circumferential debonding and smoothening of the fiber surface.

The longitudinal and cross-sectional micrographs of control and conditioned type II GFRP bar are shown in Figure 4.11 and 4.12, respectively. Figure 4.11(a) of the unconditioned control sample identifies a rough surface of fibers with adequate bond to the resin. Specimens placed in water at 20°C for 15 months (Figure 4.11 (b)) experience formation of a gap along the length of the fiber-matrix interface. The increase in temperature to 40°C increased the gap opening as shown in Figure 4.11(c). Yet, it should be pointed out that the glass fiber remained unaffected by the water solution. The micrograph of samples immersed in water at 60°C is presented in Figure 4.11(d). Substantial disintegration of the matrix is noticed at the fiber-matrix interface with further crack development to a width of 6 μm . In addition, deterioration of type II GFRP bar is observed in cross-sectional micrographs of Figure 4.12. Although conditioning at 20°C initiated a gap at the fiber-matrix interface, the separation between the fiber and matrix was intensified at the higher temperatures. Figure 4.12(d) shows that specimens conditioned at 60°C experienced complete fiber debonding and severe matrix disintegration. Conditioning of GFRP bars at temperatures above 50°C promotes expansion of matrix in the transverse direction which could jeopardize the integrity of GFRP bars and impair the bond at the fiber-matrix interface [41,42,43].

GFRP samples subject to 40°C conditioning were compared at different ages of 5, 10 and 15 months as shown in Figure 4.13 and 4.14. Figure 4.15 and 4.16 present respective longitudinal and cross-sectional micrographs of Type II GFRP exposed to 40°C over the span of 15 months. Although the specimens experienced some separation between the fiber and matrix due to conditioning, the deterioration at the fiber-matrix interface was not severely intensified with longer conditioning periods. This indicates that the deterioration at the fiber-matrix interface is less sensitive to the conditioning duration rather than the conditioning temperature.

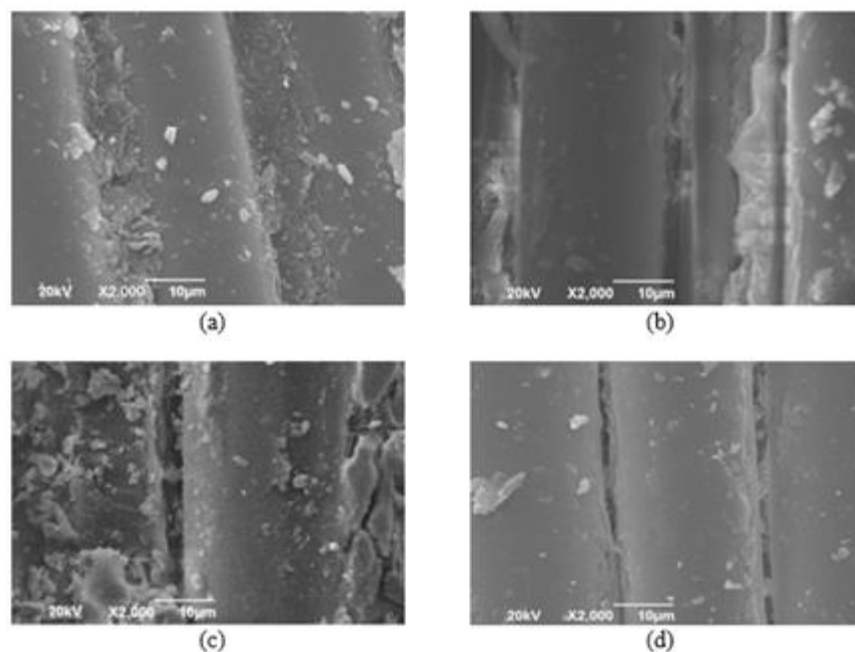


Figure 4.9: Longitudinal micrograph of Type I GFRP bars: (a) control (b) immersed for 15 months at 20°C (c) immersed for 15 months at 40°C (d) immersed for 15 months at 60°C

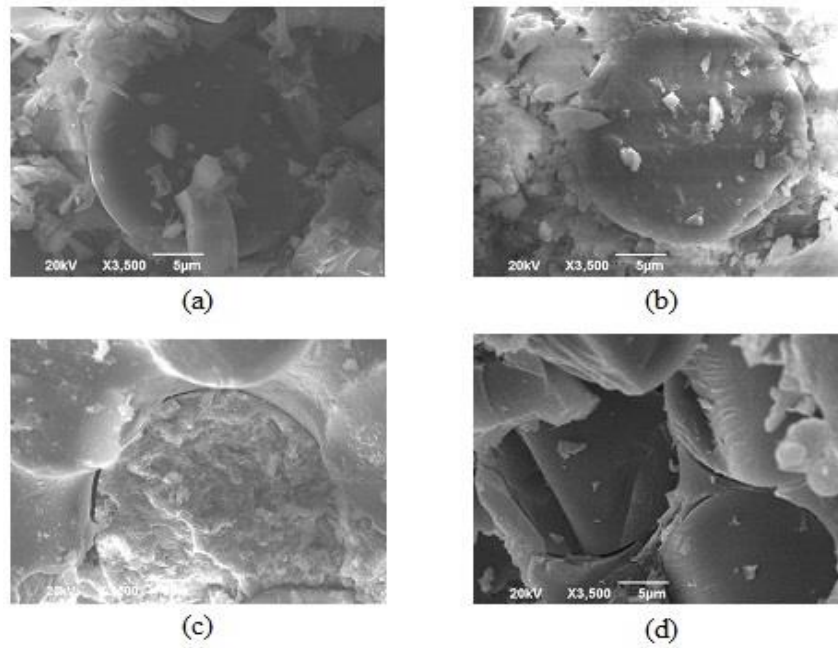


Figure 4.10: Cross-sectional micrograph of Type I GFRP bars: (a) control (b) immersed for 15 months at 20°C (c) immersed for 15 months at 40°C (d) immersed for 15 months at 60°C

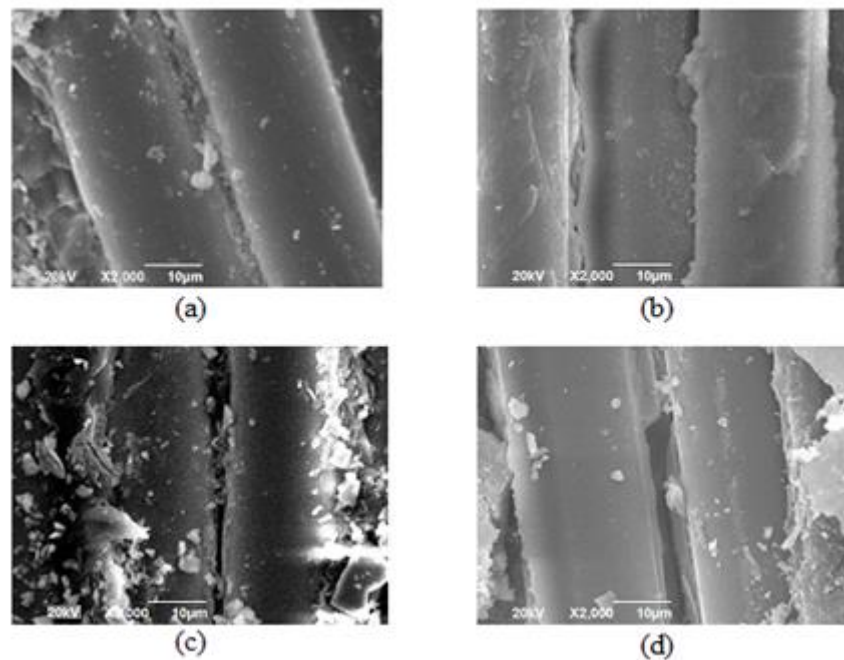


Figure 4.11: Longitudinal micrograph of Type II GFRP bars: (a) control (b) immersed for 15 months at 20°C (c) immersed for 15 months at 40°C (d) immersed for 15 months at 60°C

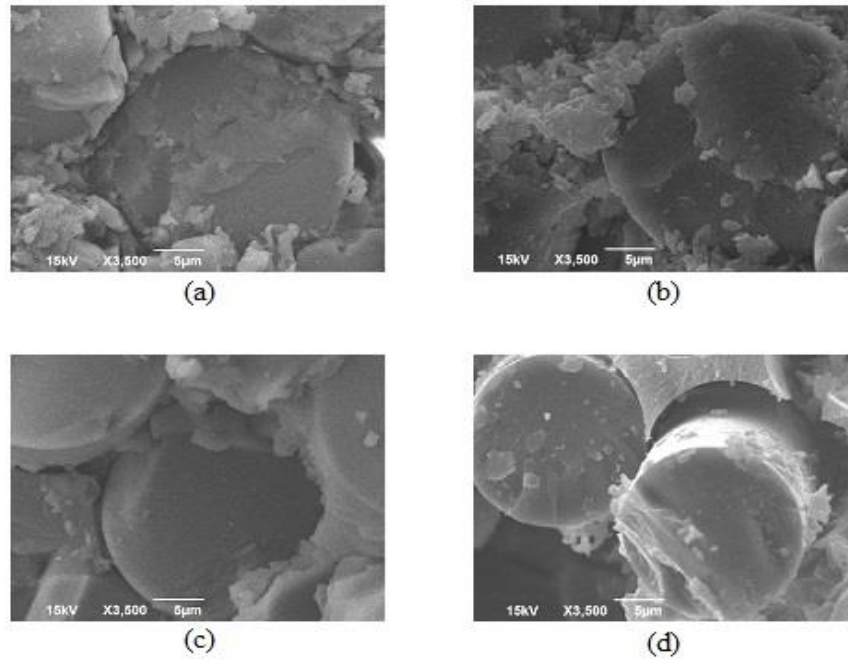


Figure 4.12: Cross-sectional micrograph of Type II GFRP bars: (a) control (b) immersed for 15 months at 20°C (c) immersed for 15 months at 40°C (d) immersed for 15 months at 60°C

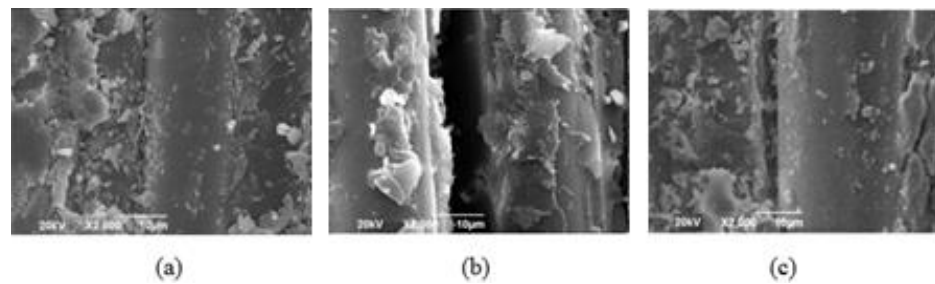


Figure 4.13: Longitudinal micrograph of Type I GFRP bars: (a) immersed for 5 months at 40°C (b) immersed for 10 months at 40°C (c) immersed for 15 months at 40°C

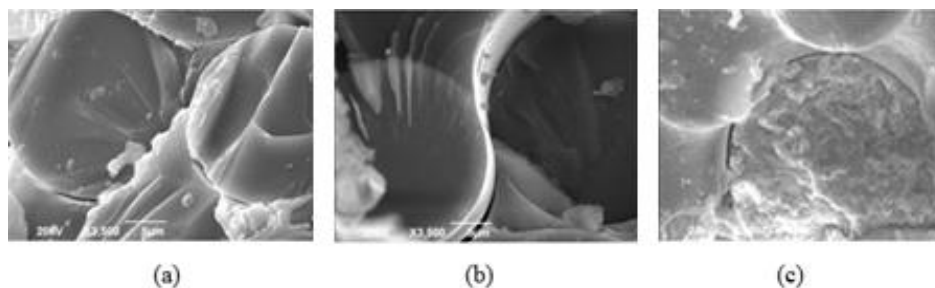


Figure 4.14: Cross-sectional micrograph of Type I GFRP bars: (a) immersed for 5 months at 40°C (b) immersed for 10 months at 40°C (c) immersed for 15 months at 40°C

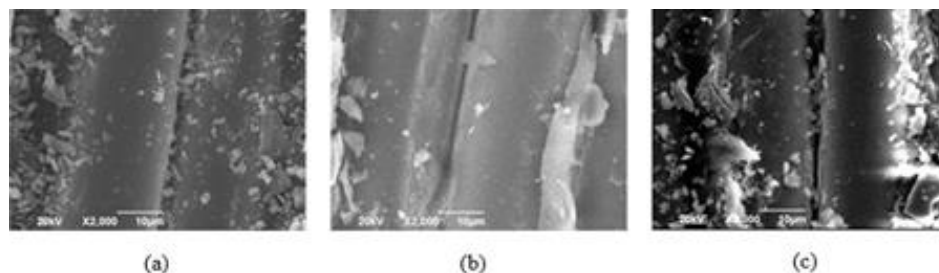


Figure 4.15: Longitudinal micrograph of Type II GFRP bars: (a) immersed for 5 months at 40°C (b) immersed for 10 months at 40°C (c) immersed for 15 months at 40°C

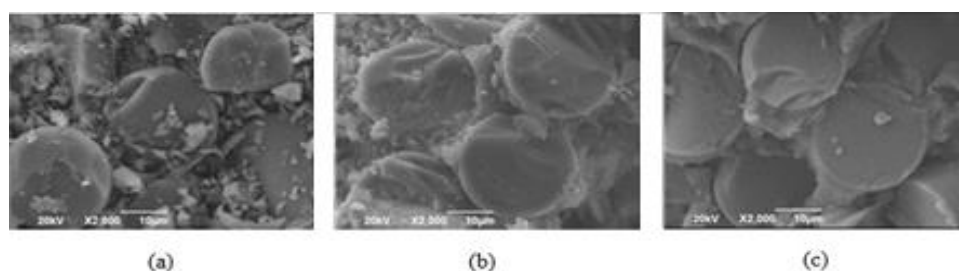


Figure 4.16: Cross-sectional micrograph of Type II GFRP bars: (a) immersed for 5 months at 40°C (b) immersed for 10 months at 40°C (c) immersed for 15 months at 40°C

4.4.1.5 Matrix Digestion Analysis

To evaluate the degradation mechanism and assess the conditioning-induced mass dissolution of GFRP bars, matrix digestion by nitric acid was used. Figure 4.17 and 4.18 show the respective matrix retention of Type I and II GFRP bars after conditioning. Longer conditioning and higher temperatures tended to reduce the matrix retention. This is consistent with other published test results [44]. At 5 months of conditioning, Type II GFRP was more prone to matrix mass loss than Type I, especially at higher temperatures. After 15 months of exposure at 60°C, Type I GFRP experienced a matrix retention of 83%, while Type II GFRP featured a matrix retention of 62% only. It is possible that the degraded matrix dissolved in the water solution or remained inside the concrete after conditioning. In fact, SEM micrographs of GFRP provide evidence of a relatively clean fiber surface after conditioning for 15 months at

60°C, indicating lesser polymer matrix at the fiber-matrix interface. A similar phenomenon was reported elsewhere with samples exhibiting a weight loss of 18% after a 2-hour exposure at 350°C [45]. TGA was also employed to validate the results obtained by matrix digestion, according to ASTM E1868-10 [27]. Table 4.2 summarizes the matrix content (% by mass) measured using both methods for control samples and those conditioned at 60°C for 15 months. Matrix retention results obtained by both test methods were similar. The matrix retention after 15 months of conditioning at 60°C was approximately 83-86% for Type I GFRP bars and 62-64% for Type II bars. Obviously, Type II GFRP bars were more susceptible to degradation during conditioning as manifested by their lower matrix retention.

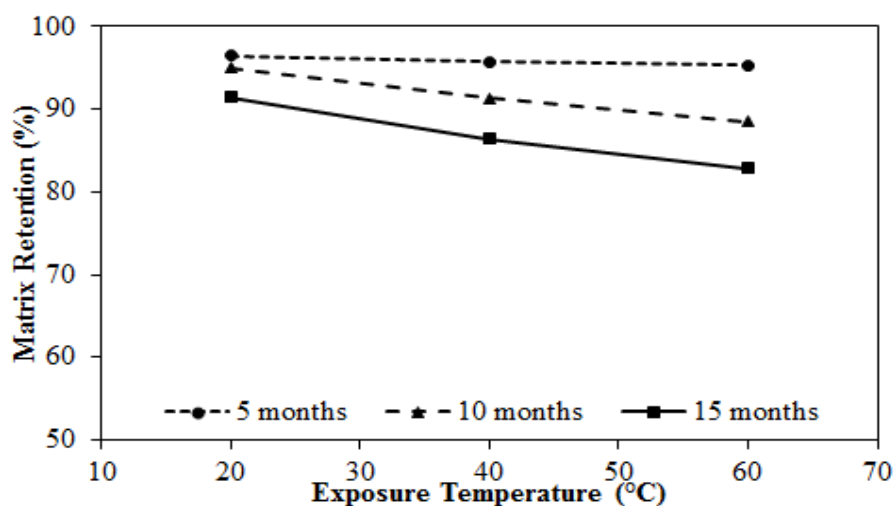


Figure 4.17: Matrix retention of Type I GFRP bar as a function of exposure temperature

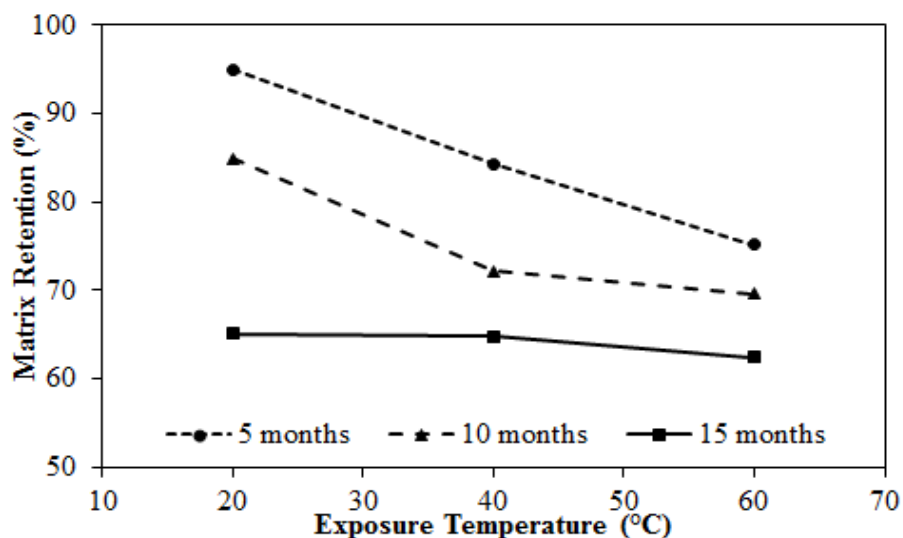


Figure 4.18: Residual matrix of Type II GFRP bar with respect to exposure temperature

Table 4.2: Fiber and matrix content of Type I and II GFRP bars using matrix digestion and TGA

GFRP Type	Conditioning		Content by Matrix Digestion (%)		Content by TGA (%)		Matrix retention (%)	
	Duration (month)	Temperature (°C)	Fiber	Matrix	Fiber	Matrix	Nitric acid	TGA
Type I	Control	Ambient	78.3	21.7	79.0	21.0	-	-
	15	60	82.0	18.0	82.0	18.0	83	86
Type II	Control	Ambient	75.5	24.5	75.0	25.0	-	-
	15	60	84.7	15.3	84.0	16.0	62	64

4.4.1.6 FTIR Analysis

FTIR spectra of Type I and Type II control samples along with those of specimens conditioned for 10 months at different temperatures are plotted in Figure 4.19 and 4.20, respectively. FTIR was used to evaluate the degree of hydrolysis reaction. In the process, two regions were studied: 2800–3000 cm^{-1} , representing carbon-hydrogen groups, and 3200–3600 cm^{-1} , representing hydroxyl groups. While hydrolysis leads to

an increase in the infrared band of OH, it does not affect that of CH. The ratio of maximum peaks in each of OH and CH band characterizes the relative amount of hydroxyl groups in the specimen [18,24]. The OH-to-CH ratio (OH/CH) of conditioned samples are presented in Table 4.3. Increasing the conditioning temperature and/or duration led to an increase in the OH/CH, signifying the progression of hydrolysis reaction. For instance, the increase in temperature from 40 to 60°C for Type I GFRP conditioned for 10 months resulted in an increase in OH/CH from 1.14 to 1.24. Alternatively, when the time of conditioning was extended from 10 to 15 months, Type I GFRP samples exposed to 40°C experienced an increase in ratio from 1.14 to 1.16. This shows that increasing the conditioning temperature could further enhance the reaction. It is conclusive that temperature is the more governing parameter compared to time of conditioning. This is consistent with similar observations reported by other researchers [18, 22].

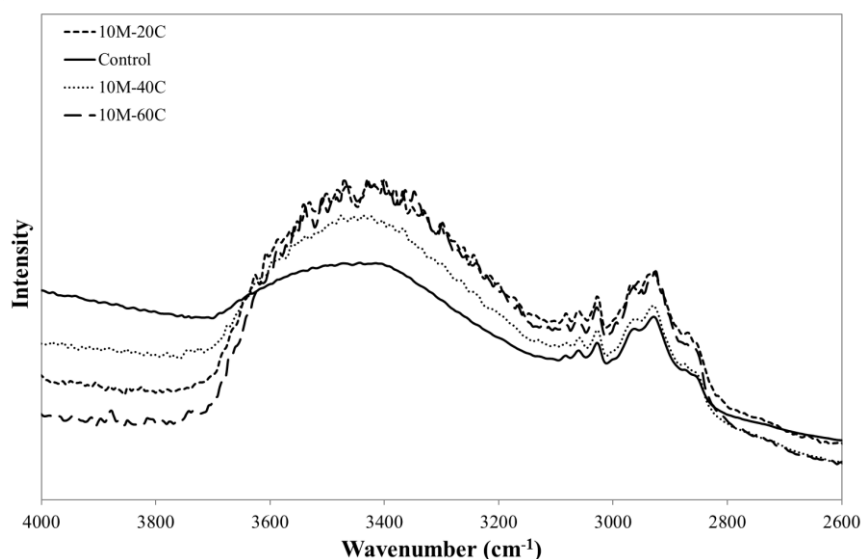


Figure 4.19: FTIR spectra of 10-month conditioned Type I GFRP bars

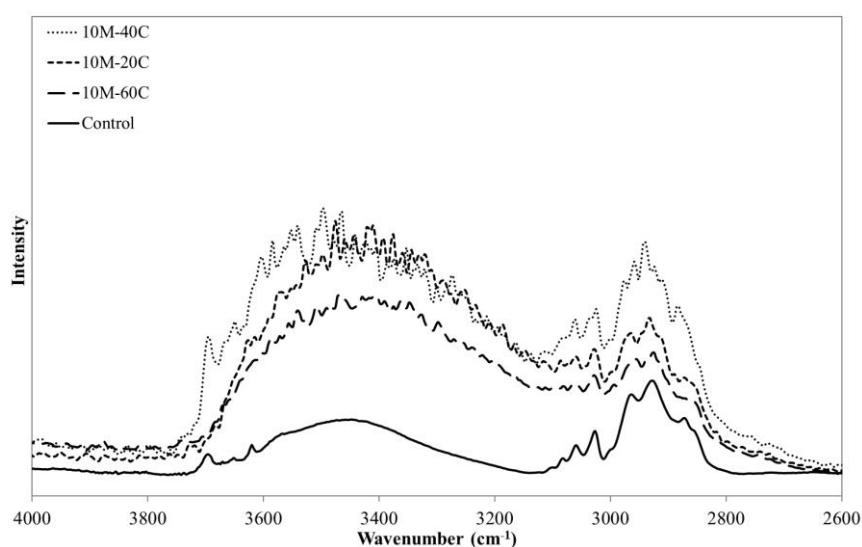


Figure 4.20: FTIR spectra of 10-month conditioned Type II GFRP bars

The percent increase in OH/CH is presented in Table 4.3. In comparison to Type I GFRP bars, Type II counterparts developed much more conditioning-induced hydroxyl groups. As a result, the tensile strength retention was much lower in Type II GFRP. It should be noted that the major increase in OH band for Type II samples was at 5 months. No or insignificant additional increase in the band was recorded over the subsequent 10 months.

Table 4.3: Band ratios of conditioned and control samples

GFRP Type	Conditioning		OH/CH	Increase (%) [*]
	Duration (months)	Temperature (°C)		
Type I	Control	Ambient	1.07	-
	5	20	1.09	2
		40	1.12	5
		60	1.36	27
	10	20	1.11	4
		40	1.14	7
		60	1.24	16
	15	20	1.12	5
		40	1.16	8
		60	1.24	16

Table 4.3: Band ratios of conditioned and control samples (Cont.)

GFRP Type	Conditioning		OH/CH	Increase (%) [*]
	Duration (months)	Temperature (°C)		
Type II	Control	Ambient	0.59	-
	5	20	1.1	86
		40	1.12	90
		60	1.14	93
	10	20	1.08	83
		40	1.12	90
		60	1.15	95
	15	20	1.11	88
		40	1.16	97
		60	1.15	95

^{*} Increase (%) represents the change in OH/CH ratio of conditioned samples with respect to that of control sample

4.4.1.7 DSC Analysis

The glass transition temperature (T_g) of control and conditioned type I and II GFRP samples are presented in Table 4.4. Two scans were conducted for each specimen. Samples subjected to higher conditioning temperatures recorded lower T_g values after the first scan. This could be due to matrix plasticization. Evidenced by SEM imaging, higher temperatures intensified the separation between the fiber and matrix and resulted in more voids in the matrix microstructure. Such voids increased the exposure of the polymer matrix to water/moisture and increased the water absorption. As a result, the polymer structure was modified and the T_g was reduced. Additionally, extending the conditioning time from 5 to 10 months while maintaining the same temperature resulted in a lower T_g after the first scan, particularly for Type II GFRP bars. T_g values recorded after 15 months of conditioning, however, were almost equal to those recorded after 10 months, indicating that water absorption had reached an equilibrium state after 10 months of conditioning.

As the samples were heated during the second scan, water evaporation reversed the plasticizing effect, and hence, T_g values of conditioned samples recorded in the second

scan became equal to those of the control ones. It should be noted that, for the Type I GFRP control specimen, the T_g corresponding to the second scan was slightly higher than that of the first scan. This could be due to a post-curing phenomenon during the second heating scan. Similar observations were reported by other researchers [21]. Since the T_g after the second scan remained unaffected by conditioning, the polymer matrix did not undergo irreversible chemical degradation. It can be concluded that the deterioration mechanisms of the GFRP bars tested in this study are degradation at the fiber-matrix interface, plasticization, hydrolysis and loss of the polymer matrix during conditioning.

Table 4.4: Glass transition temperature of GFRP bars using DSC analysis

GFRP Type	Conditioning		T_g (°C)	
	Duration (months)	Temperature (°C)	1 st Run	2 nd Run
Type I	Control	Ambient	101	106
	5	20	98	107
		40	98	106
		60	96	107
	10	20	98	107
		40	97	106
		60	95	106
	15	20	98	105
		40	98	106
		60	95	105
Type II	Control	Ambient	125	125
	5	20	115	126
		40	105	126
		60	98	126
	10	20	100	125
		40	94	126
		60	90	126
	15	20	99	126
		40	94	128
		60	90	126

4.4.2 Loaded Specimens

The loaded specimens were conditioned under a sustained load that corresponded to a 25% of their initial tensile strength. None of the loaded specimens conditioned at 20°C were creep-ruptured during conditioning. In contrast, many bars were creep-ruptured during conditioning at the higher temperatures of 40 and 60°C. This possibly occurred because the applied stress level in the bars conditioned under a sustained load (25% of the ultimate strength) was 25% higher than that specified by the ACI 440.1R-15 [46] (20% of the ultimate tensile strength).

Specimens that were creep-ruptured during conditioning as given in Table 4.5. The ruptured specimens were used to conduct the moisture uptake, SEM, DSC and FTIR tests.

Table 4.5: Ruptured bars of Type I and Type II GFRP

Duration (month)	Temp. (°C)	Number of Ruptured Bars	
		GFRP Type I	GFRP Type II
5	20	-	-
	40	2	-
	60	1	1
10	20	-	-
	40	-	-
	60	3	2
15	20	-	-
	40	-	2
	60	2	2

4.4.2.1 Moisture Absorption

Table 4.6 shows the moisture absorption of type I and II GFRP bars subjected to sustained load, different conditioning temperatures, and durations. The cracks in the concrete surrounding the loaded specimens resulted in more moisture absorption than that exhibited by the unloaded specimens.

At the same temperature, the moisture absorption increased by increasing the time of conditioning for both types of GFRP. The moisture absorption of Type I GFRP bars conditioned under sustained load for approximately 10 months at 60°C (1.95%) was higher than that of its counterpart conditioned without load (1.15%, refer to Figure 4.5). Type II GFRP bars conditioned under sustained load exhibited moisture absorption of 2.15% at 5 months of conditioning at 60°C. Its counterpart conditioned without load exhibited a moisture absorption of 0.75% only (refer to Figure 4.5). This indicates that the presence of sustained load during conditioning increased the moisture absorption.

Table 4.6: Moisture uptake of conditioned Type I and II GFRP samples (SL)

GFRP Type	Conditioning		Moisture absorption (%)
	Duration (months)	Temp. (°C)	
Type I	3.8	40	0.54
	2.5	60	1.19
	6.7	60	1.29
	7.9	60	1.36
	9.6	60	1.95
Type II	8.6	40	0.86
	13.6	40	1.56
	4.9	60	2.15
	8.6	60	2.23
	12.4	60	2.54

4.4.2.2 Tensile Strength Retention

The tensile strengths of GFRP specimens conditioned under a sustained load are compared to those of the control unconditioned counterparts in Figure 4.21. The specimen designation shown in this figure consists of four characters. The first two characters refer to the conditioning duration (5M, 10M, and 15M) whereas the last two characters refer to the conditioning temperature (20C, 40C, and 60C). The strengths

of the ruptured specimens were considered in the average strength of the three replicate specimens. The tensile strength of the ruptured specimens was taken as zero, and hence, the range of the highest and lowest values of the three replicate specimens was not included in Figure 4.21 for clarity. It is evident that the strength of the specimens conditioned under a sustained load at the higher temperatures of 40 and 60°C are significantly lower than those of the control specimens. The tensile strength retention of the Type I GFRP bars conditioned for 5 months at 60°C (5M60C) was higher than that of their counterparts conditioned for 5 months at 40°C (5M40C) because two of the three replicate specimens were ruptured at 40°C whereas only one specimen was ruptured at 60°C.

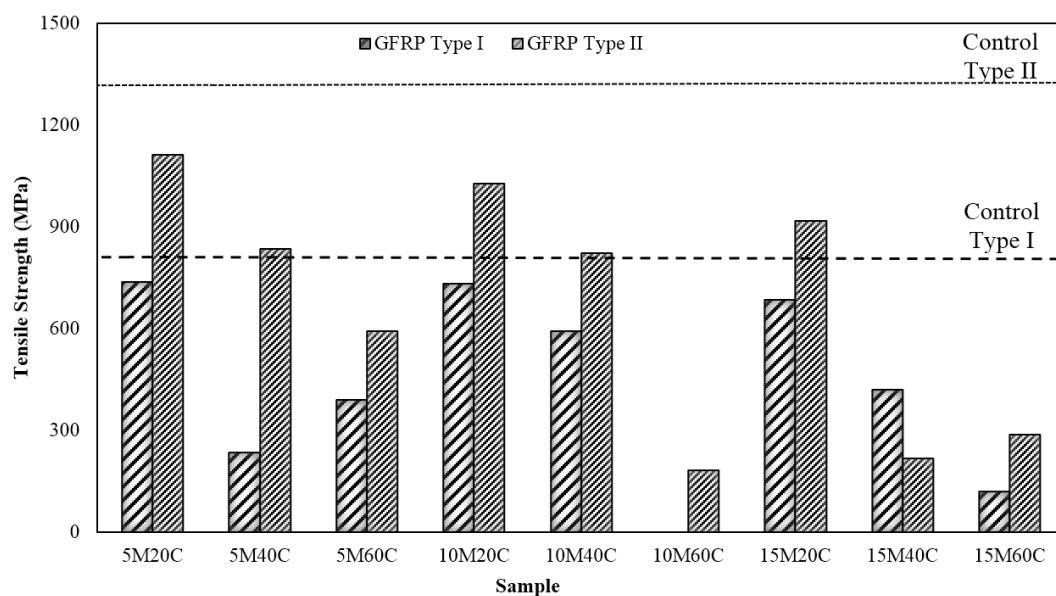
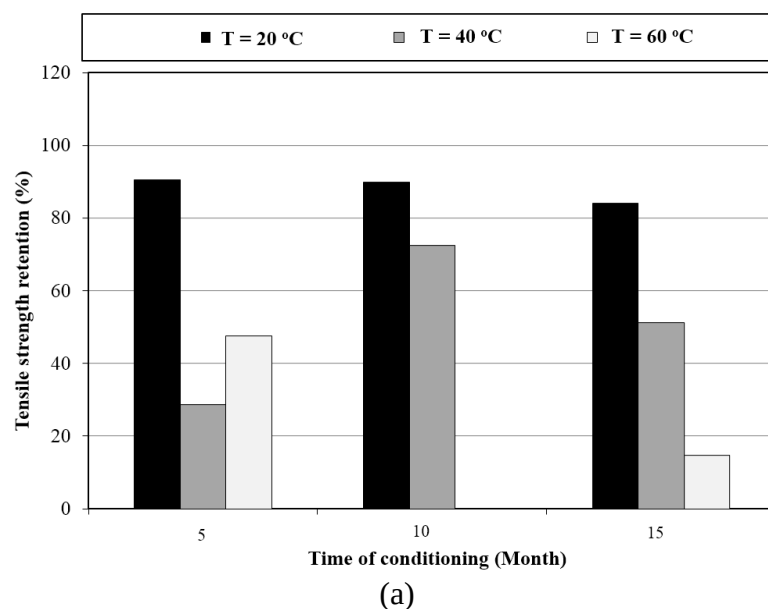


Figure 4.21: Tensile strengths of GFRP conditioned under load

Figure 4.22 shows the tensile strength retentions of the specimens conditioned under a sustained load. The tensile strength retentions of the ruptured specimens were considered in the average value of the three replicate specimens. The tensile strength retention of the ruptured specimens was taken as zero, and hence, the range of the highest and lowest values of the three replicate specimens was not included in Figure

4.22 for clarity. At a conditioning temperature of 20°C, Type I GFRP bars exhibited tensile strength retentions of 90, 90, and 84% after 5, 10, and 15 months respectively. Their Type II GFRP counterparts exhibited lower tensile strength retentions of 84, 78, and 69%, respectively, indicating that Type II GFRP bars had inferior durability performance than that of Type I.

Many GFRP bars from both types were creep-ruptured when conditioned under a sustained at temperatures of 40 and 60°C. As a result, the average tensile strength retentions of the corresponding three replicate bars were significantly reduced. For instance, at 5 months of conditioning under load at 60°C, one bar was creep-ruptured from each type and hence, Type I and Type II GFRP bars exhibited average tensile strength retentions of 48 and 45%, respectively. At 15 months of conditioning under a sustained load at 60°C, two bars were creep-ruptured from each type and hence, average tensile strength retentions of 15 and 22% only were recorded for Type I and Type II GFRP bars, respectively.



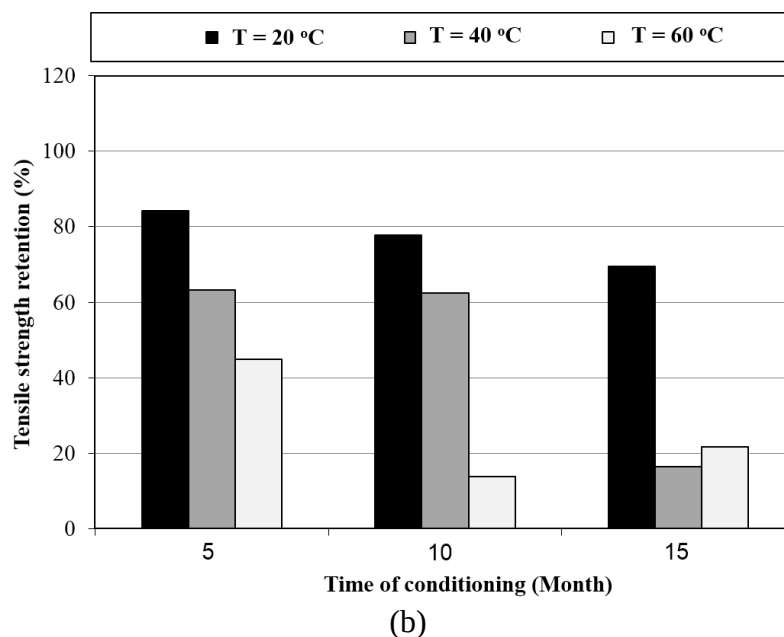


Figure 4.22: Tensile strength retention of GFRP bars conditioned under a sustained load; (a) Type I, (b) Type II

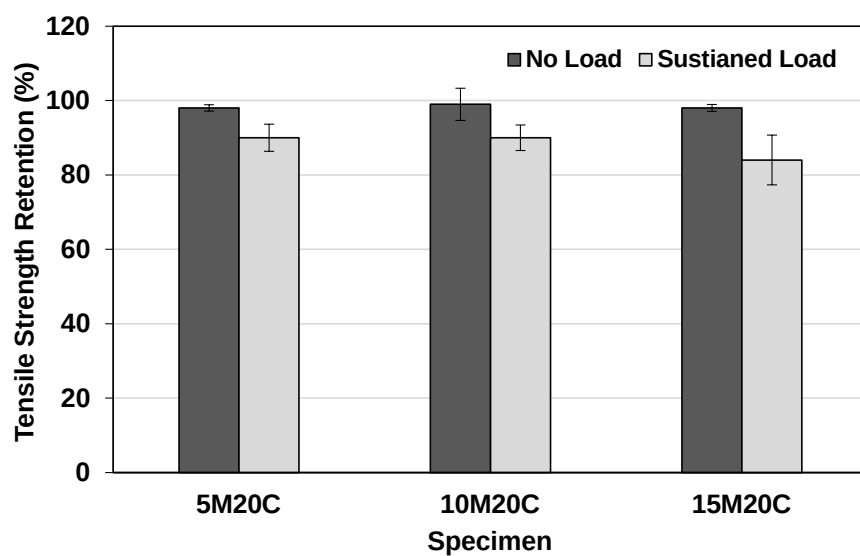
Table 4.7 compares the tensile strength of the bars conditioned under a sustained load with those conditioned without load. It can be seen that at a conditioning temperature of $20\text{ }^{\circ}\text{C}$, the ratio of the strength retention of loaded specimens to that of unloaded specimens (SR_l/SR_u) was in the range of 0.86 to 0.91 for Type I and 0.95 to 1.0 for Type II. The ratio (SR_l/SR_u) was significantly reduced at the higher temperatures of $40\text{ }^{\circ}\text{C}$ and $60\text{ }^{\circ}\text{C}$ because of creep rupture of many of the replicate specimens.

Table 4.7: Effect of sustained load on tensile strength of conditioned bars

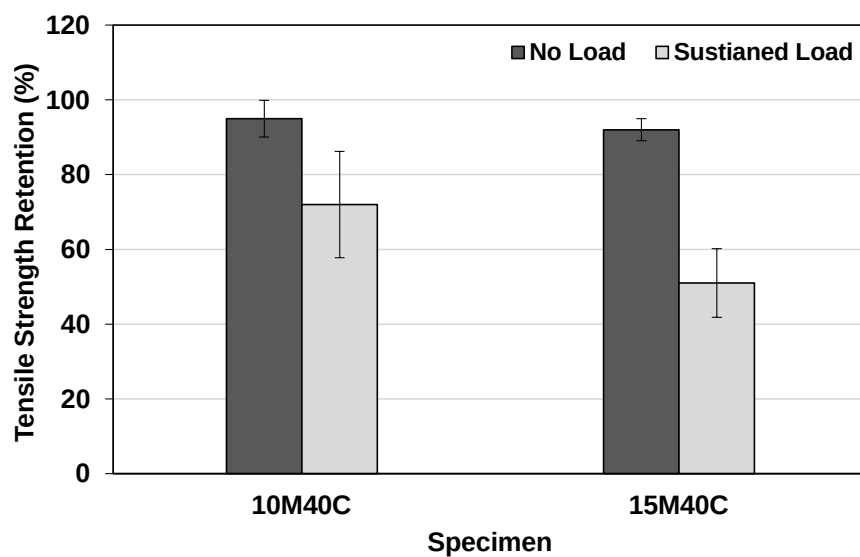
GFRP type	Time of conditioning (month)	Temperature (°C)	Strength retention (%)		Ratio (SR _l /SR _u)*	Number of ruptured bars
			No load	Sustained load		
Type I	5	20	98	90	0.92	-
		40	92	29	0.32	2
		60	90	48	0.53	1
	10	20	99	90	0.91	-
		40	95	72	0.76	-
		60	94	0	0.00	3
	15	20	98	84	0.86	-
		40	92	51	0.55	-
		60	85	15	0.18	2
Type II	5	20	81	84	1.00	-
		40	77	63	0.82	-
		60	71	45	0.63	1
	10	20	79	78	0.99	-
		40	66	62	0.94	-
		60	50	14	0.28	2
	15	20	73	69	0.95	-
		40	68	16	0.24	2
		60	53	22	0.42	2

*(SR_l/SR_u): Ratio of strength retention of loaded specimens to that of unloaded specimens.

The tensile strength retentions of Type I and Type II GFRP bars that were not creep-ruptured during conditioning under sustained load are compared to those of their counterparts conditioned without load in Figures 4.23 to 4.24, respectively. The specimen designation shown in these figures consists of four characters.



(a)

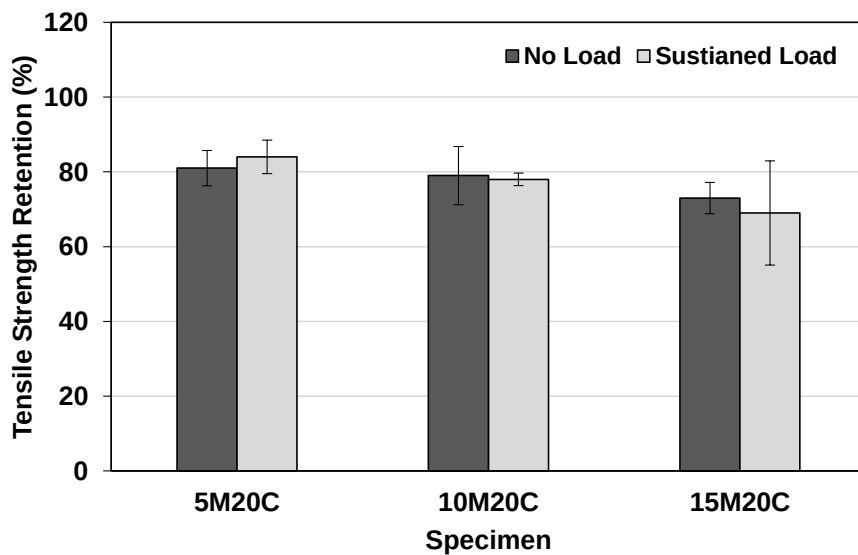


(b)

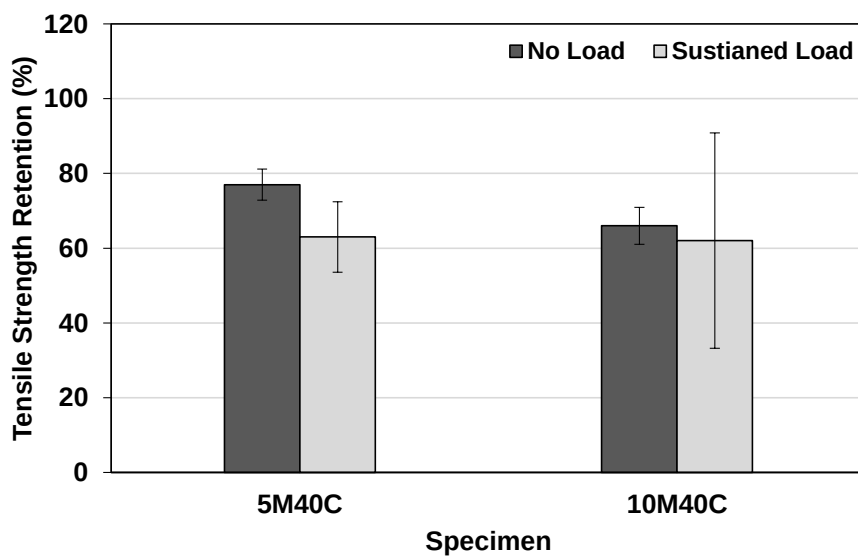
Figure 4.23: Effect of sustained load on tensile strength retention of non-ruptured Type I bars; (a) at 20°C, (b) at 40°C

From Figure 4.23, it can be seen that the tensile strength retentions of Type I GFRP bars conditioned at 20°C under a sustained load for 5, 10, and 15 months were 8, 8, and 14% lower than those of their counterparts conditioned without load. The effect of sustained load was more significant at the higher temperatures. The tensile strength retentions of Type I GFRP bars conditioned under load for 10 and 15 months at 40°C

were approximately 24 and 45% lower than those of their counterparts conditioned without load.



(a)



(b)

Figure 4.24: Effect of sustained load on tensile strength retention of non-ruptured Type II bars; (a) at 20°C, (b) at 40°C

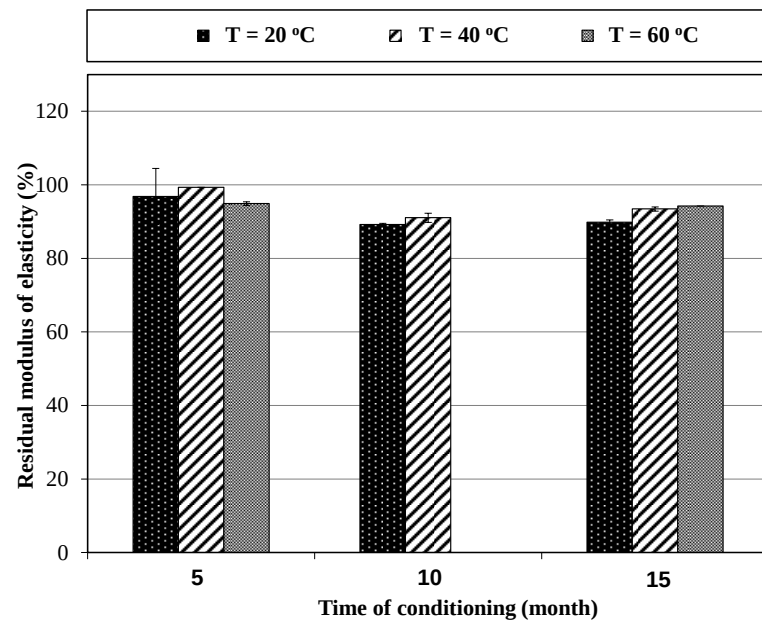
From Figure 4.23, it can be seen that the tensile strength retentions of Type II GFRP bars were not affected by the presence of sustained load during conditioning at 20°C for 5 and 10 months. At 15 months, the tensile strength retentions of Type II GFRP

bars conditioned under load at 20°C was approximately 5% lower than that of their counterparts conditioned without load. The presence of a sustained load during conditioning reduced, however, the tensile strength retentions of the specimens conditioned at the higher temperatures. The tensile strength retentions of Type II GFRP bars conditioned under load for 5 and 10 months at 40°C were approximately 18 and 6% lower than those of their counterparts conditioned without load.

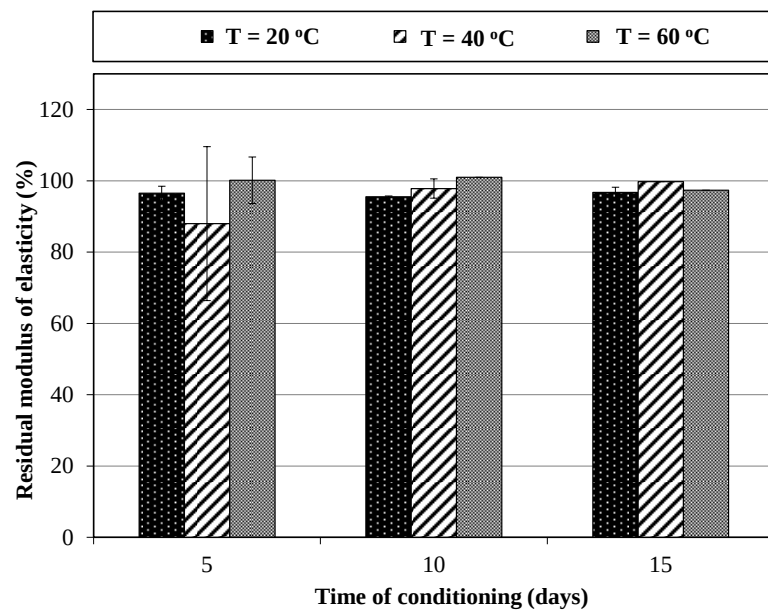
4.4.2.3 Residual Modulus of Elasticity

The effect of accelerated ageing under a sustained load on the modulus of elasticity of non-ruptured GFRP bars is shown in Figure 4.25. Type I GFRP bars exhibited insignificant reductions in the tensile modulus in the range of 1 to 5% after 5 months of exposure. At 10 months of exposure, the reduction in the tensile modulus increased to approximately 10% at 20, and 40°C. At 60°C all replicate specimens were creep-ruptured, and hence the corresponding tensile modulus was not recorded. At 15 months of exposure, the reduction of the tensile modulus of Type I was in the range of 6 to 10%. The temperature has no noticeable effect on the tensile modulus of Type I conditioned GFRP bars.

After 5 months of exposure, Type II GFRP bars exhibited a maximum reduction of 12% in the tensile modulus when conditioned at 40°C. Increasing the conditioning temperatures had no noticeable effect on the tensile modulus. The reduction in the tensile modulus at 5 months of exposure was approximately 5%. Increasing the time of conditioning did not result in a further reduction in the tensile modulus at all temperatures.



(a) Type I



(b) Type II

Figure 4.25: Residual modulus of elasticity of non-ruptured bars conditioned under load; (a) GFRP Type I, (b) GFRP Type II.

4.4.2.4 SEM Analysis

Samples taken from the middle 100 mm of some of the creep-ruptured specimens were examined by SEM. Figure 4.26 and Figure 4.27 represent the longitudinal and cross-sectional micrographs of conditioned GFRP Type I at 40°C and 60 °C. A separation between the fiber and matrix after 3.8 months of conditioning at 40°C was observed as shown in Figure 4.26(a). GFRP bars conditioned at the higher temperatures exhibited smooth fiber surface and matrix disintegration. Increasing the time of conditioning at the same elevated temperature of 60°C increased the gap between the fiber and matrix. This is an indication of matrix disintegration due to the hydrolysis reaction which accelerates in hot environment thus leading to non-uniform distribution of load among the fibers [18,39]. Cross-sectional micrographs of Figure 4.27 show signs of circumferential debonding at the fiber-matrix interface particularly at the higher temperature of 60°C.

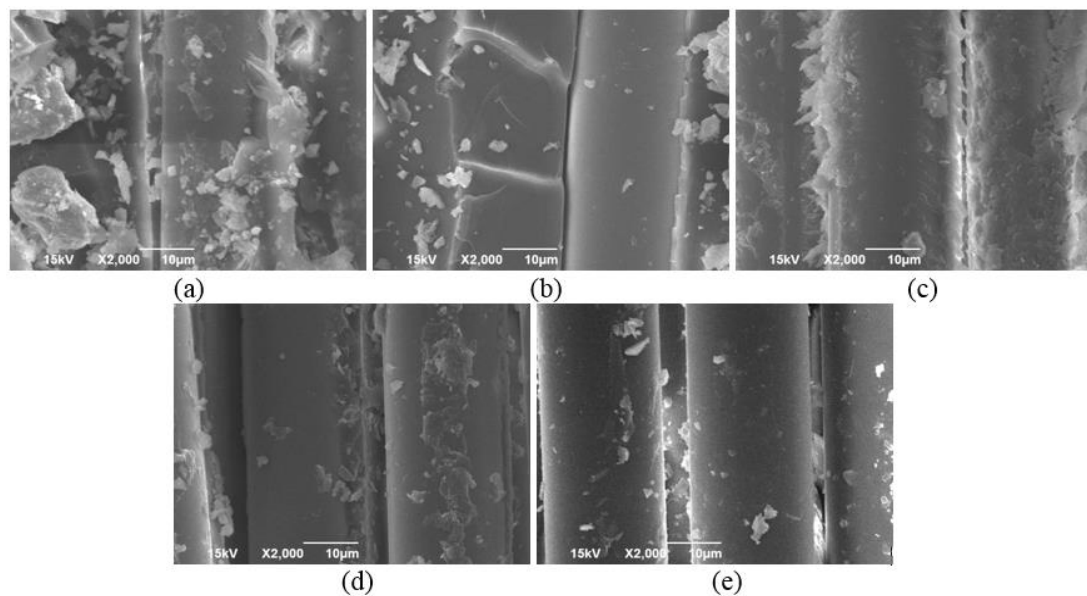


Figure 4.26: Longitudinal micrographs of Type I GFRP conditioned under load; (a) 3.8M40C, (b) 2.5M60C, (c) 6.7M60C, (d) 7.9M60C, (e) 9.6M60C

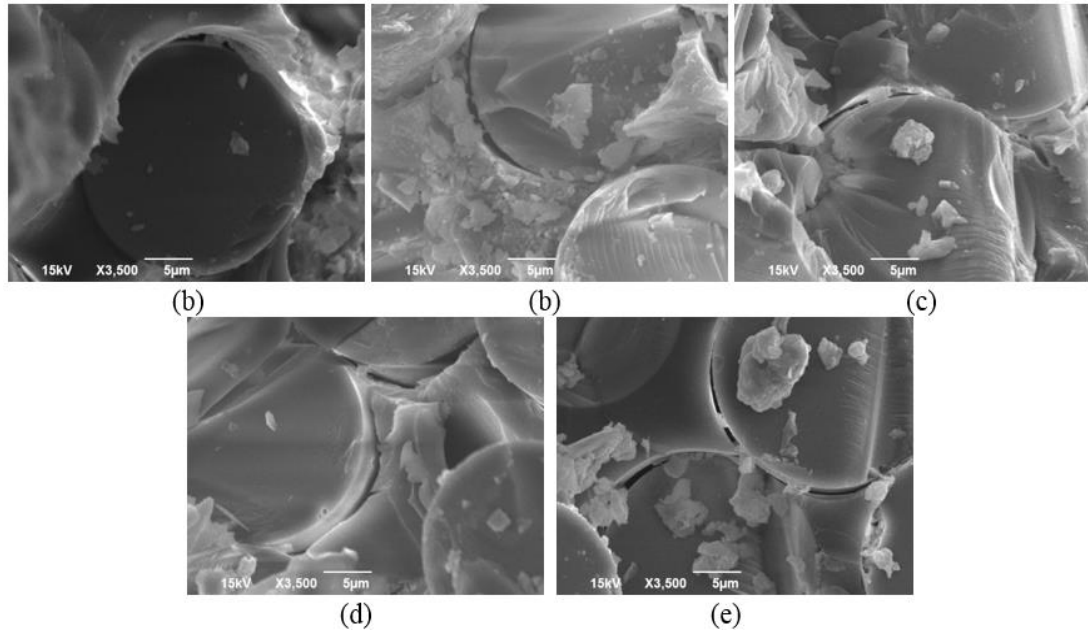


Figure 4.27: Cross-sectional micrographs of Type I GFRP conditioned under load; (a) 3.8M40C, (b) 2.5M60C, (c) 6.7M60C, (d) 7.9M60C, (e) 9.6M60C

Figure 4.28 and Figure 4.29 represent the longitudinal and cross-sectional micrographs of Type II GFRP bars conditioned at 40°C and 60°C. Matrix disintegration and debonding at the fiber-matrix interface were observed. The debonding was more obvious in specimen 13.6M40C that was creep-ruptured after 13.6 months at 40°C (Figure 4.29(c)) and specimen 12.3M60C that was creep-ruptured after 12.3 months at 60°C. These findings demonstrate that accelerated ageing weakens the bond at the fiber-matrix interface. The bond deterioration at the fiber-matrix interface was very intense to the level that many GFRP bars were creep-ruptured during conditioning under a sustained load.

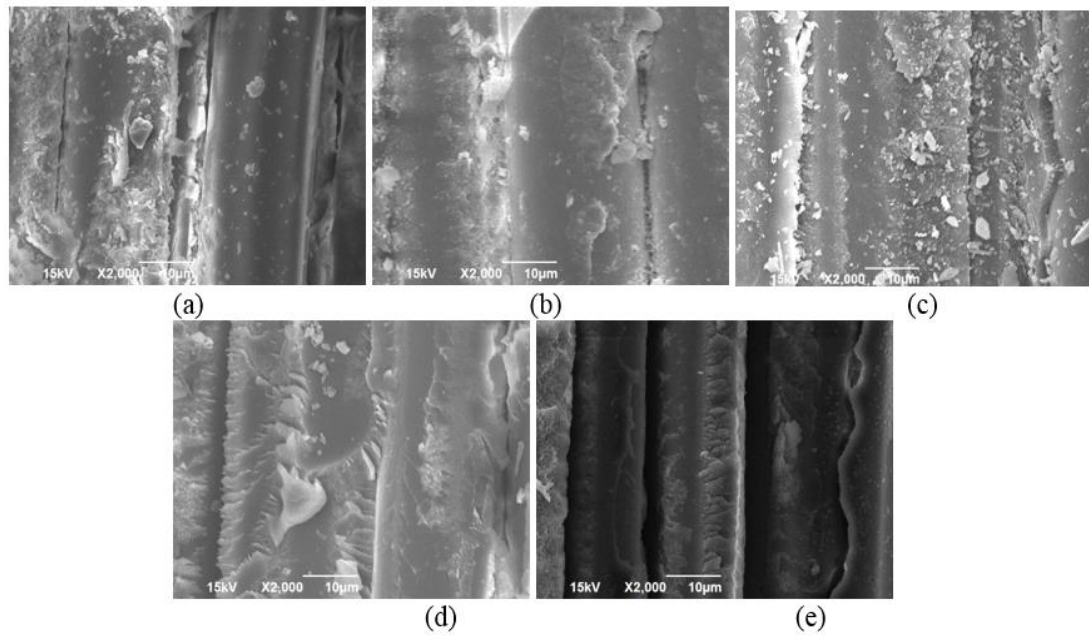


Figure 4.28: Longitudinal micrographs of Type II GFRP conditioned under load; (a) 4.9M40C, (b) 8.7M40C, (c) 13.6M40C, (d) 8.6M60C, (e) 12.3M60C

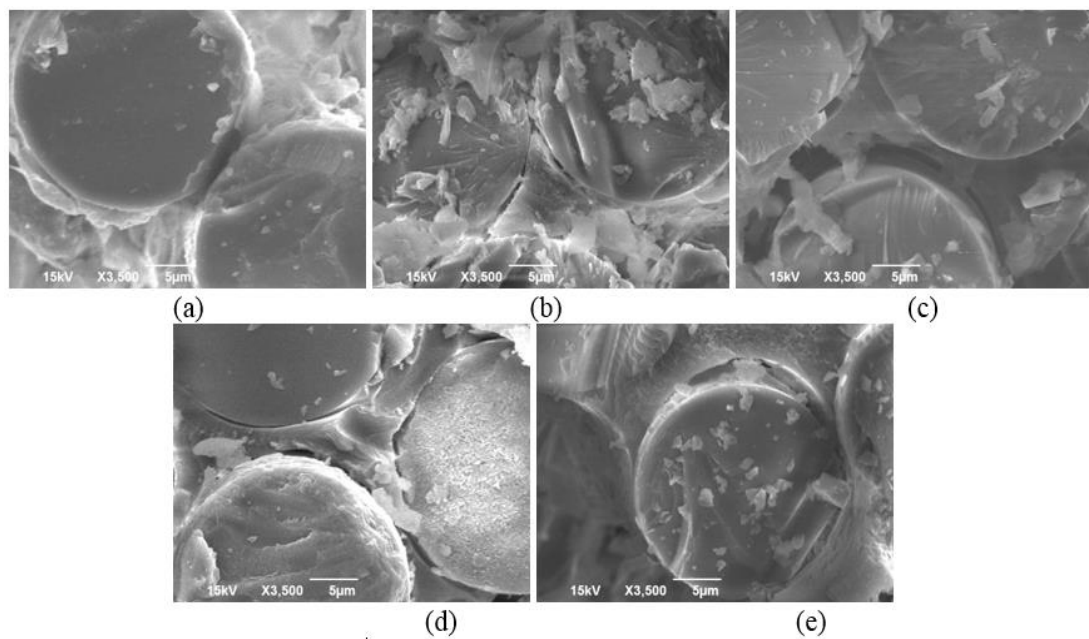


Figure 4.29: Cross-sectional micrographs of Type II GFRP conditioned under load; (a) 4.9M40C, (b) 8.7M40C, (c) 13.6M40C, (d) 8.6M60C, (e) 12.3M60C

4.4.2.5 FTIR analysis

The CH content is considered constant, any changes in the OH/CH should be a result of hydrolysis and/or saponification. Therefore, the higher OH/CH will be an indication of matrix degradation. Figure 4.30 shows the typical FTIR spectra of some of the Type I GFRP bars at the time of creep rupture. Table 4.7 shows the band ratios of conditioned and control samples; significant change in band ratio have been observed, increasing the temperature and/or duration led to an increase in OH/CH (hydrolysis reaction). Type I GFRP specimen conditioned at 60°C for 2.5 months exhibited higher OH/CH ratio than that exhibited by a counterpart specimen conditioned for a longer duration of 3.8 months but at 40°C. This further demonstrates that the progression in the hydrolysis reaction is more sensitive to the temperature rather than the conditioning duration. When Type I GFRP bars exposed to longer durations at the same temperature of 60°C, the increase in OH/CH was insignificant. Control sample of Type II GFRP bars has an OH/CH band ratio of 0.59. The OH/CH ratio increased to 1.19 to 1.27 for GFRP bars conditioned for 13.6 and 12.4 months at 40 and 60°C, respectively. The percent increase in OH/CH are also given in Table 4.8. Type II GFRP bars developed more conditioning-induced hydroxyl group, and hence, the tensile strength retention was much lower than that of Type I. Type I GFRP bars exhibited an increase in the OH/CH ratio of 19% at the 9.6 months of conditioning at 60°C only. The percent increase in the OH/CH ratio for Type II GFRP was 95% after the 8.6 months of conditioning at 60°C.

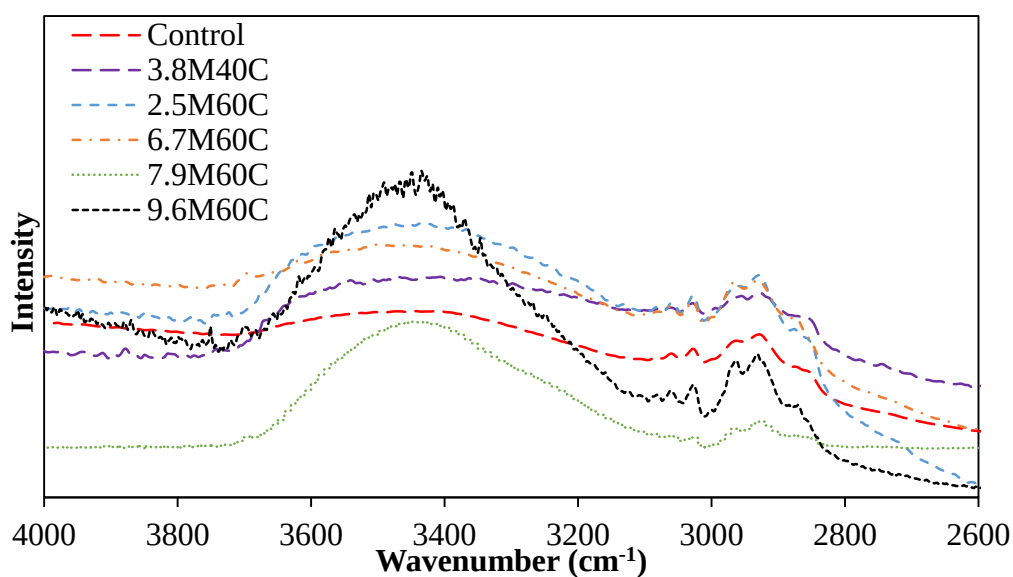


Figure 4.30: FTIR spectra of sustained load GFRP bars Type I

Table 4.8: Band ratios of conditioned and control samples, loaded GFRP bars Type I and Type II

GFRP Type	Conditioning		Peaks		OH/CH	Increase (%)*
	Duration (months)	Temp. (°C)	CH	OH		
Type I	Control	Ambient	3.62	3.38	1.07	-
	3.8	40	4.47	5.00	1.12	5
	2.5	60	3.13	3.83	1.22	14
	6.7	60	4.84	6.02	1.24	16
	7.9	60	4.92	6.20	1.26	18
	9.6	60	6.56	8.33	1.27	19
Type II	Control	Ambient	2.77	4.69	0.59	-
	8.6	40	4.30	4.91	1.14	93
	13.6	40	4.12	4.89	1.19	102
	4.9	60	4.49	5.14	1.14	93
	8.6	60	5.21	6.00	1.15	95
	12.4	60	5.58	7.11	1.27	115

*Increase (%) represents the change in OH/CH ratio of conditioned samples with respect to that of control sample

4.4.2.6 DSC Analysis

The glass transition temperature (T_g) of control and conditioned samples of GFRP bars Type I and Type II are summarized in Table 4.9. Two scans were performed for each sample; the first scan is to determine the difference in T_g between the control and conditioned samples as the decrease in T_g is an evidence of plasticizing effect of chemical degradations, while the second scan is to have information about the degradation mechanism.

Samples conditioned at the higher temperatures recorded lower T_g (GFRP bars Type I and Type II at 60°C) and this is because of the plasticization that caused chemical degradations of polymer matrix. The T_g of Type I GFRP bars conditioned at 40°C for 3.8 months was 95°C; while it decreased to 94°C when conditioned at 60°C conditioned for 2.5 months. The least T_g value of 90 °C was recorded for the specimens that were creep-ruptured after 7.9 and 9.6 months of conditioning at 60°C. Type II GFRP bars expressed significant decrease in the T_g value due to conditioning. The T_g of Type II GFRP conditioned at 40°C for 8.6 months was 98°C; while it decreased to 92°C when conditioned for the same duration at 60°C.

No major changes occurred in the T_g value after the second scan. As the samples were heated during the second scan, water evaporation reversed the plasticizing effect, and hence, T_g values of conditioned samples recorded in the second scan became equal to those of the control ones.

Table 4.9: Glass transition temperature of loaded GFRP bars using DSC analysis

GFRP Type	Conditioning		T_g (°C)	
	Duration (months)	Temp. (°C)	1st Run	2nd Run
Type I	Control	Ambient	101	106
	3.8	40	95	104
	2.5	60	94	105
	6.7	60	91	104
	7.9	60	90	105
	9.6	60	90	105
Type II	Control	Ambient	125	125
	8.6	40	98	125
	13.6	40	93	126
	4.9	60	95	125
	8.6	60	92	126
	12.4	60	90	125

Chapter 5: Durability Design Model

5.1 Introduction

The methodology adopted to develop a durability design model for the GFRP basr in moist seawater-contaminated concrete are presented in this chapter. A brief introduction about Arrhenius concept is provided. Modeling procedures are described. Durability design models for both type of GFRP conditioned in moist seawater-contaminated with and without a sustained load were produced.

5.2 Arrhenius Relationship

In the Arrhenius relationship, the degradation rate of GFRP is expressed by Equation 5.1 [13, 14, 17, 18, 23, 48].

$$k = Ae^{-E_a / RT} \quad (5.1)$$

Where:

k = degradation rate (1/time)

A = constant related to material and degradation process

E_a = activation energy

R = universal gas constant

T = temperature in kelvin

Arrhenius model assumes that the dominate degradation mechanism of the material will not change with time and temperature during exposure. Arrhenius model assumes that the rate of degradation is accelerated with an increase in temperature. Accelerated aging test results confirmed the validity of this assumption. Equation 5.1 can be expressed by Equations 5.2 and 5.3.

$$\frac{1}{K} = \frac{1}{A} e^{E_a / RT} \quad (5.2)$$

$$\ln \frac{1}{K} = \frac{E_a}{R} \frac{1}{T} - \ln A \quad (5.3)$$

Equation 5.2 shows that the degradation rate K can be transformed to the inverse of the time required for material property to reach a given value. Equation 5.3 indicates that the logarithm of the time required for material property to reach a given value is a linear function of $1/T$ with slope of E_a/R .

Degradation data for at least three different temperatures are required for prediction using Arrhenius theory. Accelerated aging test results should be employed along with Arrhenius model to exploit the temperature dependence of conditioned GFRP bars in a specific conditioning regime. The following procedure should be followed to develop a durability design model for a specific GFRP bar type in a specific loading condition and surrounding media.

Step 1: Plot the tensile strength retention versus the time of accelerated aging.

Step 2: Perform regression analysis to determine the best-fit linear relationship through each set of data at a specific temperature. An acceptable regression shall have an R^2 value of at least 0.8.

Step 3: Plot $(\ln)$ the time needed to reach particular tensile strength retention versus the inverse of temperature (i.e. Arrhenius-type relationship). An acceptable linear regression shall have an R^2 value of at least 0.8.

Step 4: Determine (E_a/R) from the Arrhenius-type relationships developed in Step 3.

Step 5: Calculate the time shift factor (TSF) relative to a reference temperature.

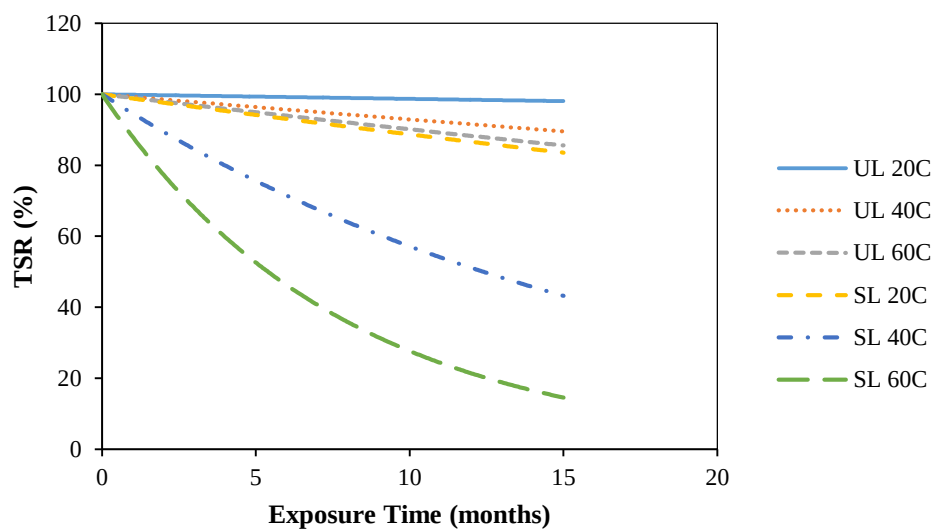
Step 6: Develop master curves for prediction of service life of GFRP bars in moist seawater-contaminated concrete by plotting the tensile strength retention versus the anticipated service life at a specific temperature expected in natural weathering.

5.3 Model Development

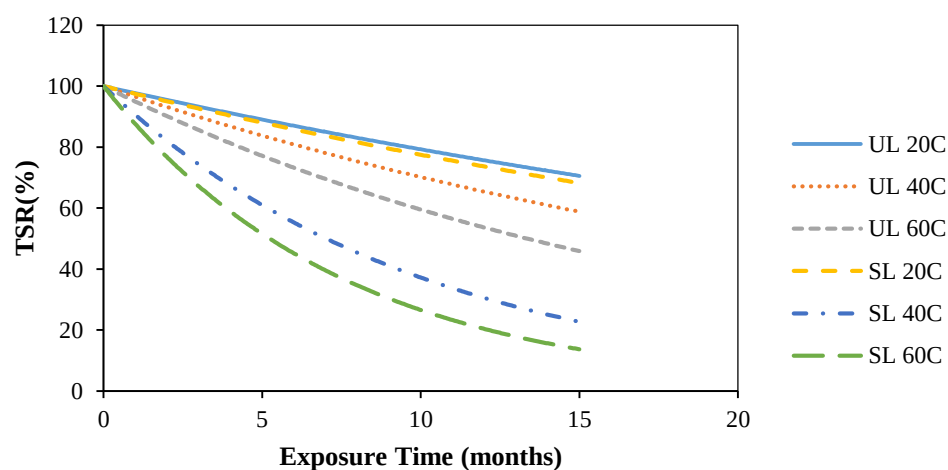
Plots of the tensile strength retention versus the time of accelerated aging is given in Figure 5.1, where UL and SL refer to no loading and sustained loading conditions, respectively. The degradation model given in Equation 5.4 has been adopted to plot best-fit curves presented in Figure 5.1, where Y = tensile strength retention (%), t = exposure time, and τ = fitted parameter. An acceptable regression line should have R^2 value of at least 0.8. Tensile strength retention data of some points were chosen in the standard-deviation range to accommodate the minimum acceptable R^2 value of 0.8.

$$Y = 100\exp(-t/\tau) \quad (5.4)$$

Table 5.1 summarizes the exponential equations for both types of GFRP with their R^2 values conditioned with and without a sustained load. These exponential equations were used to produce the times needed for unloaded and loaded specimens to reach tensile strength retentions of 40, 50, 60, 70 80, and 90% as shown in Tables 5.2 and 5.3. The $\ln$ of the time to reach particular tensile strength retention versus the inverse of temperature (i.e. Arrhenius-type relationship) for the unloaded and loaded specimens are plotted Figures 5.2 and 5.3, respectively. The coefficients of the Arrhenius-type relationships are given in Table 5.4.



(a)



(b)

Figure 5.1: Tensile strength retention versus time relationships; (a) Type I, (b) Type II

Table 5.1: Exponential equations with their R^2 value

Load Type	GFRP TYPE	Temperature	Equation	R^2	τ
No Load	Type I	20°C	$y = 100e^{-0.001x}$	0.89	1000
		40°C	$y = 100e^{-0.007x}$	0.83	143
		60°C	$y = 100e^{-0.01x}$	0.93	100
	Type II	20°C	$y = 100e^{-0.024x}$	0.83	42
		40°C	$y = 100e^{-0.035x}$	0.88	29
		60°C	$y = 100e^{-0.052x}$	0.81	19

Table 5.1: Exponential equations with their R^2 value (Cont.)

Load Type	GFRP TYPE	Temperature	Equation	R^2	τ
Sustained Load	Type I	20°C	$y = 100e^{-0.012x}$	0.88	83
		40°C	$y = 100e^{-0.041x}$	0.95	18
		60°C	$y = 100e^{-0.13x}$	0.99	8
	Type II	20°C	$y = 100e^{-0.025x}$	0.97	40
		40°C	$y = 100e^{-0.098x}$	0.82	10
		60°C	$y = 100e^{-0.124x}$	0.82	8

Table 5.2: Times needed to reach specific tensile strength retentions for unloaded specimens

40% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	916.3	6.82
	40	313.15	0.00319	131	4.875
	60	333.15	0.003	92	4.522
50% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	693	6.541
	40	313.15	0.00319	99	4.595
	60	333.15	0.003	69	4.234
60% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	511	6.236
	40	313.15	0.00319	73	4.29
	60	333.15	0.003	51	3.932
70% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	357	5.878
	40	313.15	0.00319	51	3.932
	60	333.15	0.003	36	3.584
80% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	223	5.407
	40	313.15	0.00319	32	3.466
	60	333.15	0.003	22	3.091

Table 5.2: Times needed to reach specific tensile strength retentions for unloaded specimens (Cont.)

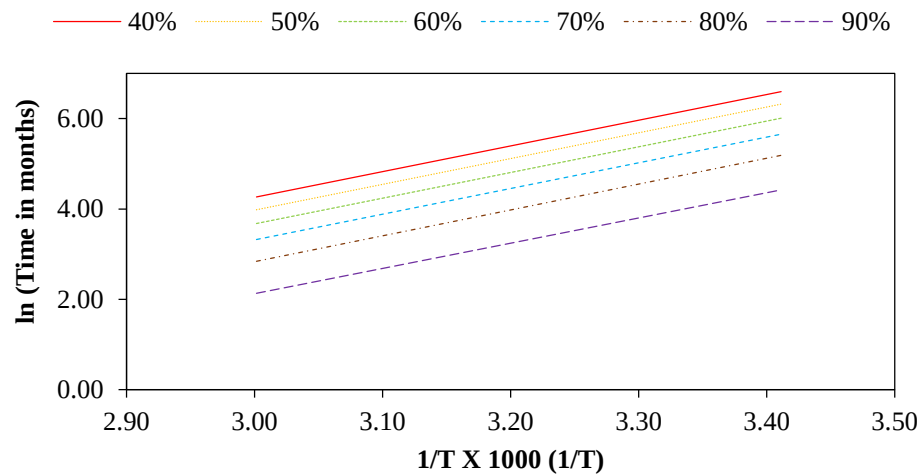
90% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	105	4.654
	40	313.15	0.00319	15	2.708
	60	333.15	0.003	11	2.398
40% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	38	3.638
	40	313.15	0.00319	26	3.258
	60	333.15	0.003	17	2.833
50% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	29	3.367
	40	313.15	0.00319	20	2.996
	60	333.15	0.003	13	2.565
60% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	21	3.045
	40	313.15	0.00319	15	2.708
	60	333.15	0.003	10	2.303
70% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	15	2.708
	40	313.15	0.00319	10	2.303
	60	333.15	0.003	7	1.946
80% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	9	2.197
	40	313.15	0.00319	6	1.792
	60	333.15	0.003	4	1.386
90% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	4	1.386
	40	313.15	0.00319	3	1.099
	60	333.15	0.003	2	0.693

Table 5.3: Times needed to reach specific tensile strength retentions for loaded specimens

40% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	76.358	4.335
	40	313.15	0.00319	22.349	3.107
	60	333.15	0.003	7.048	1.953
50% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	57.762	4.056
	40	313.15	0.00319	16.906	2.828
	60	333.15	0.003	5.332	1.674
60% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	42.569	3.751
	40	313.15	0.00319	12.459	2.522
	60	333.15	0.003	3.929	1.368
70% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	29.723	3.392
	40	313.15	0.00319	8.699	2.163
	60	333.15	0.003	2.744	1.009
80% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	18.595	2.923
	40	313.15	0.00319	5.443	1.694
	60	333.15	0.003	1.716	0.54
90% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type I	20	293.15	0.00341	8.78	2.172
	40	313.15	0.00319	2.57	0.944
	60	333.15	0.003	0.81	-0.21
40% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	36.652	3.601
	40	313.15	0.00319	9.35	2.235
	60	333.15	0.003	7.389	2

Table 5.3: Times needed to reach specific tensile strength retentions for loaded specimens (Cont.)

50% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	27.726	3.322
	40	313.15	0.00319	7.073	1.956
	60	333.15	0.003	5.59	1.721
60% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	20.433	3.017
	40	313.15	0.00319	5.213	1.651
	60	333.15	0.003	4.12	1.416
70% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	14.267	2.658
	40	313.15	0.00319	3.64	1.292
	60	333.15	0.003	2.876	1.057
80% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	8.926	2.189
	40	313.15	0.00319	2.277	0.823
	60	333.15	0.003	1.8	0.588
90% Retention					
GFRP	Temp. c	Temp (K)	1/T	Time (months)	ln (Time)
Type II	20	293.15	0.00341	4.214	1.439
	40	313.15	0.00319	1.075	0.072
	60	333.15	0.003	0.85	-0.163



(a)

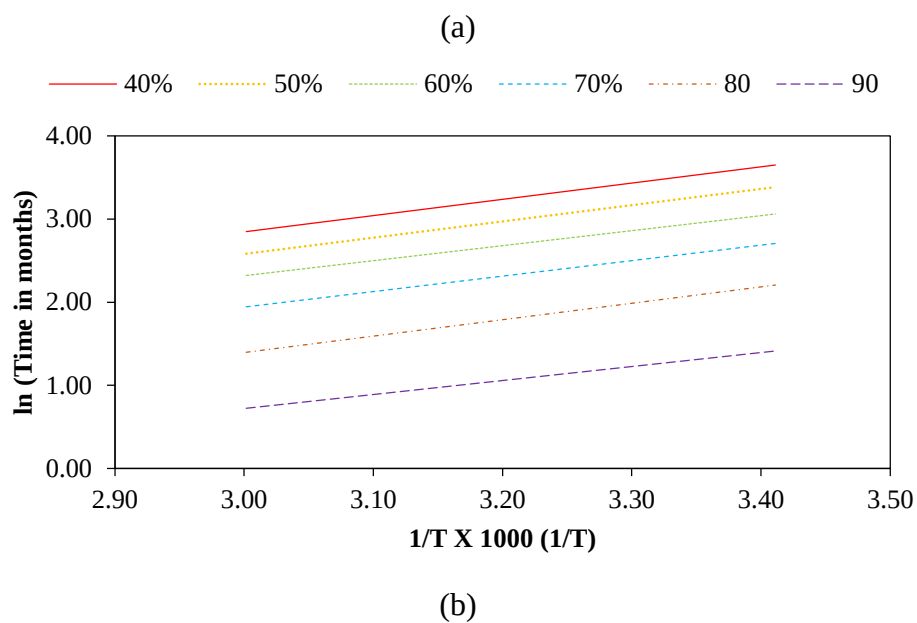
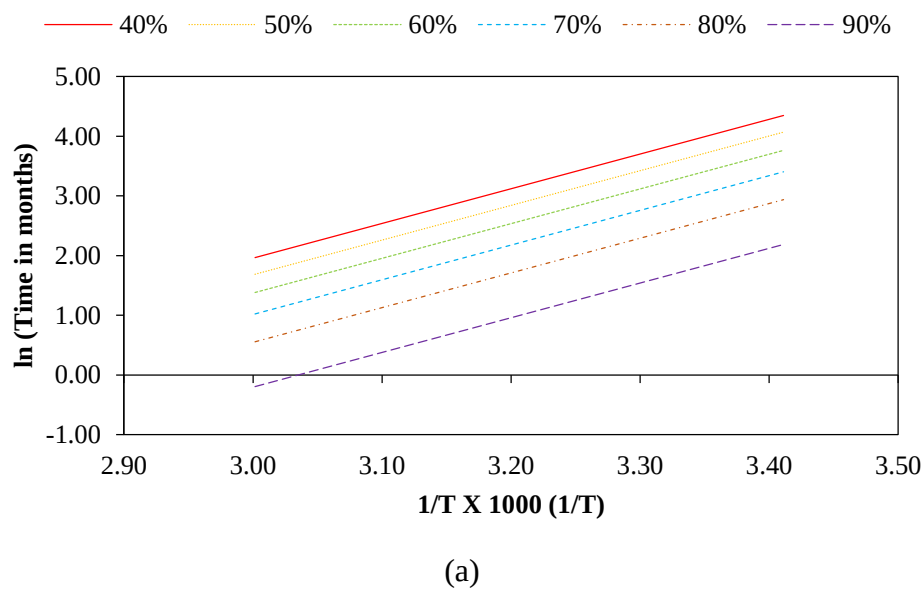
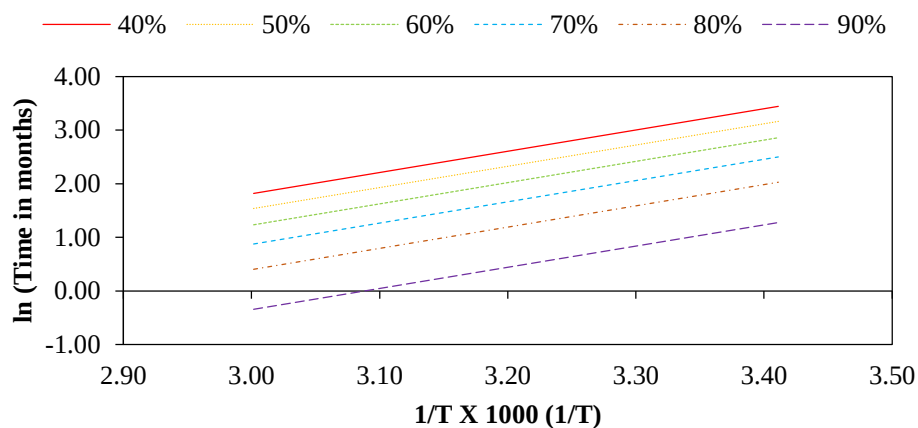


Figure 5.2: Arrhenius-type relationships for unloaded specimens; (a) Type I; (b) Type II





(b)

Figure 5.3: Arrhenius-type relationships for loaded specimens; (a) Type I; (b) Type II

Table 5.4: Coefficients of Arrhenius-type relationships

Load Type	GFRP Type	TSR* (%)	E_a/R	R^2
No Load	Type I	40	5687	0.88
		50	5707	0.89
		60	5702	0.89
		70	5677	0.89
		80	5729	0.89
		90	5586	0.88
	Type II	40	1959	0.99
		50	1954	0.99
		60	1806	0.99
		70	1861	1.00
		80	1977	0.99
		90	1684	0.98
Sustained Load	Type I	40	5831.3	1.00
		50	5831.3	1.00
		60	5831.3	1.00
		70	5831.3	1.00
		80	5831.3	1.00
		90	5831.3	1.00
	Type II	40	3963.3	0.88
		50	3963.3	0.88
		60	3963.3	0.88
		70	3963.3	0.88
		80	3963.3	0.88
		90	3963.3	0.88

*TSR = Tensile strength retention

Values of the (E_a/R) determined from the Arrhenius-type relationships can be used to calculate the time shift factor (TSF) for any reference temperature T_o using Equation 5.5. The TSF for reference temperatures T_o of 20, 40 and 60°C are given in Table 5.5.

$$TSF = e^{\left[\frac{E_a}{R} \left(\frac{1}{T_o} - \frac{1}{T_1} \right) \right]} \quad (5.5)$$

Table 5.5: Values of time shift factor (TSF) for Type I and II GFRP bars

Load Type	GFRP	Temp. c	Temp (K)	1/Temp	$T_o = 20^\circ\text{C}$	$T_o = 40^\circ\text{C}$	$T_o = 60^\circ\text{C}$
Unloaded	Type I	20	293.15	0.0030	1.00	0.29	0.10
		40	313.15	0.0030	3.40	1.00	0.34
		60	333.15	0.0030	10.20	2.97	1.00
	Type II	20	293.15	0.0030	1.00	0.66	0.46
		40	313.15	0.0030	1.50	1.00	0.70
		60	333.15	0.0030	2.15	1.43	1.00
Sustained Load	Type I	20	293.15	0.0034	1.00	0.28	0.09
		40	313.15	0.0032	3.56	1.00	0.33
		60	333.15	0.0030	10.90	3.06	1.00
	Type II	20	293.15	0.0034	1.00	0.42	0.20
		40	313.15	0.0032	2.37	1.00	0.47
		60	333.15	0.0030	5.07	2.14	1.00

The tensile strength retention data shown in Figure 5.1 can be transformed to calculate the equivalent conditioning times at a reference temperature by multiplying the exposure times by the corresponding TSF values. Master curve data at reference temperatures of 20, 40, and 60°C were generated for loaded and unloaded specimens in Tables 5.6 and 5.7, respectively. The master curves of Type I and Type II GFRP at these reference temperatures are plotted Figures 5.4 and 5.5, respectively.

From these figures, it can be seen that the presence of a sustained load of 25% of ultimate strength during conditioning can significantly reduce the service life of GFRP bars. The effect of sustained load was more pronounced at the higher temperatures. The effect of sustained load on the service life of Type II GFRP bars was less

significant compared with its effect on the service life Type I GFRP bars. This occurred possibly because Type II GFRP bars had already exhibited a significant degradation in the tensile strength when conditioned without load due to a severe deterioration of the chemical bonds at the fiber-matrix interface. It seemed that the severe deterioration of chemical bonds at the fiber-matrix interface occurred during conditioning without load was not aggravated much when Type II GFRP bars were conditioned under the sustained load.

Table 5.6: Master curve data for unloaded specimens at reference temperatures of 20, 40, and 60°C

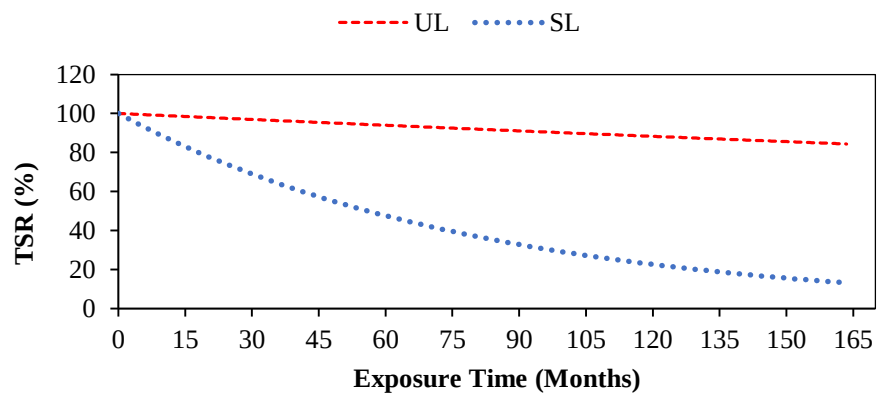
GFRP	Temp. °C	Time (month)	TSR * (%)	TSF			Factored aging time (month)		
				20°C	40°C	60°C	20°C	40°C	60°C
Type I	20	0	100	1.00	0.29	0.10	0.00	0.00	0.00
		5	99	1.00	0.29	0.10	5.00	1.45	0.50
		10	99	1.00	0.29	0.10	10.00	2.90	1.00
		15	98	1.00	0.29	0.10	15.00	4.35	1.50
	40	0	100	3.40	1.00	0.34	0.00	0.00	0.00
		5	94	3.40	1.00	0.34	17.00	5.00	1.70
		10	95	3.40	1.00	0.34	34.00	10.00	3.40
		15	89	3.40	1.00	0.34	51.00	15.00	5.10
	60	0	100	10.20	2.97	1.00	0.00	0.00	0.00
		5	93	10.20	2.97	1.00	51.00	14.85	5.00
		10	92	10.20	2.97	1.00	102.0	29.70	10.00
		15	85	10.20	2.97	1.00	153.0	44.55	15.00
Type II	20	0	100	1.00	0.66	0.46	0.00	0.00	0.00
		5	81	1.00	0.66	0.46	5.00	3.30	2.30
		10	79	1.00	0.66	0.46	10.00	6.60	4.60
		15	73	1.00	0.66	0.46	15.00	9.90	6.90
	40	0	100	1.50	1.00	0.70	0.00	0.00	0.00
		5	77	1.50	1.00	0.70	7.50	5.00	3.50
		10	66	1.50	1.00	0.70	15.00	10.00	7.00
		15	63	1.50	1.00	0.70	22.50	15.00	10.50
	60	0	100	2.15	1.43	1.00	0.00	0.00	0.00
		5	71	2.15	1.43	1.00	10.75	7.15	5.00
		10	50	2.15	1.43	1.00	21.50	14.30	10.00
		15	53	2.15	1.43	1.00	32.25	21.45	15.00

*TSR = Tensile strength retention

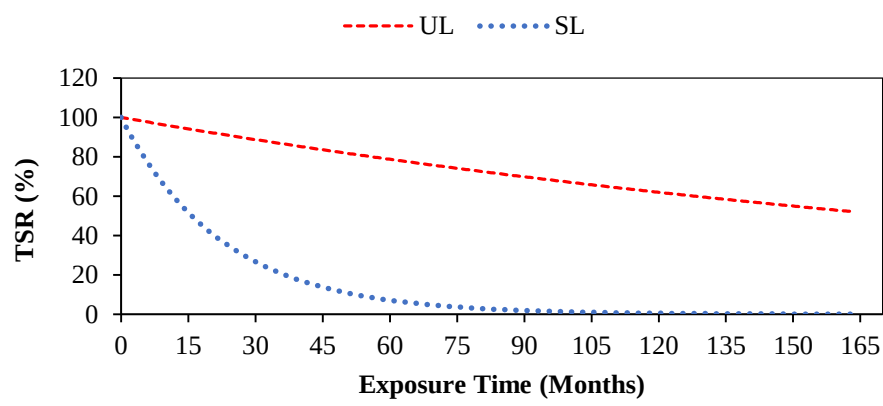
Table 5.7: Master curve data for loaded specimens at reference temperatures of 20, 40, and 60°C

GFRP	Temp. °C	Time (month)	TSR * (%)	TSF			Factored aging time (month)		
				20°C	40°C	60°C	20°C	40°C	60°C
Type I	20	0	100	1.00	0.28	0.09	0.00	0.00	0.00
		5	90	1.00	0.28	0.09	5.00	1.40	0.45
		10	90	1.00	0.28	0.09	10.00	2.80	0.90
		15	84	1.00	0.28	0.09	15.00	4.20	1.35
	40	0	100	3.60	1.00	0.33	0.00	0.00	0.00
		5	78	3.60	1.00	0.33	18.00	5.00	1.65
		10	72	3.60	1.00	0.33	36.00	10.00	3.30
		15	51	3.60	1.00	0.33	54.00	15.00	4.95
	60	0	100	10.90	3.06	1.00	0.00	0.00	0.00
		5	48	10.90	3.06	1.00	54.50	15.30	5.00
		10	-	10.90	3.06	1.00	109	30.60	10.00
		15	15	10.90	3.06	1.00	163.5	45.90	15.00
Type II	20	0	100	1.00	0.42	0.20	0.00	0.00	0.00
		5	84	1.00	0.42	0.20	5.00	2.10	1.00
		10	78	1.00	0.42	0.20	10.00	4.20	2.00
		15	69	1.00	0.42	0.20	15.00	6.30	3.00
	40	0	100	2.37	1.00	0.47	0.00	0.00	0.00
		5	63	2.37	1.00	0.47	11.85	5.00	2.35
		10	60	2.37	1.00	0.47	23.70	10.00	4.70
		15	16	2.37	1.00	0.47	35.55	15.00	7.05
	60	0	100	5.07	2.14	1.00	0.00	0.00	0.00
		5	45	5.07	2.14	1.00	25.35	10.70	5.00
		10	19	5.07	2.14	1.00	50.70	21.40	10.00
		15	22	5.07	2.14	1.00	76.05	32.10	15.00

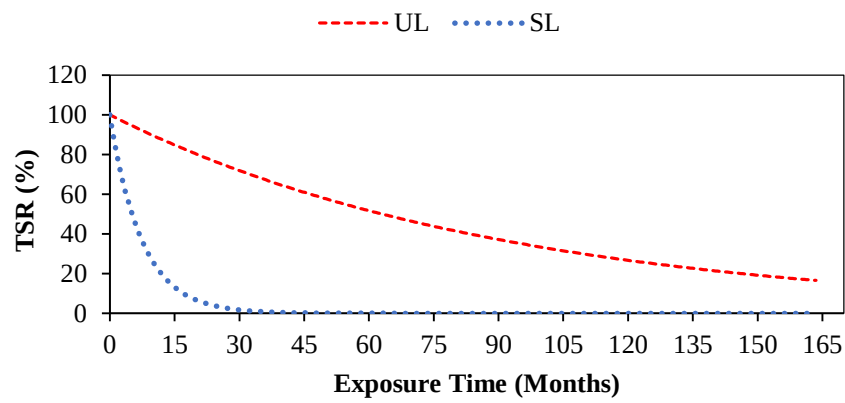
*TSR = Tensile strength retention



(a)

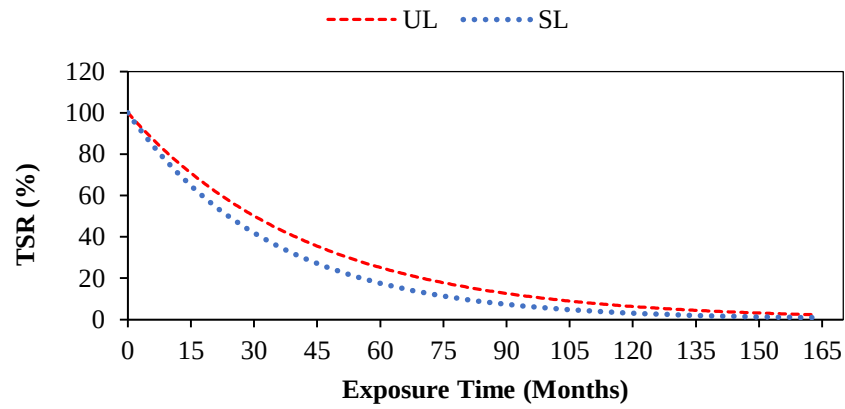


(b)

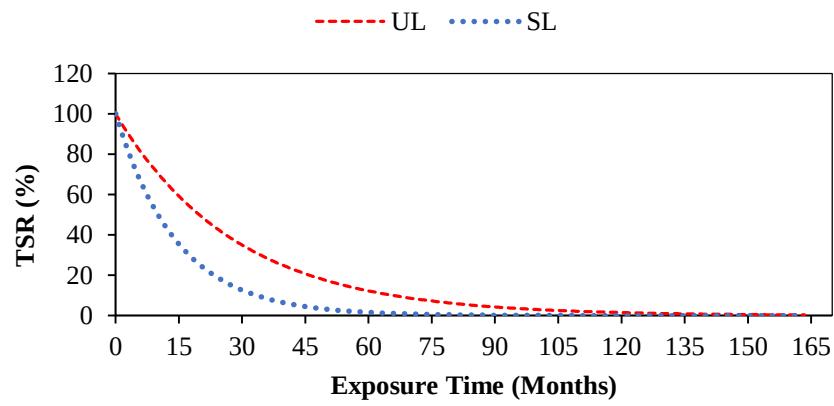


(c)

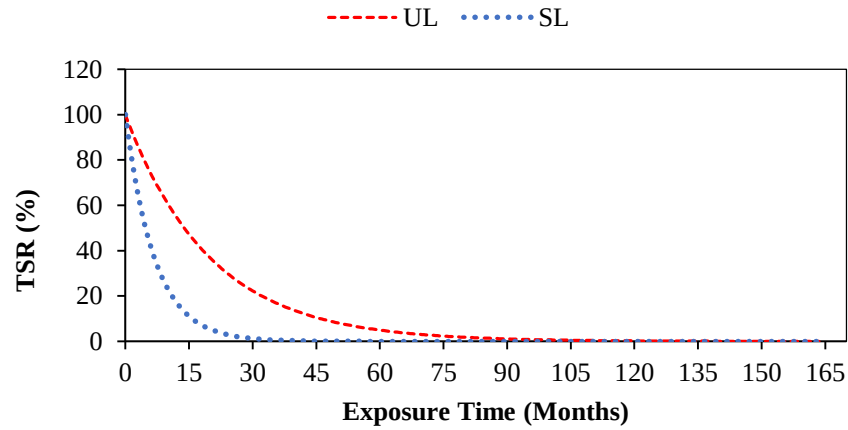
Figure 5.4: Master curves of Type I GFRP bars; (a) at 20°C, (b) at 40°C, (c) at 60°C



(a)



(b)



(c)

Figure 5.5: Master curves of Type II GFRP bars; (a) at 20°C, (b) at 40°C, (c) at 60°C

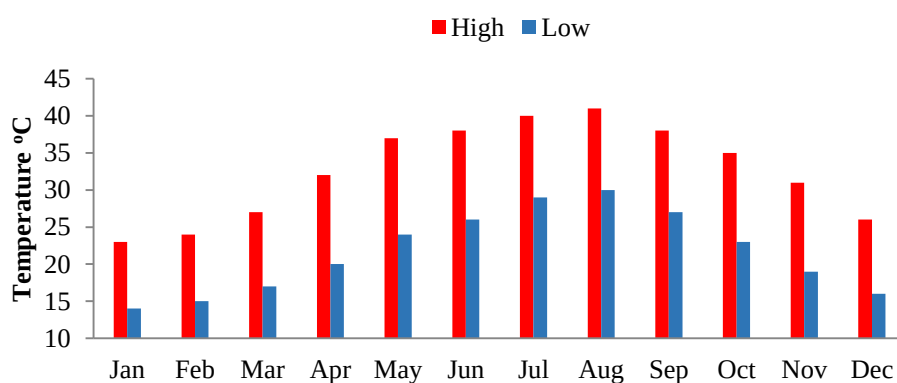
Table 5.8 and Figure 5.6, respectively, show values and plots of the temperature in Dubai and Abu Dhabi over the year (day and night).

Table 5.8: Temperature over the year (day and night) in Dubai and Abu Dhabi
Dubai City [48]

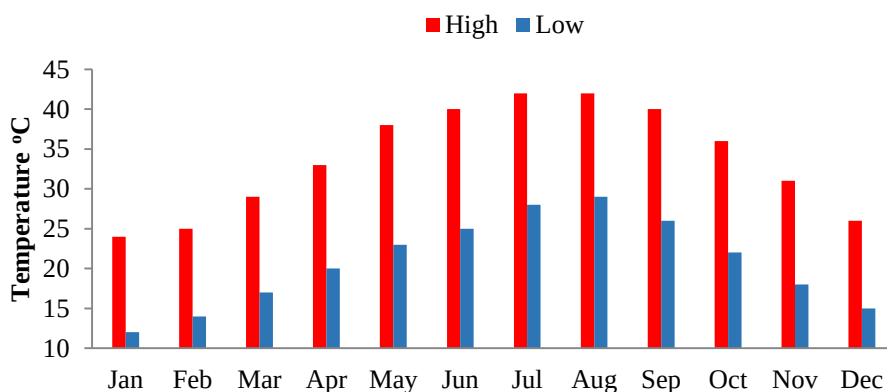
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
High	23	24	27	32	37	38	40	41	38	35	31	26
Low	14	15	17	20	24	26	29	30	27	23	19	16

Abu Dhabi city [49]

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
High	24	25	29	33	38	40	42	42	40	36	31	26
Low	12	14	17	20	23	25	28	29	26	22	18	15



(a) Dubai City



(b) Dubai City

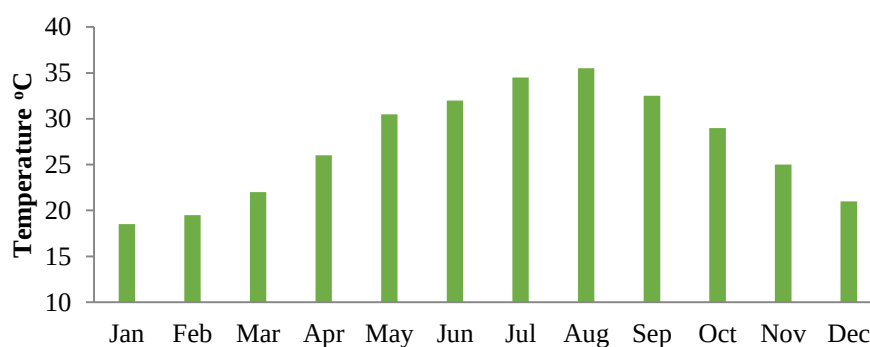
Figure 5.6: Average high and low temperature over the year; (a) Dubai, (b) Abu Dhabi

The average annual temperature in Dubai and Abu Dhabi is approximately over the 27°C (see Table 5.9) according to references [48] and [49]. This annual temperature was used to produce a durability design model for both types of GFRP bars in moist seawater-contaminated concrete structures built in either Dubai or Abu Dhabi.

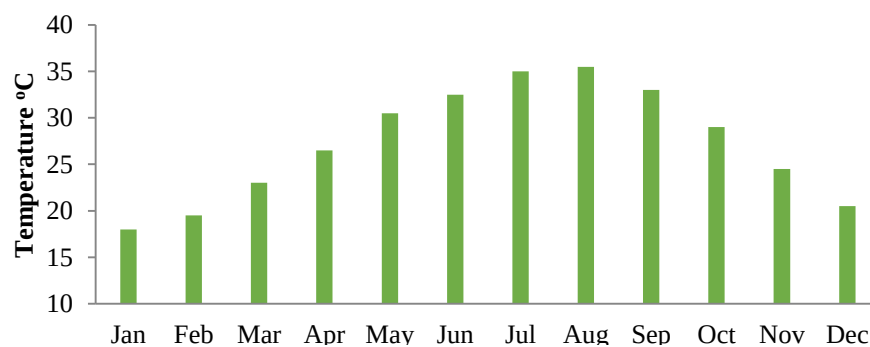
Table 5.9: Average monthly temperature in Dubai and Abu Dhabi over the year

Dubai city												
Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Avg.
19	20	22	26	31	32	35	36	33	29	25	21	27

Abu Dhabi City												
Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Avg.
18	20	23	27	31	33	35	36	33	29	25	21	27



(a) Dubai City



(b) Abu Dhabi City

Figure 5.7: Average temperatures; (a) Dubai, (b) Abu Dhabi

Values of the time shift factor for a reference temperature T_o of 27°C representing the average annual temperature in Dubai and Abu Dhabi City are listed in Table 5.10 for both GFRP bar types. The corresponding durability design models for Type I and Type II GFRP bars are plotted in Figures 5.8 and 5.9, respectively.

From the master curves, it can be seen that the tensile strength retention of Type I GFRP bars exposed to moist seawater concrete without load would drop to 80% after about 10 years. For Type I GFRP bars exposed to moist seawater concrete under a sustained load of 25% of the ultimate strength, the tensile strength would drop to 50% in about 3 years. This demonstrates the detrimental effect of sustained loading on the service life of GFRP bars. The tensile strength retention of Type II GFRP bars would drop to 50% in about 2.5 years when exposed to moist seawater concrete without load and in about 1.5 years when exposed to moist seawater concrete under a sustained load.

GFRP bars tested in the present study were encased in seawater contaminated concrete then continuously immersed in tap water during conditioning. In practical settings, GFRP-reinforced concrete elements are not continuously immersed in water. Typically, they are exposed to different levels of relative humidity. As a result, the master curves developed in this research are considered conservative. The concrete in ambient air conditions is expected to be less aggressive than concrete continuously immersed under water. Mufti et al. [50] reported no degradation in tensile properties of GFRP bars remained under field conditions for up to 8 years. The effect of the relative humidity on the service life of GFRP bars should be taken into consideration in future studies. Field data should also be collected. A relationship between accelerated aging data obtained from laboratory tests and data obtained from real-life service environment should be established.

Table 5.10: Values of TSF for a reference temperature T_o of 27°C

Load Type	GFRP Type	Temp (c)	TSF ($T_o = 27^\circ\text{C}$)
No load	Type I	20	0.64
		40	2.19
		60	6.52
	Type II	20	0.86
		40	1.30
		60	1.86
Sustained Load	Type I	20	0.63
		40	2.24
		60	6.85
	Type II	20	0.73
		40	1.73
		60	3.70

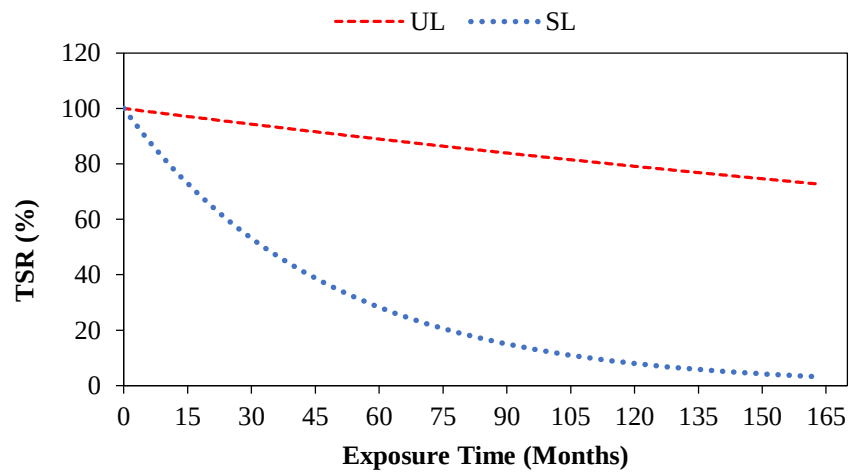


Figure 5.8: Durability design model of Type I GFRP bars in moist seawater-contaminated concrete located in Dubai or Abu Dhabi

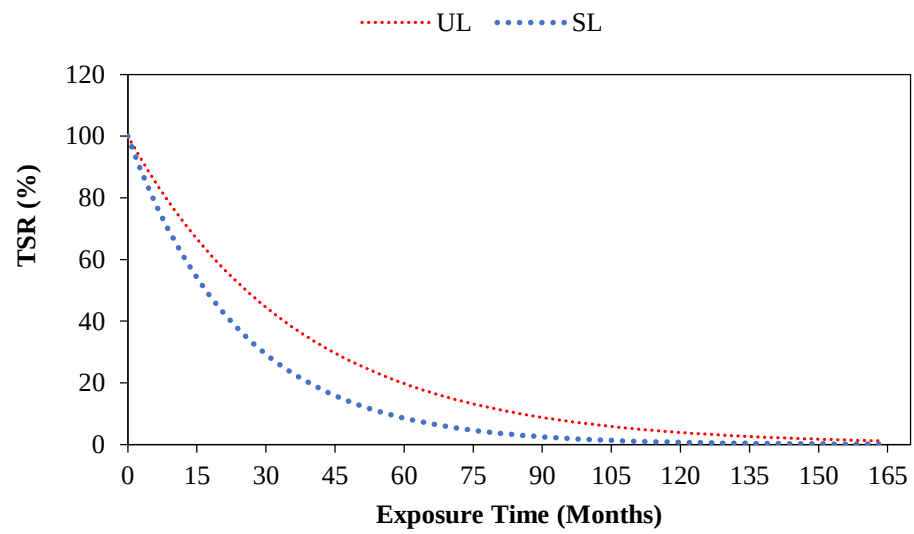


Figure 5.9: Durability design model of Type II GFRP bars in moist seawater-contaminated concrete located in Dubai or Abu Dhabi

Chapter 6: Conclusion and Remarks

6.1 Introduction

The durability performance of two types of commercially-produced GFRP bars conditioned in moist seawater-contaminated concrete has been examined in this work through microstructural characterization and measurements of tensile properties of conditioned and unconditioned specimens. Half of the specimens were conditioned under a sustained load that corresponded to 25% of the ultimate strength whereas the other half were conditioned without load. All specimens failed by rupture of fibers accompanied by fiber debonding at the fiber-matrix interface within the test region. Master curves and durability design models that can predict the tensile strength retention of both types of GFRP bars in moist seawater-contaminated concrete were developed. Findings of the work along with recommendations for future studies are presented in this chapter. Test results and durability design models are limited to the bar types, sizes, and environmental conditions adopted in the current study. Any variation in the specimen's size, manufacturer, and/or surrounding media could change the results, and hence, findings of the work. Although conclusions of this work are limited to the specimens tested in the present study, the methodology adopted can be used to develop master curves and durability design models of other GFRP bar types.

6.2 Conclusions

Results of the present study provided insight into the durability performance of two different types of commercially-produced GFRP bars in moist seawater-contaminated concrete. Aging-related degradation of GFRP composite bars in moist seawater-contaminated concrete is highly dependent on the void content and moisture

absorption properties which are affected by manufacturing procedure, chemical composition of the matrix, characteristics of the interface, and interfacial imperfections that may develop during the manufacturing process. Tensile strength results reported in the present study are consistent with those of the moisture uptake, matrix digestion, and FTIR tests. The agreement between tensile strength results and those of other microstructural tests conducted on both types of GFRP confirms the credibility and validity of the tensile strength test results. Based on the test results, the following conclusions are drawn:

1. The aging-related degradation in the tensile strength of GFRP bars in moist seawater-contaminated concrete was dependent on the void content and moisture absorption properties. Type II GFRP bars had higher void content and moisture uptake than those of Type I. The increased moisture absorption facilitated progression of the hydrolysis reaction, reduced matrix retention, impaired the bond at the fiber-matrix interface, and hence, Type II exhibited inferior durability performance than that of Type I.
2. Superior short-term tensile properties were not indicative of better durability characteristics. Type II GFRP bars, with the higher initial tensile strength, showed inferior tensile strength retentions than those of Type I, with the lower initial tensile strength. The tensile strength reduction caused by accelerated aging was in the range of 2 to 15% for the GFRP bar Type I, and 19 to 50% for GFRP bar Type II.
3. The degradation in properties of GFRP bars was more sensitive to the conditioning temperature rather than conditioning duration. Increasing the conditioning temperature reduced the tensile strength retention at all times of conditioning for both types of GFRP. Increasing the conditioning duration had

an almost no effect on the tensile strength retention of Type I GFRP bars, except at the higher temperature of 60°C where a minor additional strength reduction of 5% was recorded at 15 months of exposure. For Type II, increasing the conditioning time from 5 to 10 months reduced the tensile strength retention. Further increase in the conditioning duration had no or insignificant effect on the tensile strength retention of Type II.

4. Interfacial separation, matrix disintegration, and microgaps were detected in both types of GFRP bars due to conditioning. The matrix disintegration and fiber debonding were intensified with an increase in the conditioning temperature. Type II GFRP bars were more prone to matrix mass loss than Type I, especially at the higher temperatures. After 15 months of conditioning at 60°C, Type I GFRP experienced a matrix retention of 83%, while Type II GFRP featured a matrix retention of 62% only.
5. Absorbed water reduced the T_g corresponding to the first scan due to a plasticizing effect. Plasticization, however, was more affected by temperature than time of conditioning. The T_g of conditioned specimens after the second scan was almost equal to that of the control specimens. This provided an evidence to the absence of irreversible chemical degradation.
6. None of the loaded specimens conditioned at 20°C were creep-ruptured during conditioning. In contrast, many bars were creep-ruptured during conditioning at the higher temperatures of 40 and 60°C.
7. The moisture absorption of GFRP bars conditioned under a sustained load was higher than that of their counterparts conditioned without load. At the same temperature, the moisture absorption of the bars conditioned under load increased by increasing the time of conditioning.

8. At a conditioning temperature of 20°C, Type I GFRP bars exhibited tensile strength retentions of 90, 90, and 84% after 5, 10, and 15 months respectively. Their Type II GFRP counterparts exhibited lower tensile strength retentions of 84, 78, and 69%, respectively, indicating that Type II GFRP bars had inferior durability performance than that of Type I.
9. Many GFRP bars from both types were creep-ruptured when conditioned under a sustained at temperatures of 40 and 60°C. As a result, the average tensile strength retention of the corresponding three replicate bars were significantly reduced. For instance, at 5 months of conditioning under load at 60°C, one bar was creep-ruptured from each type and hence, Type I and Type II GFRP bars exhibited average tensile strength retentions of 48 and 45%, respectively. At 15 months of conditioning under a sustained load at 60°C, two bars were creep-ruptured from each type and hence, average tensile strength retentions of 15 and 22% only were recorded for Type I and Type II GFRP bars, respectively.
10. The tensile strength retentions of Type I GFRP bars conditioned at 20°C under a sustained load for 5, 10, and 15 months were 8, 8, and 14% lower than those of their counterparts conditioned without load. The effect of sustained load was more significant at the higher temperatures. The tensile strength retentions of Type I GFRP bars conditioned under load for 10 and 15 months at 40°C were approximately 24 and 45% lower than those of their counterparts conditioned without load. Type II GFRP bars were not affected by the presence of sustained load during conditioning at 20°C for 5 and 10 months. At 15 months, the tensile strength retentions of Type II GFRP bars conditioned under load at 20°C was approximately 5% lower than that of their counterparts conditioned without load. The presence of a sustained load during conditioning reduced, however,

the tensile strength retentions of the specimens conditioned at the higher temperatures. The tensile strength retentions of Type II GFRP bars conditioned under load for 5 and 10 months at 40°C were approximately 18 and 6% lower than those of their counterparts conditioned without load.

11. Master curves that can predict the tensile strength retentions at 20, 40, and 60°C for the two types of GFRP bars tested in the current study were developed. Durability design models that can predict the long-term performance of both types of GFRP bars in moist seawater-contaminated concrete located in coastal cities were developed. The durability design models can be used to predict the tensile strength retention of both GFRP bar types in moist concrete structures subjected to seawater splash in coastal cities.

6.3 Recommendations for Future Studies

Future research should focus on studying the fire resistance of both types of GFRP bars. The durability performance and microstructural characteristics of GFRP bars conditioned under different levels of sustained loads should be rigorously assessed before GFRP bars can be routinely used as reinforcement in concrete structures exposed to severe environment.

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List of Publications

1. El-Hassan, H., El-Maaddawy, T., Al-Sallamin, A., and Al-Saidy, A. (2017). "Performance evaluation and microstructural characterization of GFRP bars in seawater-contaminated concrete." *Construction and Building Materials*, 147, 66-78.